modenanalyse_2fe2s

Version 1.2.1

Manual and Theoretical Documentation

Mode-resolved bonding reorganization analysis
of [2Fe-2S] iron-sulfur proteins
from Gaussian-16 and ORCA-6 frequency calculations

Lukas Knauer
AG Schünemann
RPTU Kaiserslautern-Landau

Cluster type: [2Fe-2S], reduced or oxidized,
for Rieske-, ferredoxin- and mitoNEET-type ligation

Input: Gaussian-16 `.log` (`freq=hpmodes` preferred;
standard block as fallback) or ORCA-6 `.hess`
PDB file with the same cluster (optional, recommended)

Output: Excel reports (mode analysis, SCSD,
reorganization energies, modulation spectra,
UMAP cluster), report (plain text), plots (PNG)

External packages: `numpy`, `scipy`, `pandas`,
`matplotlib`, `openpyxl`,
`scikit-learn`, `umap-learn`,
`hdbscan`, `scsdpy`

References: Marcus, *J. Chem. Phys.* **24**, 966 (1956)
Marcus, *Angew. Chem. Int. Ed. Engl.* **32**, 1111 (1993)
Hammes-Schiffer & Soudackov,
J. Phys. Chem. B **112**, 14108 (2008)
Huang & Rhys,
Proc. R. Soc. Lond. A **204**, 406 (1950)
Kingsbury & Senge, *Chem. Sci.* **15**, 13638 (2024)

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Part I

Introduction and Theory

1 Introduction

`modenanalyse_2fe2s` is a scientific Python tool for analyzing normal modes from DFT frequency calculations (`freq=hpmodes`) of iron-sulfur proteins containing [2Fe-2S] clusters. It combines geometric classification, rigorous symmetry decomposition, and mode-resolved bond reorganization analysis in a unified workflow. Application areas include mechanistic studies of proton-coupled electron transfer (PCET), comparison of Rieske- and ferredoxin-type systems, and the analysis of mutational effects (e.g., HH→CC).

1.1 What `modenanalyse_2fe2s` provides

The tool focuses on the structural analysis of vibrational modes in DFT frequency calculations of [2Fe-2S] clusters. The main per-mode results are:

- **Cluster-specific classification**: detection of the [2Fe-2S] core, classification of each mode according to out-of-plane / in-plane character, and symmetry decomposition into $D_{2\mathrm{h}}$ irreducible representations.
- **Ligand resolution**: PDB-based identification of coordinating His, Cys, and Asp/Glu residues, with per-mode Fe-N, Fe-S, and Fe-O stretching contributions.
- **Marcus-Hush reorganization energies**: per-mode Δr_X and λ_X for the bonding channels FeFe, FeN, FeS, NH, and HA. System-level aggregates (total reorganization, modulation spectra, cumulative $\Lambda_X(\omega)$) for band identification, processed externally e.g. in Origin.
- **Multi-cluster support**: detection of multiple [2Fe-2S] clusters in the same system (e.g., glutaredoxin dimer) and their parallel evaluation as a first-class feature.
- **Embeddings** (UMAP, HDBSCAN clustering): two-dimensional projection of the mode feature space for exploratory analysis.
- **ORCA support** (standalone `.hess` parser): mode analysis directly from ORCA frequency calculations.

NIS spectra, Fe-pDOS, the Lamb-Mößbauer factor, and Fe-projected thermodynamics are *not* part of this package. These quantities can be computed from the same DFT frequency calculation using specialized external tools.

Changelog highlights since v1.0.0. This release of the manual reflects the package at version v1.2.1 (see `CHANGELOG.md` for the full list). The substantive deltas since v1.0.0 are:

- **v1.0.2**: `_build_group_map` fix – prevents silent zero rows in the `Groups_OOP/INP/Tors` sheets when residue numbers overlap with non-protein entries (e.g. HOH waters). If you are running protein systems that contain solvent residues, you need v1.0.2 or later.
- **v1.0.3**: new diagnostic outputs – the `Coordination` sheet documents the ligand and group atom mapping (§E.1); two additional UMAP embeddings (SSE-UMAP and Ca-UMAP) target protein-domain modes and spatially localised modes respectively (§E.2, §E.2).
- **v1.0.4**: audit-driven uncertainty propagation fix (§2.3). Values unchanged; the `s_*`-columns for N-H stretches, SSE-element amplitudes, and SCSD amplitudes shift in magnitude. Half-open window boundaries and defense-in-depth warnings are also part of this release.
- **v1.0.5**: physics corrections to the secondary-structure element analysis – the SSE-UMAP `bending_*` features (which were always zero) were replaced by the real `lateral_*` keys; `tilting_angle` is now a genuine rigid-body tilt (rocking rotation perpendicular to the axis); `com_amplitude` (per element and for the cluster core) is mass-weighted; and the SSE principal axis is taken from the Cα backbone. The SSE sheets and their UMAP change, so affected systems must be re-run once; the SSE-UMAP feature set was also reduced to a decorrelated five-metric subset.
- **v1.1.0**: the abbreviation *SS* (secondary structure) was renamed project-wide to *SSE* – sheet names, the output filename (`*_analysis_SSE.xlsx`), configuration keys and function names. This removes the clash with disulfide (S–S) and matches the conventional term *secondary structure element*. Legacy TOML keys (`analyze_ss`, `ss_chain`) remain accepted for backward compatibility.
- **v1.2.0**: critical-review release. Multi-cluster runs no longer abort the whole process on the first failing cluster; the SCSD symmetry decomposition (which had become a silent no-op against the published `scsdpy`) is restored; PCET hydrogen-bond detection gains a D–H$\cdots$A angle criterion ($\geq 120°$) and a corrected acceptor-distance weighting; His protonation is restricted to the correct residue; Kabsch cluster alignment uses element-matched anchors with an RMSD acceptance gate; and mass weighting now uses the QM program's own atomic masses. Several of these change numerical output, so affected systems should be re-run and headline reorganization numbers re-validated. A default-run regression pins $\Lambda_{\mathrm{FeFe}}/\Lambda_{\mathrm{FeS}}$ for the reference fixture.
- **v1.2.1**: ORCA reduced masses are reconstructed as $\mu_k = \sum_i m_i\,|l_{i,k}|^2$ (verified against real ORCA `.hess` files and confirmed to match the Gaussian "Red. mass" convention), replacing a `1.0` amu placeholder that made absolute ORCA amplitudes too large by $\sqrt{\mu}$. The Gaussian path is unaffected.

1.2 Prerequisites

- Python ≥ 3.11 with `numpy`, `scipy`, `openpyxl`, `matplotlib`, `pandas`, `scikit-learn`, `umap-learn`, `hdbscan`, and `scsdpy`. All are installed *automatically* via the wheel.
- A **Gaussian-16 log file** from a `freq=hpmodes` calculation of the optimized cluster (with at least four optimized heavy atoms: two Fe and two bridging S) **or** an **ORCA-6 `.hess` file**.
- Optional: a **PDB file** containing the same cluster, for ligand resolution and secondary-structure analysis.

1.3 Conventions used in this manual

- **Axes**: wherever possible, the canonical cluster orientation is used: $\hat{\mathbf{x}}$ along Fe-Fe, $\hat{\mathbf{y}}$ along S-S, $\hat{\mathbf{z}} = \hat{\mathbf{n}}$ as the cluster normal. This is consistent with the SCSD reference geometry (see §2.6).
- **Units**: cm^{-1} for frequencies, Å for lengths, AMU for masses, meV for energies, Kelvin for temperatures. For Boltzmann factors we use $1/k_\mathrm{B} = 11.6048\,\mathrm{K/meV}$.
- **Heuristic vs. rigorous**: whenever a quantity is derived from a heuristic (e.g., the motion-pattern scores breathing, hinge, ...), it is explicitly labeled as *heuristic*. Rigorous symmetry decomposition is provided by SCSD.

Reading tip. First-time users should read §1 and §3, and start with the example in §4. The theoretical chapters (§2) can be consulted as needed but are not required for routine use.

2 Theoretical foundations

2.1 Normal modes from Gaussian: the `hpmodes` format

A DFT frequency calculation in Gaussian-16 yields, after a successful geometry optimization, the second derivative of the energy with respect to the Cartesian atomic coordinates – the mass-weighted Hessian $\mathbf{F}$. Diagonalizing $\mathbf{F}$ gives the eigenvectors $\mathbf{L}$ and eigenvalues ω_l^2 of the normal modes [11]:

$$\mathbf{F}\,\mathbf{L}_l = \omega_l^2\,\mathbf{L}_l, \qquad l = 1, \ldots, 3N-6 \tag{1}$$

with N the number of atoms in the cluster (more precisely, in the optimized model region). The Cartesian displacements per atom i in mode l are obtained by mass-unweighting [11]:

$$\mathbf{e}_{l,i} \;=\; \frac{1}{\sqrt{m_i}}\,\mathbf{L}_{l,i}, \tag{2}$$

where m_i is the atomic mass of atom i.

HP block versus standard block

With the keyword `freq=hpmodes`, Gaussian writes the eigenvectors in a second block (*High-Precision Modes*) with five decimal places. The *standard block*, by contrast, uses only two decimal places and is insufficient for very small displacements (e.g., modes below 100 cm^{-1} in large systems).

`modenanalyse_2fe2s` reads both blocks. The HP eigenvectors are preferred; only when a single HP block is missing (rare, e.g., in mixed-version log files) does the reader fall back to the standard block. This fallback logic is implemented in `logio.read_blocks`.

Consistency check between HP and standard

Both blocks output the frequencies ω_l independently. `modenanalyse_2fe2s` checks at run time whether the HP and standard frequencies agree for matching mode indices (`logio.check_hp_std_frequency_co`
tolerance $\Delta\omega \leq 0{,}01\ \mathrm{cm}^{-1}$). A discrepancy indicates either:

- an incompletely written log file (job aborted),
- a mismatch between the read offset and the actual block start (e.g., when a job was continued via `Restart`),
- or two log files mixed up.

Consistency errors are written as warnings to the report; the program continues running but flags the affected modes in the report.

Eigenvector scaling

For all subsequent analyses, the raw Gaussian eigenvectors are scaled by the RMS amplitude $u_{\mathrm{rms}}(\omega_l, m_{\mathrm{red}}, T)$ (`core.compute_thermal_amplitude`) [5, 11]:

$$u_{\mathrm{rms}}(\omega, m_{\mathrm{red}}, T) \;=\; \sqrt{\frac{\hbar}{2\,m_{\mathrm{red}}\,\omega}\,\coth\!\left(\frac{\hbar\omega}{2k_{\mathrm{B}}T}\right)}. \quad \rightsquigarrow \texttt{core.compute\_thermal\_amplitude} \tag{3}$$

Here m_{red} is the reduced mass that Gaussian provides for each mode. At $T = 0$, equation (3) reduces to the zero-point amplitude $u_{\mathrm{rms}} = \sqrt{\hbar/(2m_{\mathrm{red}}\omega)}$.

The eigenvectors multiplied by u_{rms} have units of Å and describe the thermally averaged (or, at $T \to 0$, purely quantum-mechanical) displacement per mode. They are the input for all geometric analyses from §2.4 onwards.

2.2 Uncertainty propagation

Every numerical observable that the tool reports (`stretch`, `bend`, `kern_d`, `amplitude_mean`, `hn_stretch`, the SCSD-amplitudes, ...) comes with a companion uncertainty (`s_stretch`, `s_bend`, `s_kern_d`, ...). These are first-order Gaussian error propagations from two underlying input noise sources:

- $\sigma_e :=$ `cfg.sigma_eigvec` (default $5 \cdot 10^{-4}$): numerical noise per eigenvector component, in Cartesian units of the un-scaled Gaussian eigenvector. Controls the "insignificant contribution" threshold for displacement-based observables.
- $\sigma_c :=$ `cfg.sigma_coord` (default 10^{-3} Å): coordinate noise in Å, used for SCSD reference uncertainty.

Convention: thermally scaled sigma

All geometric observables consume the *thermally scaled* eigenvector $\mathbf{e}_{l,i}\, u_{\mathrm{rms}}(\omega_l)$, which has units of Å. The companion sigma is propagated at the same scaling:

$$\sigma_e^{\mathrm{therm}}(l) \;=\; \sigma_e \cdot u_{\mathrm{rms}}(\omega_l), \qquad \rightsquigarrow \texttt{core.analyze\_his\_hn, core.analyze\_fe\_ligand} \tag{4}$$

so that the dimensionless *significance ratio* $|\text{observable}|\,/\,\sigma$ is invariant under the choice of u_{rms} for a given physical quantity. This convention is essential for the value/sigma ratios used in the colour coding of the Excel sheets.

Worked examples. For a typical low-frequency mode with $u_{\mathrm{rms}} \approx 0.05$ Å:

- Fe-X stretch (i = ligand, j = Fe): $|\mathbf{e}_i - \mathbf{e}_j|$ is a difference of two independent Gaussian-noise components, so $\sigma(\text{stretch}) = \sigma_e^{\mathrm{therm}}\sqrt{2} \approx 3.5 \cdot 10^{-5}$ Å.
- Bend INP/OOP: same factor $\sqrt{2}$ (Pythagorean decomposition does not change the noise propagation along any single axis), so $\sigma_{\mathrm{bend\_inp}} = \sigma_{\mathrm{bend\_oop}} = \sigma_e^{\mathrm{therm}}\sqrt{2}$.
- N-H stretch (`analyze_his_hn`, `include_hn_vibration`) follows the same convention: $\sigma_{\mathrm{hn}} = \sigma_e^{\mathrm{therm}}\sqrt{2}$.
- SSE-element amplitude: $\sigma_{\mathrm{amp}} = \sigma_e^{\mathrm{therm}}\sqrt{N_{\mathrm{atoms}}}$, the $\sqrt{N}$ growth coming from the averaging over N_{atoms} in the secondary-structure element.
- SCSD amplitudes: distinct "reference" and "distortion" sigmas. The reference is the static geometry uncertainty $\sigma_c\sqrt{2}$, the distortion combines reference and thermal noise: $\sigma_{\mathrm{SCSD,dist}} = \sqrt{2}\sqrt{\sigma_c^2 + (\sigma_e \cdot u_{\mathrm{rms}})^2}$.

What changed in v1.0.4. Three of these sigma propagations were previously implemented inconsistently (§2.3): N-H stretch lacked the thermal scaling, SSE-element used a hard-coded constant in place of `cfg.sigma_eigvec`, and SCSD used hard-coded reference constants in place of `cfg.sigma_coord` and `cfg.sigma_eigvec`. With v1.0.4, all sigmas honour Equation 4 and the `cfg.sigma_eigvec/sigma_coord` settings from the TOML.

2.3 Migration notes for v1.0.4

Version 1.0.4 is a bug-fix release that affects the *reported uncertainties* of several observables but leaves the underlying *values* unchanged. If you have v1.0.2 or v1.0.3 results in hand (e.g. for a paper or thesis), please re-read this section before drawing significance conclusions from the older output.

What did not change. All geometric and energetic observables – `stretch`, `bend`, `kern_d`, `lambda_X`, SCSD amplitudes, `hn_stretch`, SSE-element `amplitude_mean`, embedding coordinates – produce *numerically identical* values across v1.0.2, v1.0.3 and v1.0.4 for the same input log. No physical interpretation of existing values is invalidated.

What changed.

- `s_hn_stretch`: now scales with u_{rms} (Eq. 4). For typical low-frequency modes ($u_{\text{rms}} \sim$ 0.05 Å) this makes `s_hn_stretch` about **20 × smaller** than in v1.0.2/v1.0.3. Consequence: many N-H stretches that appeared "below noise" under the old convention are now correctly identified as significant PCET signals.
- `s_amplitude_mean` / `s_axial_amplitude` / `s_tilting_angle` ... in `analyze_sse_element`: pre-v1.0.4 used a hard-coded $1 \cdot 10^{-4}$ in place of `cfg.sigma_eigvec`. With the default $\sigma_e = 5 \cdot 10^{-4}$, the new sigmas are about **5 × larger**. Re-classification: SSE-element entries that were marked "significant" under the old code may now read as "marginal".
- `s_SCSD_*`: pre-v1.0.4 used hard-coded $5 \cdot 10^{-7}$ / $5 \cdot 10^{-6}$ in place of `cfg.sigma_coord` and `cfg.sigma_eigvec`. New sigmas are about **1000 × larger**; the previous values were physically implausible.
- Window-edge handling: multi-window mode (`freq_windows`) used *closed-closed* intervals $[lo, hi]$, so a mode with frequency exactly equal to a window boundary appeared in *two* adjacent windows. As of v1.0.4 the boundaries are *half-open* $[lo, hi)$, with the last window remaining $[lo, hi]$. Practical effect: none for typical float-frequency runs, but B-factor and $\Lambda(\omega_{\text{cut}})$ are now guaranteed double-count free.
- Defense-in-depth warnings: a number of internal lookup failures (Fe ligand *c2l* miss, His-H *c2l* miss, missing PCET acceptor index, all-zero output rows in the Excel sheets, ambiguous PDB-Gaussian atom matches) used to be silent fall-throughs. With v1.0.4 each of these emits a `UserWarning` the first time it occurs in a run (deduplicated per session). If you see

new warnings in a previously-quiet log, check the affected ligand or atom mapping before trusting the corresponding Excel rows.

- Interpolation boundary anchor (FUND 12, user-reported): when `freq_max` is set and the next mode above it lies farther than `interp_context_cm1` (default 5 cm^{-1}), pre-v1.0.4 loaded no context mode and the interpolated pDOS dropped to zero just below `freq_max`. With v1.0.4, exactly *one* additional mode – the closest above `freq_max` – is loaded as a fallback anchor, so `np.interp` always has a right-edge anchor regardless of mode density. The same fallback applies symmetrically at `freq_min`. The run-log records when the fallback fires.
- Synthetic zero anchor when the DFT spectrum ends inside the analysis range (FUND 13): an extension of the above. If *no* real mode exists above `freq_max` at all (the DFT spectrum genuinely terminates), v1.0.4 inserts a *synthetic null-mode* at `freq_max`+`interp_context_cm1` with all observable fields set to zero. This gives the interpolated pDOS a smooth decay across the buffer zone instead of a step at the upper edge. The synthetic mode is flagged with `mode_type = "synthetic_zero"` and `precision = "synthetic"` so it is unambiguous in any per-mode listing. By construction it contributes zero to B-factors, $\Lambda_X^{\mathrm{total}}$, and the modulation spectra, making the fix conservative.

Post-release audit patches (May 2026). After v1.0.4 was released, an audit on a real-world [2Fe-2S]-Cys_2His_2 batch run (analysing both the protonated and the deprotonated states of a Rieske-type system) identified four additional warning-and-reporting issues. None of them affects numerical output. Full descriptions are in the supplement (FUND 14–17); briefly:

- FUND 14: The “`His_HN NICHT erkannt`” warning fired twice per deprotonated His instead of once. Fixed by hoisting the warning out of the PDB-vs-Gaussian-fallback loop.
- FUND 15: For systems with deprotonated His ligands (e.g. Rieske-type Cys_2His_2 in the deprotonated state, or mitoNEET-type Cys_3His after deprotonation), the PCET-disabled message stated “`no His ligands at cluster`” even though `Fe_N_His` sheets and `Lambda_FeN` confirmed His ligands were present. The decision text now distinguishes “no His” from “His present but all deprotonated”.
- FUND 16: The “`interp_boundary_mode='context' but no context modes available`” warning fired even for full-spectrum runs (`freq_min`/`freq_max` both `None`), where no context-loading is attempted. The warning is now suppressed unless the user has explicitly set a window.
- FUND 17: REPORT.txt showed `freq_filter:  - - - cm-1` when only `freq_windows` was set, hiding the active filter from the run audit trail. The report now emits a separate `freq_windows:  [...]` line.

Re-running existing data. The recommended path: rerun your existing TOML configurations under v1.0.4 and compare the `s_*` columns of `Fe_S_Cys.xlsx`, `His_HN.xlsx`, `SCSD.xlsx`, and the SSE-Element sheets. Values are identical; only sigmas and the significance-colouring change.

2.4 OOP/INP classification

A central question for every cluster mode is: *how much of the vibration is perpendicular to the cluster plane (out-of-plane, OOP), and how much lies in the plane (in-plane, INP)?* Pure rocking motions of the four cluster atoms (umbrella, hinge) are OOP-dominated, whereas stretching modes (Fe-Fe, S-S, breathing) live in the plane.

Definition of the cluster normal

Let $\mathbf{r}_i$ ($i \in \{\mathrm{Fe}_1, \mathrm{Fe}_2, \mathrm{S}_1, \mathrm{S}_2\}$) be the equilibrium positions of the four cluster atoms. The normal direction of the plane is the direction that minimizes the sum of squared perpendicular components. Concretely, it is obtained from the singular-value decomposition of the centered cluster coordinates $\tilde{\mathbf{r}}_i = \mathbf{r}_i - \bar{\mathbf{r}}$ [17]:

$$\tilde{\mathbf{R}} \;=\; \mathbf{U}\,\boldsymbol{\Sigma}\,\mathbf{V}^{\top}, \qquad \hat{\mathbf{n}} \;=\; \mathbf{V}_{:,3}\,. \quad \rightsquigarrow \texttt{geometry.cluster\_normal} \tag{5}$$

This is implemented as `geometry.cluster_normal`. By convention, $\hat{\mathbf{n}}$ points in the $+\hat{\mathbf{z}}$ direction of the Gaussian frame, so that a uniform sign convention applies across all modes.

Mode component perpendicular to the plane

For mode l and an atom i, the OOP component is the projection of the displacement onto $\hat{\mathbf{n}}$ (standard decomposition, cf. [7]):

$$\mathbf{e}_{l,i}^{\mathrm{OOP}} \;=\; (\mathbf{e}_{l,i} \cdot \hat{\mathbf{n}})\,\hat{\mathbf{n}}, \qquad \mathbf{e}_{l,i}^{\mathrm{INP}} \;=\; \mathbf{e}_{l,i} - \mathbf{e}_{l,i}^{\mathrm{OOP}}. \tag{6}$$

The *OOP fraction* of the mode on an atom group G is the ratio of the squared norms [7]:

$$\alpha_{\mathrm{OOP}}(G;l) \;=\; \frac{\sum_{i\in G}\|\mathbf{e}_{l,i}^{\mathrm{OOP}}\|^2}{\sum_{i\in G}\|\mathbf{e}_{l,i}\|^2} \cdot 100\,\%\,. \quad \rightsquigarrow \texttt{core.classify\_oop\_inp} \tag{7}$$

Note on the physical interpretation. Gaussian's `hpmodes` block writes *Cartesian-displacement* eigenvectors $\mathbf{e}_{l,i}$ (already mass-unweighted via Eq. 2), normalised so that $\sum_i \|\mathbf{e}_{l,i}\|^2 \approx 1$ rather than the mass-weighted norm $\sum_i m_i \|\mathbf{e}_{l,i}\|^2$. Equation 7 therefore measures the *geometric Cartesian out-of-plane fraction*: it tells you what share of the squared Cartesian displacement amplitudes points out of the cluster plane. It does *not* equal the kinetic-energy fraction of the mode flowing out of the plane, which would require the mass-weighted form $\sum_i m_i \|\mathbf{e}_{l,i}^{\mathrm{OOP}}\|^2 / \sum_i m_i \|\mathbf{e}_{l,i}\|^2$. For a [2Fe-2S] cluster the two quantities are numerically similar (only Fe and S contribute, with comparable masses), but the distinction matters when comparing α_{OOP} values across chemically different systems.

Binary and fine classification

Classically, $\alpha_{\mathrm{OOP}}(G_{\mathrm{cluster}}; l)$ is subjected to a threshold test:

$$\text{mode\_type}(l) = \begin{cases} \text{OOP}, & \alpha_{\mathrm{OOP}} \geq \theta_{\mathrm{OOP}} \\ \text{INP}, & \alpha_{\mathrm{OOP}} \leq 100\,\% - \theta_{\mathrm{OOP}} \\ \text{mixed}, & \text{otherwise} \end{cases} \tag{8}$$

with default $\theta_{\mathrm{OOP}} = 60\,\%$ (configurable via `cfg.mode_type_threshold`).

In addition, there is a *fine classification* with three thresholds $(\theta_1, \theta_2, \theta_3)$ (`core.classify_oop_inp`), default $(60, 75, 90)$:

Level	Definition
Pure OOP	$\alpha_{\mathrm{OOP}} \geq \theta_3 = 90\,\%$
Strong OOP	$\theta_2 \leq \alpha_{\mathrm{OOP}} < \theta_3$, i.e. $75 - 90\,\%$
Majority OOP	$\theta_1 \leq \alpha_{\mathrm{OOP}} < \theta_2$, i.e. $60 - 75\,\%$
Mixed	$\lvert \alpha_{\mathrm{OOP}} - 50\,\% \rvert < \theta_1 - 50\,\%$
Majority INP	$100\,\% - \theta_2 < \alpha_{\mathrm{OOP}} \leq 100\,\% - \theta_1$, i.e. $25 - 40\,\%$
Strong INP	$100\,\% - \theta_3 < \alpha_{\mathrm{OOP}} \leq 100\,\% - \theta_2$, i.e. $10 - 25\,\%$
Pure INP	$\alpha_{\mathrm{OOP}} \leq 100\,\% - \theta_3 = 10\,\%$

The fine classification provides a more precise statement about the character of a mode in the Excel sheet – particularly in the OOP-hinge regime, where a binary 60 % threshold is either too permissive (modes between 60 and 75 % look very different from each other) or returns a fuzzy answer.

Which threshold for which question? For comparison with NRVS spectra in polarization geometry, the binary 60 % threshold provides sufficient discrimination. For manuscript tables of mode properties (e.g., D_{2h} symmetry candidates), the fine 7-level classification is more informative; with it, "pure" OOP / INP modes can be reliably identified.

Worked example: computing OOP% for a cluster breathing mode

To illustrate, we compute α_{OOP} explicitly for a hypothetical cluster breathing mode in a Cys_4 model cluster.

Step 1 – geometry and cluster normal. The four cluster atoms sit, in idealized form, at (units of Å):

$$\begin{aligned}\mathbf{r}_{\mathrm{Fe}_1} &= (+1.50,\ 0.00,\ 0.00)\\ \mathbf{r}_{\mathrm{Fe}_2} &= (-1.50,\ 0.00,\ 0.00)\\ \mathbf{r}_{\mathrm{S}_1} &= (0.00,\ +1.10,\ 0.00)\\ \mathbf{r}_{\mathrm{S}_2} &= (0.00,\ -1.10,\ 0.00)\end{aligned}$$

Centroid $\bar{\mathbf{r}} = \mathbf{0}$. The centered coordinates all lie in the xy plane; SVD yields $\hat{\mathbf{n}} = (0, 0, 1)$ (cluster normal parallel to the z axis).

Step 2 – mode eigenvectors (thermally scaled, $T = 5\,\mathrm{K}$). A pure **breathing mode** stretches and contracts both Fe-S diagonals synchronously in the plane. A typical thermally scaled eigenvector looks as follows (after $\mathbf{e} \to \mathbf{e} \cdot u_{\mathrm{rms}}$):

$$\begin{aligned}\mathbf{e}_{\mathrm{Fe}_1} &= (+0.012,\ +0.009,\ 0.000)\\ \mathbf{e}_{\mathrm{Fe}_2} &= (-0.012,\ -0.009,\ 0.000)\\ \mathbf{e}_{\mathrm{S}_1} &= (0.000,\ -0.018,\ 0.000)\\ \mathbf{e}_{\mathrm{S}_2} &= (0.000,\ +0.018,\ 0.000)\end{aligned}$$

Step 3 – extract the OOP components. For each atom i: $\mathbf{e}_i^{\mathrm{OOP}} = (\mathbf{e}_i \cdot \hat{\mathbf{n}})\,\hat{\mathbf{n}}$. Since all eigenvectors have z component 0, $\mathbf{e}_i^{\mathrm{OOP}} = \mathbf{0}$ for all i.

Step 4 – evaluate α_{OOP}.

$$\begin{aligned}\sum_{i\in G} \|\mathbf{e}_i^{\mathrm{OOP}}\|^2 &= 0\\ \sum_{i\in G} \|\mathbf{e}_i\|^2 &= (0.012^2 + 0.009^2)\cdot 2 + (0.018^2)\cdot 2\\ &= 2\cdot 0.000225 + 2\cdot 0.000324 = 0.001098\end{aligned}$$

$$\alpha_{\mathrm{OOP}} = \frac{0}{0.001098}\cdot 100\,\% = 0\,\%$$

Classification: **Pure INP** ($\alpha_{\mathrm{OOP}} \leq 10\,\%$). The cluster breathing motion is an in-plane mode – as expected.

Step 5 – comparison with the hinge mode. A cluster hinge mode (the "book-opening" of the two Fe-S bonds about an S-S axis) has, by contrast, the form:

$$\begin{aligned}\mathbf{e}_{\mathrm{Fe}_1} &= (0.000,\ 0.000,\ +0.020)\\ \mathbf{e}_{\mathrm{Fe}_2} &= (0.000,\ 0.000,\ +0.020)\\ \mathbf{e}_{\mathrm{S}_1} &= (0.000,\ 0.000,\ -0.020)\\ \mathbf{e}_{\mathrm{S}_2} &= (0.000,\ 0.000,\ -0.020)\end{aligned}$$

Here $\mathbf{e}_i^{\mathrm{OOP}} = \mathbf{e}_i$ for all i, so $\alpha_{\mathrm{OOP}} = 100\,\%$ – classified as **Pure OOP**.

Step 6 – what happens in a real DFT run. In a real Cys_4 or Cys_2His_2 cluster, pure breathing or hinge modes are rare – components typically mix, and α_{OOP} lies between 20 and 80 %. Classification according to the default thresholds $(60, 75, 90)$ produces a seven-level distribution, which for a typical [2Fe-2S] run might look something like:

- Pure INP: $\sim 30\,\%$ of cluster modes
- Majority/Strong INP: $\sim 35\,\%$
- Mixed: $\sim 20\,\%$
- Majority/Strong OOP: $\sim 12\,\%$
- Pure OOP: $\sim 3\,\%$

(These numbers vary considerably between systems – in rigid cluster geometries, INP modes dominate; in flexible ligand spheres, mixed modes are more common.)

2.5 Fe-ligand bond decomposition: stretch, bend INP, bend OOP

The OOP/INP decomposition of §2.4 applies to the *cluster core* (Fe_1, Fe_2, S_1, S_2). A second decomposition is performed for every Fe-ligand bond (Fe-S_{Cys}, Fe-N_{His}, Fe-$O_{\mathrm{Asp/Glu}}$): the relative displacement of the two bond partners is split into a *stretch* component (along the bond axis) and a *bend* component (perpendicular to it), the bend then being split further into INP and OOP relative to the cluster plane.

Let $\mathbf{e}_{\mathrm{Fe}}$ and $\mathbf{e}_{\mathrm{lig}}$ be the thermally scaled displacements of the Fe and the ligand-donor atom, and let

$$\hat{\mathbf{b}} = \frac{\mathbf{r}_{\mathrm{lig}} - \mathbf{r}_{\mathrm{Fe}}}{\|\mathbf{r}_{\mathrm{lig}} - \mathbf{r}_{\mathrm{Fe}}\|} \tag{9}$$

be the unit vector pointing from Fe to the ligand donor in the equilibrium geometry. Define the relative displacement

$$\Delta\mathbf{e} = \mathbf{e}_{\mathrm{lig}} - \mathbf{e}_{\mathrm{Fe}}. \tag{10}$$

Decomposition along and perpendicular to the bond. The signed stretch component is the projection on $\hat{\mathbf{b}}$; the bend vector is what remains:

$$\Delta r_{\text{stretch}} = \Delta\mathbf{e} \cdot \hat{\mathbf{b}}, \qquad \Delta\mathbf{e}_{\perp} = \Delta\mathbf{e} - \Delta r_{\text{stretch}}\hat{\mathbf{b}}. \tag{11}$$

Splitting the bend into INP and OOP parts. The cluster normal $\hat{\mathbf{n}}$ (§2.4) further decomposes $\Delta\mathbf{e}_{\perp}$ into a component along $\hat{\mathbf{n}}$ (out of the cluster plane) and a remainder (in the cluster plane):

$$\Delta r_{\text{bend\_oop}} = \Delta\mathbf{e}_{\perp} \cdot \hat{\mathbf{n}}, \qquad \text{(signed)} \tag{12}$$

$$\Delta r_{\text{bend\_inp}} = \|\Delta\mathbf{e}_{\perp} - \Delta r_{\text{bend\_oop}}\hat{\mathbf{n}}\|. \quad \rightsquigarrow \texttt{core.analyze\_fe\_ligand} \tag{13}$$

The reported `bend`, `bend_inp`, and `bend_oop` columns in `Fe_S_Cys.xlsx` / `Fe_N_His.xlsx` are the *absolute values* (lengths) of these three quantities, all in Å. By Pythagoras,

$$\|\Delta\mathbf{e}_{\perp}\|^2 = (\Delta r_{\text{bend\_inp}})^2 + (\Delta r_{\text{bend\_oop}})^2, \tag{14}$$

which is the natural energy decomposition of the bend mode.

Bend INP / OOP percentages. For comparing modes with very different overall amplitudes, the absolute lengths are converted to *squared-amplitude fractions* relative to the total bend:

$$\text{bend\_inp\_pct} = 100 \cdot \frac{(\Delta r_{\text{bend\_inp}})^2}{(\Delta r_{\text{bend\_inp}})^2 + (\Delta r_{\text{bend\_oop}})^2}, \tag{15}$$

and analogously `bend_oop_pct`. The two add to 100 % exactly. For very small overall bends (below the noise level $2\sigma_e^{\text{therm}}\sqrt{2}$, §2.2), the percentages are set to NaN to avoid amplifying numerical noise.

Sigma propagation. All three quantities (stretch, bend_inp, bend_oop) are differences of two independent Gaussian-noise eigenvector components and therefore carry the standard $\sqrt{2}\,\sigma_e^{\text{therm}}$ sigma:

$$\sigma_{\text{stretch}} = \sigma_{\text{bend\_inp}} = \sigma_{\text{bend\_oop}} = \sqrt{2}\,\sigma_e \cdot u_{\text{rms}}(\omega_l). \tag{16}$$

The sigmas are written into the `Fe_S_Cys.xlsx` / `Fe_N_His.xlsx` files under `s_stretch`, `s_bend_inp`, `s_bend_oop` and are used for the significance colouring of the bend cells.

Physical interpretation. For PCET applications, the INP/OOP split of the Fe-N(His) bend is particularly informative:

- *bend_inp dominant* (>70 %): the His ring rocks in the cluster plane – a motion that strongly modulates the Fe-Fe distance and the iron-sulfur core, classic ET-active.

- *bend_oop dominant* (>70 %): the His ring tilts out of plane, which modulates the N-H bond direction – coupling channel for PT to a buried acceptor.
- *mixed* (30–70 %): the mode contributes to both ET and PT pathways simultaneously, the typical PCET fingerprint.

2.6 SCSD after Kingsbury & Senge

The OOP/INP decomposition of the previous section is a *geometric heuristic* – it knows only the cluster plane, not the symmetry. A rigorous symmetry decomposition is provided by the *Symmetry-Coordinate Structural Decomposition* (SCSD) of Kingsbury & Senge [1].

Idea

A [2Fe-2S] cluster with ideal rhombic geometry belongs to the point group $D_{2\mathrm{h}}$. The eight possible displacement patterns of the four atoms (1 translation × 3 axes + 1 rotation × 3 axes + 2 stretches) distribute over the eight $D_{2\mathrm{h}}$ irreducible representations: $A_\mathrm{g}, B_{1\mathrm{g}}, B_{2\mathrm{g}}, B_{3\mathrm{g}}, A_\mathrm{u}, B_{1\mathrm{u}}, B_{2\mathrm{u}}, B_{3\mathrm{u}}$.

A real cluster geometry has finite distortions, however – the cluster is never ideally rhombic. SCSD projects the actual atomic coordinates onto the eight irreps of the point group and yields a contribution amplitude per irrep. Applied to a normal mode, SCSD answers: *what symmetry does this displacement actually have?*

$D_{2\mathrm{h}}$ character table and symmetry operations

The point group $D_{2\mathrm{h}}$ has eight symmetry operations $\{E, C_2(z), C_2(y), C_2(x), i, \sigma_h, \sigma_v(xz), \sigma_v(yz)\}$ and correspondingly eight one-dimensional irreps. The character table:

D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	σ_h	$\sigma_v(xz)$	$\sigma_v(yz)$	**Linear basis**
A_g	1	1	1	1	1	1	1	1	x^2, y^2, z^2
B_{1g}	1	1	−1	−1	1	1	−1	−1	xy
B_{2g}	1	−1	1	−1	1	−1	1	−1	xz
B_{3g}	1	−1	−1	1	1	−1	−1	1	yz
A_u	1	1	1	1	−1	−1	−1	−1	—
B_{1u}	1	1	−1	−1	−1	−1	1	1	z
B_{2u}	1	−1	1	−1	−1	1	−1	1	y
B_{3u}	1	−1	−1	1	−1	1	1	−1	x

In the [2Fe-2S] context, the axis convention (Kingsbury 2024) is: $\hat{\mathbf{x}}$ along Fe-Fe, $\hat{\mathbf{y}}$ along S-S (bridging-sulfur to bridging-sulfur), $\hat{\mathbf{z}} = \hat{\mathbf{n}}$ (cluster normal).

Meaning of the individual irreps for cluster motion

Directly from the character table one can read off which motions correspond to which irrep:

Irrep	Cluster motion in [2Fe-2S] convention
A_g	Symmetric stretches along the principal axes. Examples: Fe-Fe breathing, S-S stretch, cluster-breathing mixture. The classical "reorganization mode" for ET.
B_{1g}	In-plane bending with simultaneous distortion of both diagonals.
B_{2g}	In-plane motion with unequal Fe-S distances on the two cluster sides.
B_{3g}	In-plane motion with rotation of the rhombus center relative to the atoms.
A_u	"Central inversion mode" – rare in rigid clusters; can appear in strongly distorted geometries.
B_{1u}	Pure OOP displacement of all cluster atoms in the $\hat{\mathbf{z}}$ direction. Hinge mode.
B_{2u}	OOP displacement with twist about the $\hat{\mathbf{x}}$ axis.
B_{3u}	OOP displacement with twist about the $\hat{\mathbf{y}}$ axis.

The g irreps are *gerade* (in-plane, parity-preserving), the u irreps *ungerade* (typically OOP-active).

Canonical reference and sign convention

The projection requires an unambiguous axis choice. `scsdpy` uses the Kingsbury canonical convention:

1. $\hat{\mathbf{z}}$ is the cluster normal.
2. The cluster's principal axis (longest interatomic distance) is placed along $\hat{\mathbf{x}}$.
3. Signs are chosen so that the cluster atoms fall in a specific order onto positive quadrants.

In `modenanalyse_2fe2s`, the reference geometry is built once per run and stored (`core._get_scsd_model`). It is independent of the orientation in which Gaussian optimized the cluster.

Application to normal modes

For mode l, the cluster (four atoms) is displaced by $\mathbf{e}_{l,i}\, u_{\text{rms}}$; the difference between this displaced geometry and the reference geometry yields a 12-dimensional displacement vector. The SCSD projection produces eight amplitudes [1]:

$$\mathbf{a}_l \;=\; \mathbf{P}_{D_{2\mathrm{h}}}\big(\mathbf{e}_l^{\mathrm{cluster}}\, u_{\mathrm{rms}}\big), \qquad \mathbf{a}_l \in \mathbb{R}^8. \quad \rightsquigarrow \texttt{core.compute\_scsd\_for\_mode\_full} \tag{17}$$

The squared sum $\sum_\Gamma a_{l,\Gamma}^2$ equals the unweighted sum over the squared atomic displacements – the SCSD decomposition is an *orthogonal* basis transformation.

Dominant irrep – determined rigorously

The rigorous function `core.extract_dominant_scsd_irreps` identifies the dominant irrep as follows. It:

1. sums the squared amplitudes per irrep,
2. normalizes to the total sum,
3. returns the *primary* and *secondary* irrep, together with their percentage contributions.

This classification can disagree with the OOP/INP heuristic: a mode can, for example, be "majority OOP" (heuristic) and at the same time "primarily B_{1g}" (SCSD) – B_{1g} is a specific OOP symmetry, which the heuristic does not distinguish.

The `SCSD` sheet in the Excel output lists, per mode, the eight amplitudes plus the two dominant irreps with their percentage contributions. Modes with consistent (>80 %) primary irrep are those that should be cited in the symmetry table of a manuscript.

When SCSD is disabled

The SCSD analysis requires the external package `scsdpy`. If it is not installed, the SCSD sheet is skipped and a note is written to the report. All other analyses (OOP/INP heuristic, kernel classification, reorganization energies) are unaffected.

2.7 Heuristic kernel classification

In addition to the rigorous SCSD decomposition, there is a *heuristic kernel classification* that assigns each mode to a motion type based on its geometric displacement pattern (`core.classify_kernel_mode_fro`
It is robust even without `scsdpy` and provides an intuitive categorization. The heuristic is explicitly labeled as such and should not be confused with the rigorous SCSD answer.

Motion scores

The heuristic classifier computes eight motion scores per mode:

Score	Definition
Translation	Center-of-mass displacement of the cluster (should be small for an intramolecular mode)
Umbrella	Mean OOP displacement of all four cluster atoms (all moving collectively out of the plane)
Hinge-Fe	OOP difference Fe_1 minus Fe_2 (Fe atoms rocking against each other)
Hinge-S	OOP difference S_1 minus S_2 (S atoms rocking against each other)
OOP-Twist	Combination of both hinge modes (twisting)
Fe stretch	In-plane component along the Fe-Fe axis
S stretch	In-plane component along the S-S axis
Breathing	Breathing motion: all four atoms simultaneously moving away from / toward the centroid
Rhombus shear	Antisymmetric distortion of the rhombus (Fe-S bond angles)
Rotation-ip	In-plane rotation of the cluster about its normal

Each score is normalized to the total displacement of the four cluster atoms and lies in $[0, 1]$.

Classification

The two highest scores are returned as the *primary* and *secondary* motion patterns. In the Excel sheet `Kernel_Scores`, all eight scores plus the two top labels are listed. This heuristic provides a pre-classification that works *without* a D_{2h} symmetry assumption (also in distorted geometries).

Heuristic vs. SCSD: when do they disagree?

In ideally rhombic clusters the two agree:

Heuristic	SCSD irrep	Note
Umbrella	B_{2g} or B_{3g}	depending on the axis convention
Hinge-Fe	A_u	antisymmetric about the cluster center
Breathing	A_g	fully symmetric
Fe stretch	B_{1g}	antisymmetric along Fe-Fe
Rotation-ip	B_{1g} (in-plane)	depending on the axis choice

In the distorted clusters typical of proteins, motions often split across two heuristic scores (e.g., a mode with Hinge-Fe *and* rhombus shear of comparable magnitude). Here the SCSD answer is more conclusive: it assigns a unique symmetry that does not depend on the axis choice.

Caveat – do not confuse. The heuristic labels (umbrella, hinge, breathing, ...) are *motion descriptors*, not symmetry labels. A "Hinge-Fe" mode can carry symmetry A_u, B_{1u}, or B_{2u} depending on the axis choice. For statements about mode symmetry in a manuscript, always use the SCSD answer, not the heuristic.

2.8 Embeddings: UMAP and HDBSCAN

For a [2Fe–2S] system with 5 000–19 000 modes, displaying all mode properties simultaneously on two-dimensional plots is not feasible. `modenanalyse_2fe2s` uses UMAP (Uniform Manifold Approximation and Projection) for dimensionality reduction, optionally followed by HDBSCAN clustering.

Choice of method: why UMAP?

Compared to alternatives (PCA, t-SNE), UMAP has, after test runs on protonated and deprotonated Cys_2His_2 model systems as well as an HHCC mutant, proven to be the only method that consistently yields interpretable results for DFT mode data.

- **PCA** [31, 32, 33] is linear and fast, but cannot resolve nonlinear manifolds. For DFT modes, the data points typically lie on curved submanifolds in feature space (frequency-OOP-INP-score relations are nonlinear); PCA shows only a shadow projection. PC1 in our data typically explains only 25–35 % of the variance – too little for a stand-alone visualization.
- **t-SNE** [34, 35] preserves local neighborhoods, but global distances are meaningless: an apparently large gap between two cluster regions in a t-SNE plot does not necessarily correspond to a large distance in feature space [34]. Furthermore, t-SNE is stochastic and the results depend on the random seed. For cluster identification, t-SNE often creates the appearance of clusters even when the underlying data are continuously distributed.
- **UMAP** [36, 37] preserves both local neighborhood and global topology in a controllable way and is significantly faster than t-SNE at comparable quality [37]. The `n_neighbors` parameter allows explicit control of the locality–globality balance, which is essential for mode analysis (small n_neighbors emphasize finer symmetry classes, large n_neighbors the global frequency structure).

A PCA-style user perspective (variance explained per axis) is no longer required here, because the meaning of each feature is made explicit upstream of the embedding by the curated selection (see section *Feature matrix*). The other sheets in the Excel output (`Mode_analysis`, `Reorganization_energy`, `Reorg_total`) provide the quantitative information that PCA would have contributed in addition.

UMAP: mathematical foundation

UMAP constructs a low-dimensional embedding in two steps, based on the theory of topological manifolds and fuzzy simplicial sets [36].

Step 1: high-dimensional neighborhood graph. In the original feature space $\mathbb{R}^p$ (with $p = 31$ features), a local neighborhood graph is constructed for each data point x_i. Per point, the k nearest neighbors are determined (parameter `n_neighbors`). Distances to these neighbors are converted via an adaptive metric $d_i(x_i, x_j)$ into a probability $\mu_{ij} \in [0, 1]$:

$$\mu_{ij} = \exp\left(\frac{-\max(0,\, d(x_i, x_j) - \rho_i)}{\sigma_i}\right), \tag{18}$$

where ρ_i is the distance to the nearest neighbor of x_i (this guarantees that every point is connected to its immediate neighbor with probability 1) and σ_i is chosen such that $\sum_j \mu_{ij} = \log_2(k)$ (local-scale adaptation).

The asymmetry of μ_{ij} is combined into a symmetric adjacency matrix w_{ij} via a *fuzzy union* [36]:

$$w_{ij} = \mu_{ij} + \mu_{ji} - \mu_{ij}\,\mu_{ji}. \tag{19}$$

Step 2: low-dimensional embedding. In the 2D target space (`n_components=2`), positions y_i are sought that produce a similar neighborhood graph v_{ij}. The probability in the target space is

$$v_{ij} = \left(1 + a\,\|y_i - y_j\|^{2b}\right)^{-1}, \tag{20}$$

with constants $a \approx 1.577$ and $b \approx 0.895$ from the default configuration (`min_dist=0.1`). The loss function is the fuzzy cross-entropy:

$$\mathcal{L} = \sum_{ij}\left[w_{ij}\log\frac{w_{ij}}{v_{ij}} + (1 - w_{ij})\log\frac{1 - w_{ij}}{1 - v_{ij}}\right], \tag{21}$$

minimized by stochastic gradient descent.

Preserved properties. In contrast to t-SNE, UMAP preserves not only local but also global structure [37]. Distances between cluster regions in the UMAP projection correlate with distances in feature space – with the caveat that large distances are slightly *compressed* (UMAP is not distance-preserving, but topology-preserving).

Parameters

- `n_neighbors` (auto heuristic: $\max(5, \min(15, N/100))$): controls the local-vs.-global balance. Small values (5–10) preserve the finest local structure; large values (20–50) emphasize global topology. For a typical [2Fe-2S] protein with $N \approx 5\,000$–$10\,000$ modes, the auto heuristic yields `n_neighbors` $= 15$, which matches the paper standard [36].

- `min_dist` = 0.1 (default): minimum distance between points in the embedding. Smaller values produce more compact clusters.
- `n_components` = 2: 2D visualization.
- `random_state` = 42 (deterministically reproducible).

Configurable via `cfg.umap_n_neighbors` (manual override).

Feature matrix

Per mode, 31 numerical features are extracted. Two originally planned `Cys *_OOP` features are by geometric construction always zero (Cys-S_γ lies in the cluster plane and has no out-of-plane component) and were excluded. The remaining 31 features are:

ter-ring fractions (3 features, see section 2.6):

- `lig_OOP%` – fraction of the mode motion in the ligand sphere that is OOP relative to the cluster plane
- `kern_OOP%` – the same for the cluster core (Fe + cluster-S)
- `kern_|d|` – absolute cluster displacement in Å (norm)

cores / D_{2h} irreps (10 features, see section 2.6):

- `Translation`, `Umbrella`, `Hinge-Fe`, `Hinge-S`, `OOP-Twist`, `Fe-stretch`, `S-stretch`, `Breathing`, `Rhombus-shear`, `Rotation-ip`

These are projections of the mode onto the irreducible representations of the ideal D_{2h} cluster symmetry.

d OOP fractions (2 features):

- `His 255_OOP`, `His 259_OOP`

Distribution of the ligand OOP motion across the two His residues. The values satisfy `His 255_OOP + His 259_OOP = 1`; they are perfectly anticorrelated. UMAP, however, does not weight anticorrelated features more strongly, and treating both ligands symmetrically is conceptually cleaner. *Not* included: `Cys 207_OOP`, `Cys 216_OOP` – see above.

nd INP fractions (4 features):

- `Cys 207_INP`, `Cys 216_INP`, `His 255_INP`, `His 259_INP`

The same for in-plane fractions. Here too perfect pairwise anticorrelation, but all four are kept.

nd atom motions (12 features, see sheets `Fe_S_Cys`, `Fe_N_His`):

- Per Cys (207, 216): `stretch`, `bend_inp`, `bend_oop`

- Per His (255, 259): `stretch`, `bend_inp`, `bend_oop`

Stretch and bend of the Fe-S / Fe-N bond vector with the bend component split into INP and OOP parts relative to the cluster plane.

Before UMAP, the feature matrix is standardized (`sklearn.preprocessing.StandardScaler`; per feature mean 0, variance 1). This ensures that no individual variable dominates solely because of its absolute value range.

What is deliberately excluded from the matrix.

- *Frequency* (ω_l): not used as a feature, since otherwise the UMAP map would trivially sort by frequency. Frequency is kept as a metadata column and can be used for coloring the UMAP plot.

- *Reorganization lambda* (λ_X): not introduced as a feature, because the underlying geometric modulations (`dr_FeFe`, `dr_NH`, `dr_FeN`) are already implicitly captured in the per-ligand stretch/bend features. Including λ_X in addition would over-emphasize reorganization-relevant modes.

- *Secondary structure* (HELIX/SHEET fractions): handled in a *separate* secondary-structure UMAP (sheet `SSE_UMAP_Cluster`). The mode UMAP and SSE UMAP are conceptually different analyses.

HDBSCAN: density-based cluster detection

HDBSCAN [38, 39] is a hierarchical density-based clustering algorithm. Unlike k-means, it does not require a predetermined number of clusters; it identifies clusters as connected regions of high density and labels isolated points as noise (label -1).

min_cluster_size. The only important parameter: the minimum cluster size. `modenanalyse_2fe2s` uses an auto heuristic

$$\texttt{mcs} = \max(5, \min(30, \sqrt{N})),$$

For typical [2Fe-2S] protein systems ($N \approx 5\,000$–$10\,000$), this gives $\texttt{mcs} = 30$.

If HDBSCAN finds many clusters or much noise – what does that mean?
UMAP on a typical Cys_2His_2 system with 31 features and `n_neighbors=15` typically yields 5–8 clusters with no noise. If HDBSCAN finds significantly more clusters (e.g., 80–100), this is usually due to too small an `mcs` value or a fragmented UMAP topology. If HDBSCAN finds 0 clusters (everything noise), the mode distribution in feature space is too smooth – which is expected behavior for DFT mode data without sharp transitions between mode classes [38].
Consequence. The UMAP visualization itself is informative even without HDBSCAN clusters – one can see mode densities and transitions through coloring by frequency, mode type, or kernel score. HDBSCAN provides additional information, but is not a required component of the analysis.

Required. `hdbscan` and `umap-learn` must be installed. Without these packages the UMAP sheets are skipped.

Practical recommendation

1. **Inspect the UMAP plot**: the sheet `Projections` contains the UMAP coordinates for all modes. Color in Origin/matplotlib by frequency, mode type, or kernel-primary score.

2. **Examine the cluster sheets**: the sheet `UMAP_Cluster` shows HDBSCAN labels and feature values per cluster. `UMAP_Cluster_Profile` shows Z-score profiles and the top modes per cluster.

3. **Comparison between structures** (e.g., prot vs. deprot, or vs. a mutant variant) is performed externally: two UMAP plots for the two structures side by side, colored on the same frequency scale. Within a single run no direct comparison is performed – one system per run.

2.9 Marcus-Hush theory of reorganization

The reorganization chapter §2.12 computes per-mode contributions to the reorganization energy. Before going there, we need classical Marcus-Hush theory as a *benchmark anchor* – so that it is clear what our mode-resolved variant has in common with historical Marcus theory and where the two differ.

Classical picture: two diabatic energy surfaces

Marcus' theory of electron transfer [19, 20] describes a reaction between a reactant and a product state via two *diabatic* energy surfaces $U_R(\mathbf{Q})$ (reactant) and $U_P(\mathbf{Q})$ (product). The reaction coordinates $\mathbf{Q} = (Q_1, Q_2, \ldots, Q_N)$ are the vibrational modes of the reactive complex – *all* of them, not selected ones.

In the harmonic approximation, both surfaces are paraboloids:

$$U_R(\mathbf{Q}) = U_R^0 + \frac{1}{2}\sum_i \mu_i\,\omega_i^2\,(Q_i - Q_i^R)^2 \tag{22}$$

$$U_P(\mathbf{Q}) = U_P^0 + \frac{1}{2}\sum_i \mu_i\,\omega_i^2\,(Q_i - Q_i^P)^2 \tag{23}$$

where μ_i is the reduced mass, ω_i the eigenfrequency, and Q_i^R, Q_i^P are the equilibrium displacements of mode i in the two states.

Marcus' central assumption is: **the frequencies ω_i are equal in both states**, only the displacements $Q_i^R \neq Q_i^P$. This is the "equal-harmonic" approximation; it is valid when reactant and product are electronically similar (e.g., two oxidation states of the same complex).

Reorganization energy as the central quantity

The **reorganization energy** λ is defined as the energy the system pays if it remains in the reactant state but is displaced to the product geometry:

$$\lambda = U_R(\mathbf{Q}^P) - U_R(\mathbf{Q}^R) = \frac{1}{2}\sum_i \mu_i\,\omega_i^2\,(\Delta Q_i)^2 \quad \text{with} \quad \Delta Q_i = Q_i^P - Q_i^R. \tag{24}$$

This is the famous Marcus-Hush standard form. It is an *equilibrium property* of the system – not a mode property, not an activation energy, not a kinetic quantity. The thermal population of individual modes plays no role in (24).

Activation energy and rate

From λ and the free reaction energy $\Delta G^0 = U_P^0 - U_R^0$, a short geometric argument leads to the Marcus activation energy:

$$\Delta G^\ddagger = \frac{(\lambda + \Delta G^0)^2}{4\lambda} \tag{25}$$

and consequently to the Marcus rate:

$$k_{\mathrm{ET}} = \frac{2\pi}{\hbar}|H_{RP}|^2 \cdot \frac{1}{\sqrt{4\pi\lambda k_B T}} \cdot \exp\left(-\frac{\Delta G^\ddagger}{k_B T}\right) \tag{26}$$

where H_{RP} is the electronic coupling element between the reactant and product diabats. Three characteristic features:

- For small $|\Delta G^0|$: $\Delta G^\ddagger \approx \lambda/4$. *The reorganization energy is the measure of the reaction barrier.*
- For $\Delta G^0 = -\lambda$: barrierless, k is maximal ($\Delta G^\ddagger = 0$).
- For $|\Delta G^0| > \lambda$: the *Marcus inverted regime*, where the rate decreases again because the reaction becomes too exothermic.

In iron-sulfur proteins, all three regimes are relevant: ferredoxin-type ET steps are often in the Marcus normal regime; PCET steps in mitoNEET and Rieske can touch the inverted region.

What this theory provides – and what it does not

What Marcus-Hush delivers:

- A complete analytical description of the ET/PCET activation energy, provided the harmonic approximation holds.
- A directly measurable prediction of the rate k_{ET} (eq. 26), if λ, ΔG^0, and $|H_{RP}|^2$ are known.
- A justification for why λ is system-specific (no universal value: it depends on the solvation environment, the cluster geometry, and the ligands).

What Marcus-Hush does not directly provide:

- *Which* modes contribute substantially to λ. The sum in (24) is global; promoter modes (see below) had to be teased out in the 1990s by Hammes-Schiffer and others.
- *Direct access from DFT frequency calculations.* Classical computation of λ requires two separate frequency calculations (reactant + product) plus a robust mode mapping between them – a non-trivial effort.
- *Reorganization energy as a spectroscopic observable.* λ from (24) is an activation energy for a concrete reaction, not a band frequency or a line profile.

These three gaps motivate extending the theory in two directions:

1. **Mode-resolved view** (Hammes-Schiffer, Cukier, Stuchebrukhov): decomposes λ into contributions of individual modes and identifies promoter modes with particular significance for the rate. The multi-mode view turns (24) into frequency-resolved information.
2. **Spectroscopically accessible variant**: replaces the static displacements ΔQ_i with thermal bond modulations $\Delta r_X(i)$. This turns λ into a quantity that follows directly from a single frequency calculation – and that assigns bands in NRVS spectra.

The second extension is exactly what `modenanalyse_2fe2s` computes in §2.12. What we call λ_X there is *not* the classical Marcus-Hush λ from (24), but a related yet distinct quantity – the thermal bond reorganization energy per mode and per bond. The connection to Huang-Rhys theory makes clear why this quantity is a well-defined spectroscopic observable (next section).

Hammes-Schiffer extension: multi-mode PCET

Hammes-Schiffer and Soudackov [49] showed that PCET in complex systems such as proteins *cannot* be described by a single promoter mode. Instead, the reaction is coupled to many modes simultaneously; the spectrum of contributions $\lambda_i = \frac{1}{2}\mu_i\omega_i^2(\Delta Q_i)^2$ is broadly distributed.

Standard Marcus theory can capture this multi-mode reality through an *effective* reorganization energy $\lambda_{\text{eff}} = \sum_i \lambda_i$, but the mode-resolved distribution is then lost. For NRVS band identification, that distribution is precisely what matters:

- Which frequency ranges contribute most strongly to the reorganization?
- Which bonds are active in those ranges?
- Which mutations should change which band structure?

The multi-mode view is also why global "top-mode classification" approaches (PCET score, L_{PCET}, filter-variant) were conceptually unsatisfying: they searched for *one* dominant mode, while reality is a broad distribution. The mode-resolved reorganization pipeline makes that distribution explicit.

What does ΔQ mean vs. Δr_X?

A central subtlety in preparation for the next chapter: the classical Marcus displacement ΔQ_i (mass-weighted mode coordinate) is *not* the same as a bond modulation Δr_X (Cartesian distance change in Å).

- ΔQ: displacement of the *eigenmode coordinate* between two equilibrium geometries. Has units of $\sqrt{\text{amu}} \cdot$ Å. Appears in (24).
- $\Delta r_X(i)$: *bond modulation* along the X bond, induced by mode i with thermally scaled displacements. Has units of Å. Computed in `reorganization.compute_mode_modulations`.

Both have the same *form* in the standard equation $\lambda = \frac{1}{2}\mu\omega^2(\cdot)^2$, but distinct *physical meanings*. The bridge between the two is Huang-Rhys theory – see the next section.

2.10 Huang-Rhys theory and our λ_X

In §2.12 we compute $\lambda_X(i) = \frac{1}{2}\mu_i\omega_i^2(\Delta r_X(i))^2$ with *thermally scaled* bond modulations. This section shows that this is exactly the **Huang-Rhys factor** (in a particular convention), and that this quantity has well-defined spectroscopic meaning.

What Huang and Rhys originally described

Huang and Rhys [50] computed optical line profiles of color centers in solids. Their question: when an electronic transition shifts the equilibrium position of a harmonic oscillator, what does the resulting vibronic spectrum look like?

Answer: the spectrum is a Poisson distribution

$$P(n) = \frac{S^n e^{-S}}{n!} \tag{27}$$

over phonon number n, parametrized by the **Huang-Rhys factor**

$$S = \frac{\mu\,\omega\,(\Delta q)^2}{2\hbar}, \tag{28}$$

where Δq is the transition-induced equilibrium shift of a mode with reduced mass μ and frequency ω.

Meaning of S:

- $S \ll 1$: the transition is approximately vertical; the line spectrum is dominated by the 0–0 band.
- $S \approx 1$: strong mode coupling; vibronic band with several phonon excitations.
- $S \gg 1$: Gaussian-shaped band structure (Stokes shift equal to $S\hbar\omega$).

In modern language: S measures mode coupling *quantitatively*, in units of the zero-point amplitude $u_{\mathrm{ZP}} = \sqrt{\hbar/(2m\omega)}$:

$$S = \frac{(\Delta q)^2}{2u_{\mathrm{ZP}}^2}. \tag{29}$$

Connection to Marcus-Hush

Compare (28) with the Marcus reorganization energy of a single mode (from (24) with $N = 1$):

$$\lambda_{\mathrm{single}}^{\mathrm{Marcus}} = \frac{1}{2}\mu\,\omega^2\,(\Delta q)^2 = S \cdot \hbar\omega. \tag{30}$$

Thus: $\lambda^{\mathbf{Marcus}} = S \cdot \hbar\omega$. The two quantities are interconvertible.

For a multi-mode distribution: $\lambda_{\mathrm{total}}^{\mathrm{Marcus}} = \sum_i S_i \cdot \hbar\omega_i$.

What is Δq in our case?

The crux: in classical Marcus theory, Δq is a *static displacement* of the mode coordinate between two electronic states. `modenanalyse_2fe2s`, however, has *only one* electronic state per run – the single DFT frequency calculation provides only one equilibrium geometry and its eigenmodes.

Solution: we replace the static displacement with a *thermal displacement*:

$$(\Delta q)^2 \to \langle q^2 \rangle = \frac{\hbar}{2\mu\omega}\coth\!\left(\frac{\hbar\omega}{2k_B T}\right) = u_{\mathrm{rms}}^2. \tag{31}$$

With the convention $\Delta r_X(i) = u_{\mathrm{rms}}(i) \cdot \alpha_X(i)$, where $\alpha_X(i)$ is the mode contribution along bond X, our $\lambda_X(i)$ becomes:

$$\lambda_X(i) = \frac{1}{2}\mu_i\,\omega_i^2\,u_{\mathrm{rms}}^2(i)\,\alpha_X^2(i). \tag{32}$$

Substituting (31) and using $\hbar\omega/(k_B T) \to \infty$ at $T \to 0$ (so coth $\to 1$), one obtains:

$$\lambda_X(i) \xrightarrow{T\to 0} \frac{1}{2}\mu_i\,\omega_i^2\,\frac{\hbar}{2\mu_i\omega_i}\,\alpha_X^2(i) = \frac{\hbar\omega_i}{4}\,\alpha_X^2(i). \tag{33}$$

Connection to the Huang-Rhys factor. From (30) with $(\Delta q)^2 = u_{\mathrm{rms}}^2 \cdot \alpha_X^2$:

$$S_X(i) = \frac{\mu_i\omega_i u_{\mathrm{rms}}^2(i)\,\alpha_X^2(i)}{2\hbar} \xrightarrow{T\to 0} \frac{\alpha_X^2(i)}{4}. \tag{34}$$

In the $T \to 0$ limit, the Huang-Rhys factor per bond is therefore simply *one quarter of the squared mode contribution*. For a pure stretching mode of the bond ($\alpha_X = 1$), $S_X = 1/4$ – that is, *weak coupling* in the Huang-Rhys sense, with a typical line spectrum.

Magnitude

Let us put in concrete numbers. For a typical Fe-S stretching mode at $\omega = 350$ cm^{-1} and $T = 5\,\mathrm{K}$:

- Zero-point energy: $\hbar\omega/4 = 87.5$ cm^{-1}.
- For a pure stretching mode ($\alpha_X = 1$): $\lambda_X(T \to 0) = 87.5$ cm^{-1}.
- Under thermal population at 5 K ($\ll \hbar\omega/k_B = 503\,\mathrm{K}$): coth ≈ 1, hence essentially the zero-point value.
- At room temperature (300 K, coth ≈ 1.04): $\lambda_X \approx 91$ cm^{-1}, only marginal increase.

The same mode at $\omega = 100$ cm^{-1}:

- $\hbar\omega/4 = 25$ cm^{-1}.
- At 5 K: coth ≈ 1.0, $\lambda_X \approx 25$ cm^{-1}.
- At 300 K: coth ≈ 4.2, $\lambda_X \approx 105$ cm^{-1}.

Low-frequency modes are therefore strongly thermally populated at room temperature – their λ_X contributions are amplified relative to the zero-point value by a factor $\coth(\hbar\omega/2k_B T)$.

Consequence for low-temperature NRVS. At low temperature (5–40 K, $\hbar\omega \gg k_B T$ for typical NRVS frequencies), $u_{\mathrm{rms}}^2 \approx \hbar/(2\mu\omega)$, hence T-independent. We therefore have

$$\lambda_X(i) \approx \frac{\hbar\omega_i}{4}\alpha_X^2(i) \quad \text{at low-temperature NRVS.} \tag{35}$$

This form is *purely quantum-mechanical* (zero-point energy), independent of the precise sample temperature. This is exactly what NRVS measures: the quantum-mechanical mode activity, not a thermal Boltzmann average.

What our λ_X is – final definition

With these preliminaries we can now formulate a precise definition:

What λ_X is in `modenanalyse_2fe2s`

$\lambda_X(i)$ is the **thermal bond reorganization energy** of mode i along bond class X:

- Form: identical to Marcus-Hush ($\frac{1}{2}\mu\omega^2 r^2$), but with thermally scaled r instead of a static reactant–product offset.
- Related to the Huang-Rhys factor: $S_X(i) = \lambda_X(i)/(\hbar\omega_i)$.
- In the low-temperature limit: $\lambda_X(i) \to (\hbar\omega_i/4)\,\alpha_X^2(i)$ – purely quantum-mechanical, T-independent.
- Direct spectroscopic meaning: $\lambda_X(i)$ is the energy of the bond-X modulation in mode i, measured at the QHO expectation value.

This quantity is related to, but not identical with, the classical Marcus-Hush activation energy of a reaction between two electronic states. For the latter, two separate DFT calculations would be required.

2.11 NRVS theory as relevant for band identification

The tool *does not* compute NRVS spectra – these are synthesized from the same DFT frequency calculation using specialized external tools. But the modulation spectra $M_X(\omega)$ and the cumulative reorganization curves $\Lambda_X(\omega)$ computed in §2.12 can be compared directly with NRVS bands – which is why we need NRVS theory as an interpretive framework.

What does NRVS measure?

NRVS [12] measures the ^{57}*Fe-projected vibrational density of states*:

$$D_{\mathrm{Fe}}(\omega) = \sum_i |\mathbf{e}_{\mathrm{Fe}}^{(i)}|^2 \cdot \delta(\omega - \omega_i) \tag{36}$$

weighted by the Lamb-Mößbauer factor f_{LM} and a multi-phonon convolution. For band identification it is usually sufficient to use the Fe-pDOS without multi-phonon contributions – we call this $g_{\mathrm{Fe}}(\omega)$.

Important properties:

- NRVS is **element-selective**: only modes with Fe participation appear. Modes containing pure ligand motions (e.g., C-H stretching) are not visible.
- NRVS is **not symmetry-selective**: unlike Raman/IR, all modes are visible regardless of their D_{2h} symmetry. This makes NRVS particularly useful for iron-sulfur clusters, where

many modes are dark in IR/Raman by selection rules.

- NRVS is **quantitative**: the integral of the pDOS is normalized to 1 (or $3N_{\mathrm{Fe}}$ in some conventions). Band intensities are directly mode couplings.

Band structure in [2Fe-2S] proteins

The typical [2Fe-2S] NRVS band structure (Gee 2021 [7]; a comprehensive overview of NRVS modes is given by Wang *et al.* 2025 [46]) has three main regions:

Low-frequency region (50–200 cm^{-1}). Skeletal vibrations of the entire protein. Cluster-relevant: cluster-plane motions (translational + rotational in the protein matrix), cluster-OOP buckling (acoustic mode mixing with ligands). λ_X is small in all channels here ($\lambda_X \approx \hbar\omega/4 \approx 12$–$50\,\mathrm{cm}^{-1}$ per mode), but the *number* of modes is large.

Mid-frequency region (200–400 cm^{-1}). This is where most of the cluster bond stretching modes lie:

- Fe-N(His) stretching: 260–290 cm^{-1} in mitoNEET-type $\mathrm{Cys_3His_1}$ coordination (Gee 2021 [7]: computed 223 cm^{-1} for Fe(II)-$\mathrm{N_\delta}$(His) in the reduced state, around 270–290 cm^{-1} in the oxidized state). For classical Rieske $\mathrm{Cys_2His_2}$ clusters: a doublet at 266 + 274 cm^{-1} (Kuila *et al.* 1992 [45], Fe(III)-N), assigned to the protonated (266) and deprotonated (274) imidazole, respectively.
- Fe-Fe cluster breathing: $\sim 200\,\mathrm{cm}^{-1}$, depending on the cluster geometry and ligands.
- Fe-$\mathrm{S_{bridge}}$ symmetric stretch: 300–340 cm^{-1}.
- Fe-$\mathrm{S_{Cys}}$ stretching: 280–320 cm^{-1}.

For a pure Fe-X stretch mode with $\alpha_X \approx 1$, we have $\lambda_X \approx \hbar\omega/4 \approx 60$–$100\,\mathrm{cm}^{-1}$ – a substantial contribution per mode.

High-frequency region (400–800 cm^{-1}).

- Fe-$\mathrm{S_{bridge}}$ asymmetric stretch: 380–420 cm^{-1}.
- Imidazole ring modes with Fe participation: 600–700 cm^{-1}.
- Cluster-ligand bending modes with Fe contribution: 500–700 cm^{-1}.

How to compare $\Lambda_X(\omega)$ to NRVS

The cumulative reorganization curve $\Lambda_X(\omega) = \sum_{\omega_i \le \omega} \lambda_X(i)$ is a **step function**: every mode with non-trivial λ_X contribution produces a step of height $\lambda_X(i)$ at $\omega = \omega_i$.

In comparison with the NRVS spectrum:

- Where there is a band in the Fe-pDOS, there should be a step in $\Lambda_X(\omega)$ – for the channel X that dominates this band.
- The height of the step is proportional to $\hbar\omega \cdot \alpha_X^2$. A step of $50\,\text{cm}^{-1}$ at $\omega = 350\,\text{cm}^{-1}$ corresponds to $\alpha_X^2 = 4 \cdot 50/350 = 0.57$ – a mode with about 57 % bond contribution along X.
- If the NRVS spectrum shows a band but all $\Lambda_X(\omega)$ curves stay flat there, the mode is not strongly active in any of our 5 channels – it is likely a ligand-internal mode with only minor Fe participation.

This correspondence makes $\Lambda_X(\omega)$ a direct **band-assignment tool**. When a reviewer asks "which bond is active in band X?", one plots $\Lambda_{\text{FeFe}}, \Lambda_{\text{FeN}}, \Lambda_{\text{FeS}}, \Lambda_{\text{NH}}, \Lambda_{\text{HA}}$ above the NRVS spectrum and reads off the answer.

Low-temperature consistency

NRVS measurements are typically performed at 5–40 K (liquid He) to maximize Lamb-Mößbauer resolution and minimize thermal line broadening. In this low-temperature region, our λ_X is identical with the zero-point value $\hbar\omega/4 \cdot \alpha_X^2$ – i.e., our computed λ_X *is* what NRVS measures, modulo spectral broadening and multi-phonon effects.

Consequence: for band identification one does NOT need to compute at the exact measurement temperature. A run at `temp_k = 5` yields the low-temperature values; a run at `temp_k = 300` yields the room-temperature values; in the NRVS-relevant frequency region ($\omega > 200\,\text{cm}^{-1}$) both are practically identical, because the Bose-Einstein population vanishes there.

What NRVS *does not* see

- **Pure N-H stretching modes** (imidazole NH at 3000+ cm^{-1}): outside the typical NRVS measurement range (instruments end at 800–1000 cm^{-1}), and the mode has only tiny Fe participation anyway. Consequence: Λ_{NH} in our typical filters (`freq_max = 800`) is always very small – see the report table.
- **Pure OH stretching modes**: same.
- **Backbone modes** without Fe contribution: remain dark in the pDOS.

These points are important when justifying band assignments in the report: if a mode is dark in the pDOS but our $\Lambda_X(\omega)$ shows a step there, this is a *hint* that the mode exists and has Fe contribution but a very small Lamb-Mößbauer factor (the mode is not coherent with Fe). This is particularly interesting in PCET studies.

2.12 Mode-resolved bonding reorganization energies

The package implements a *conceptual pivot*: away from classifying individual modes as "PCET candidates" (with thresholds, filters, tanh saturation), toward direct computation of mode-resolved bond reorganization contributions, in Marcus-Hush form but with thermally scaled mode amplitudes.

Before describing the pipeline, a clarification of the physical meaning of the computed quantities.

What λ_X is and what it is not

The **classical Marcus-Hush reorganization energy** [19, 20] is

$$\lambda_{\text{total}}^{\text{Marcus}} = \frac{1}{2} \sum_i \mu_i \, \omega_i^2 \, (\Delta Q_i)^2, \tag{37}$$

where ΔQ_i is the *static* displacement of the harmonic mode coordinate i between reactant and product states (activation energy of the reaction). Computing this quantity requires two separate frequency calculations (reactant and product) plus a mode mapping between them – effort that we *do not* undertake here.

What we compute instead is a related but distinct quantity: the mode-resolved contribution energy of the *thermal bond-length fluctuations* per bonding channel. From the eigenvectors of the frequency calculation, for each mode i and each bond X the thermally scaled bond modulation $\Delta r_X(i)$ is computed (displacement vectors multiplied by u_{rms} from the QHO expectation value), and from this

$$\lambda_X(i) = \tfrac{1}{2} \, \mu_i \, \omega_i^2 \, (\Delta r_X(i))^2. \tag{38}$$

The sum over all modes, $\Lambda_X^{\text{total}} = \sum_i \lambda_X(i)$, is the thermal bond reorganization energy of bond X. At $T \to 0$ this reduces (with $\mu = m_{\text{mode}}$) to

$$\lambda_X(i) \to \tfrac{\hbar\omega_i}{4} \cdot \alpha_X^2(i), \tag{39}$$

where $\alpha_X(i)$ is the dimensionless "mode contribution" to bond X, defined implicitly by $\Delta r_X(i) = u_{\text{rms}}(i) \cdot \alpha_X(i)$. For a pure stretching mode of bond X, $\alpha_X(i) = 1$ and the reorganization contribution at $T \to 0$ is $\hbar\omega/4$ (zero-point energy divided by 2). In practice $\alpha_X(i)$ can exceed 1 for modes where the two bond partners move in strongly opposite directions (relative-displacement enhancement); this is geometrically legitimate and does not violate any conservation law – the eigenvector normalisation $\sum_i \|\mathbf{e}_i\|^2 \approx 1$ is a sum over *atoms*, not over bonds.

Significance for PCET studies.

- Λ_X^{total} is a well-defined spectroscopic observable. It measures, mode-resolved, which bond fluctuates most strongly in the thermal vibrations.
- Suitable for **NRVS band identification**: frequency ranges with large $\Lambda_X(\omega)$ steps are

the bands in which bond X is strongly modulated. Direct comparison with NRVS spectra is possible.

- Suitable for **system comparisons at the same temperature**: WT vs. mutant, prot vs. deprot. Scales consistently. In particular: in the deprot variant, NH and HA vanish by channel definition; the others remain comparable.
- Related to the **Huang-Rhys factor** [50] and to Franck-Condon theory of vibronic spectra.

What the quantity does not provide.

- *No* classical Marcus-theory activation energies for a specific ET or PT reaction – this would require separate reactant and product geometries.
- *No* reaction-rate prediction.
- *No* direct comparison with literature values of the Marcus λ for a reaction (insofar as those use static reactant–product differences).

The Hammes-Schiffer view. Hammes-Schiffer and Soudackov [49] show that PCET in proteins typically activates *many modes simultaneously*. The notion of a single "promoter mode" is a special case (small models), not the rule. Our mode-resolved view makes precisely this multi-mode reality explicit: instead of a top-mode classification we provide the distribution of bond modulations across all modes.

Five bonding channels

Per mode, five bonding channels are evaluated:

- `FeFe` – Fe-Fe cluster breathing (ET reorganization).
- `FeN` – Fe-N(His) bond (PT-relevant in Rieske-type systems).
- `FeS` – Fe-S(Cys) bond and Fe-S_{bridge} within the cluster.
- `NH` – N-H stretching of the protonated His.
- `HA` – H-acceptor distance (hydrogen bond), computed in *reaction-coordinate mode* (see below).

NH and HA are skipped for deprotonated His ligands (no H available). FeS includes both Fe-Cys bonds and cluster-internal Fe-μS bonds; the per-bond breakdown is provided in the sheet `Reorg_per_bond`.

Geometric modulation Δr_X

For a mode with thermally scaled displacement vectors $\mathbf{e}_a, \mathbf{e}_b$ at the two bond partners (positioned at $\mathbf{r}_a, \mathbf{r}_b$), the signed bond-length modulation is

$$\Delta r_X^{(\text{class.})} = (\mathbf{e}_a - \mathbf{e}_b) \cdot \hat{\mathbf{n}}_{ab}, \qquad \hat{\mathbf{n}}_{ab} = \frac{\mathbf{r}_a - \mathbf{r}_b}{\|\mathbf{r}_a - \mathbf{r}_b\|}. \quad \rightsquigarrow \texttt{reorganization.signed\_dr\_along\_axis} \tag{40}$$

Convention: $\Delta r > 0$ means "the bond stretches". This classical definition is used for FeFe, FeN, FeS, and NH.

HA as a reaction coordinate. For the HA channel, equation 40 does *not* measure what is physically relevant for PT: in a high-frequency C=O vibration in the protein backbone, the acceptor moves strongly and the H stays nearly fixed, but the distance $|\mathbf{r}_H - \mathbf{r}_A|$ changes – and would be counted as an HA modulation, even though it has no PT character.

The HA modulation measures the *motion of the H relative to the donor N*, projected onto the N→acceptor axis:

$$\Delta r_{\text{HA}}^{(\text{react. coord.})} = (\mathbf{e}_H - \mathbf{e}_N) \cdot \hat{\mathbf{n}}_{NA}, \qquad \hat{\mathbf{n}}_{NA} = \frac{\mathbf{r}_A - \mathbf{r}_N}{\|\mathbf{r}_A - \mathbf{r}_N\|}. \quad \rightsquigarrow \texttt{reorganization.compute\_mode\_modulations} \tag{41}$$

If only the acceptor moves (skeletal mode), then $\mathbf{e}_H - \mathbf{e}_N \approx 0$ and hence $\Delta r_{\text{HA}} = 0$ – no PT contribution. A genuine PT motion (H moving from N toward A) yields, by contrast, $\Delta r_{\text{HA}} > 0$.

Main acceptor per H. For each His-H atom, all acceptors within `pcet_hbond_cutoff_a` (default 4 Å) are detected, but only the *main acceptor* (highest Gaussian weight $\exp(-(r_{\text{eq}} - r_0)^2/(2\sigma^2))$ with $r_0 = 2.8$ Å, $\sigma = 0.4$ Å) is used as the HA channel. This avoids double-counting acceptors – a single H physically moves in only one direction. Concretely for PCET, the logic is: per His imidazole there is one NE2-H (or ND1-H), and at this H sits exactly one dominant H-bond acceptor (typically a water, backbone carbonyl O, or an Asp/Glu O).

Effect of the Gaussian weight. The Gaussian weight $w = \exp(-(r_{\text{eq}} - r_0)^2/(2\sigma^2))$ is used *only* for main-acceptor selection – not as a damping factor in the lambda aggregation. The binary logic is now: *is an H-bond acceptor within `pcet_hbond_cutoff_a` reach or not?* If acceptors are in range, the HA mode counts fully (weight = 1.0); if not, the channel is dropped. The geometric damping of distant acceptors arises automatically from the smaller Δr_{HA} values (distant H-bonds are weakly modulated anyway); additional multiplication by $w < 1$ would be a double penalty.

Reduced mass: two conventions

For the per-mode λ contribution we have two choices for the reduced mass:

- μ_{pair} – bond-specific reduced mass, computed from the two bonded atoms ($\mu_{\text{NH}} \approx 0.94$ amu,

$\mu_{\text{FeFe}} \approx 27.92\,\text{amu}$, etc.). Closer to the *isolated-oscillator* picture: good for mode-to-mode comparison *within the same* bond.

- μ_{mode} – the mode's own reduced mass from the Gaussian output. By construction this satisfies the orthogonality of the eigenmodes, so the λ contributions add up *without double counting* – this is the **Marcus-Hush-correct** choice for the sum in eq. 37.

Both are written out (columns `lambda_X_pair_cm1` and `lambda_X_mode_cm1` in the sheet `Reorganization_energy`). For the system aggregation in `Reorg_total`, μ_{mode} is used.

Per-mode reorganization contribution λ_X

Formula (38) above, $\lambda_X(i) = \frac{1}{2}\,\mu_i\,\omega_i^2\,(\Delta r_X(i))^2$, is applied with the geometric modulations $\Delta r_X(i)$ defined above and the chosen reduced-mass convention (μ_{pair} or μ_{mode}).

Unit conversions: $\omega_{\text{SI}} = 2\pi c\,\omega_{\text{cm}^{-1}}$, $\mu_{\text{SI}} = \mu_{\text{amu}} \cdot \text{u}$, $\Delta r_{\text{SI}} = \Delta r_{\text{Å}} \cdot 10^{-10}\,\text{m}$; $\lambda_{\text{cm}^{-1}} = \lambda_{\text{J}}/(hc)$.

Important: because of the thermal scaling $\mathbf{e}_a = \mathbf{e}_a^{\text{normalized}} \cdot u_{\text{rms}}$, $\Delta r_X(i)$ lies on the scale of the QHO probability distribution, not of a reactant–product difference. For pure stretching modes of bond X, $\lambda_X(i)$ converges at $T \to 0$ to $\hbar\omega_i/4$ (which equals $\omega_i/4$ in cm^{-1}); for mixed modes, this is multiplied by the squared mode contribution α_X^2.

System aggregation

$$\Lambda_X^{\text{total}} = \sum_{i=1}^{N_{\text{modes}}} \lambda_X(i) \quad \rightsquigarrow \texttt{reorganization.compute\_total\_reorganization} \tag{42}$$

per channel $X \in \{\text{FeFe, FeN, FeS, NH, HA}\}$.

The sums are also written out *frequency-resolved*:

- **Modulation spectra** $M_X(\omega)$ – Gaussian-broadened accumulation of the per-mode bond modulations along the frequency axis:

$$M_X(\omega) = \sum_{i=1}^{N_{\text{modes}}} \Delta r_X^{\text{RSS}}(i)\,\mathcal{G}(\omega - \omega_i, \sigma_M), \quad \mathcal{G}(x, \sigma) = \frac{e^{-x^2/(2\sigma^2)}}{\sigma\sqrt{2\pi}}. \quad \rightsquigarrow \texttt{reorganization.compute\_modulation\_spe} \tag{43}$$

Default broadening: $\sigma_M = 5\,\text{cm}^{-1}$, configurable via `reorg_spectrum_sigma_cm1`.

What is $\Delta r_X^{\text{RSS}}(i)$ exactly? Each parent channel $X \in \{\text{FeFe}, \text{FeN}, \text{FeS}, \text{NH}, \text{HA}\}$ can have several *sub-channels*: e.g. FeN comprises the two Fe-N(His_a) and Fe-N(His_b) sub-bonds, FeS comprises all four Fe-S(Cys_k) sub-bonds (or two for Cys_2His_2 clusters). For mode i we therefore aggregate sub-channel modulations as the weighted root-sum-of-squares

$$\Delta r_X^{\text{RSS}}(i) = \sqrt{\sum_{s \in \text{sub-channels}(X)} w_s\,(\Delta r_X^{(s)}(i))^2}, \quad \rightsquigarrow \texttt{reorganization.aggregate\_by\_parent} \tag{44}$$

where the weights w_s are the geometric weights from `build_channels_from_coord_info` (Gaussian H$\cdots$acceptor weighting for HA, unity for FeFe/FeN/FeS/NH). For single-sub-channel parents (FeFe, NH, HA-per-acceptor) this reduces to $\sqrt{w}\,|\Delta r_X(i)|$, recovering the simple absolute modulation. The RSS aggregation is the *energetically consistent* choice: it satisfies $\lambda_X^{\text{parent}}(i) = \frac{1}{2}\mu_i\omega_i^2\,(\Delta r_X^{\text{RSS}}(i))^2$ with $\lambda_X^{\text{parent}}$ being the weighted sum of the sub-channel λ's.

- **Co-modulation spectra** – geometric mean of two M_X spectra, large only where *simultaneously* two bonds are modulated:

$$C_{\text{PCET}}(\omega) = \sqrt{M_{\text{HA}}(\omega)\,M_{\text{FeFe}}(\omega)}, \tag{45}$$

$$C_{\text{PT-FeN}}(\omega) = \sqrt{M_{\text{HA}}(\omega)\,M_{\text{FeN}}(\omega)}, \tag{46}$$

$$C_{\text{ET-FeS}}(\omega) = \sqrt{M_{\text{FeFe}}(\omega)\,M_{\text{FeS}}(\omega)}. \tag{47}$$

- **Cumulative reorganization curves** $\Lambda_X(\omega)$ – monotonically increasing, converging at the spectrum end to Λ_X^{total}:

$$\Lambda_X(\omega_{\text{cut}}) = \sum_{i\,:\,\omega_i \le \omega_{\text{cut}}} \lambda_X(i). \tag{48}$$

Steps in $\Lambda_X(\omega)$ mark reorganization-active frequency windows that can be compared directly with NRVS spectra.

Excel output

Five sheets per run or per frequency window:

- `Reorganization_energy` – per mode 21 columns: `freq_cm1` and, for each channel, four values (Δr_{rms}, $\Delta r_{\text{sum-signed}}$, λ_{pair}, λ_{mode}).
- `Reorg_total` – 5-row summary: per channel Λ_X^{total} in both μ conventions plus the number of contributing modes.
- `Reorg_per_bond` – per individual bond (FeS_Cys207, FeS_Cys216, FeS_Cluster_*, FeN_His255, etc.) the contribution, with acceptor weight and number of modes.
- `Modulation_spectra` – frequency-resolved: five $M_X(\omega)$ curves plus three co-indicator curves $C_{\text{PCET}}, C_{\text{PT-FeN}}, C_{\text{ET-FeS}}$.
- `Lambda_cumulative` – five $\Lambda_X(\omega)$ curves for direct comparison with NRVS spectra.

2.13 HP/standard frequency consistency check

As indicated in §2.1, with `freq=hpmodes` Gaussian writes the frequencies *twice* into the log file – once in the HP block (5 decimal places) and once in the standard block (2 decimal places). Both blocks refer to the same eigenvalue calculation; their frequencies *must* therefore agree (within the writing precision of $0.01\,\mathrm{cm}^{-1}$).

Implementation

`logio.check_hp_std_frequency_consistency` compares, during the run:

1. Mode by mode: for each mode index, the HP frequency is compared to the standard-block frequency.
2. Tolerance: $0.01\,\mathrm{cm}^{-1}$ (default), configurable via `cfg.hp_std_freq_tolerance_cm1`.
3. Where they disagree, the maximum difference, the mode index in question, and both frequencies are logged.

What discrepancies mean

Symptom	Possible cause
1–2 modes with $\Delta\omega > 0.01$	Numerical truncation in the Gaussian output (rare, harmless)
Many modes systematically offset	Wrong block pairing (HP block from another job, restart mismatch)
Complete disagreement	Aborted job, partial log file, or two different log files concatenated

In all cases, the discrepancy is *not* treated as fatal – the program continues and uses the HP values. The diagnosis is written to the report so that follow-up investigation is possible.

Practical note. In very old log files (Gaussian 09 or earlier), there are occasionally sign discrepancies between HP and standard block for the lowest frequencies ($\omega \to 0$): a mode that appears as $-2.3\,\mathrm{cm}^{-1}$ in the HP block and as $+2.3\,\mathrm{cm}^{-1}$ in the standard block. This is a Gaussian artifact of the translation/rotation handling. The consistency check warns about this, but the behavior is harmless – such modes are usually excluded anyway because $\omega <$ threshold.

Part II

Application

3 Configuration

All run parameters are collected in a "Config" object. "Config" is a Python "dataclass" with default values for all optional fields (`src/modenanalyse_2fe2s/config.py`). Three fields are *required*: `log_file`, `output_dir`, and – if ligand resolution is desired – `pdb_file`.

3.1 Example configuration

The typical form, runnable in Spyder (see "example_run.py"):

```
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file    = r"D:\Data\cluster_freq.log",
    pdb_file    = r"D:\Data\cluster.pdb",
    output_dir = r"D:\Data\results",
    temp_k      = 40.0,
    freq_max    = 800.0,
    freq_windows = [(0, 100), (100, 300), (300, 500), (500, 700)],
)

run_analysis(cfg)
```

All other fields keep their defaults. The following tables list the fields grouped by function.

3.2 Complete parameter reference

The following overview shows all 48 configuration fields with their default values. Detailed descriptions and recommendations appear in the topic sections below.

Table 1: Complete overview of all configuration fields with default values. An annotated example TOML file with all fields is provided as `run_example_full.toml` in the distribution directory.

Field	Default	Section
`log_file`	— (required)	Input
`output_dir`	— (required)	Input
`pdb_file`	`""`	Input
`pdb_chain`	`"A"`	Input
`logname_suffix`	`""`	Input
`freq_min`	`None`	Frequency
`freq_max`	`None`	Frequency

Field	Default	Section
`freq_windows`	`None`	Frequency
`temp_k`	`None`	Thermo
`amplitude`	`1.0`	Thermo
`cluster_index`	`0`	Cluster
`analyze_all_clusters`	`False`	Cluster
`fe_s_cutoff`	`3.0` Å	Cluster
`fe_fe_cutoff`	`3.5` Å	Cluster
`strict_cluster`	`False`	Cluster
`fe_coord_cutoff_n`	`2.50` Å	Ligand
`fe_coord_cutoff_s`	`2.70` Å	Ligand
`fe_coord_cutoff_o`	`2.50` Å	Ligand
`include_hn_vibration`	`True`	Ligand
`mode_type_threshold`	`60.0`%	OOP
`mode_type_detail_thresholds`	`(60, 75, 90)`	OOP
`analyze_scsd`	`True`	SCSD
`analyze_sse`	`True`	SCSD
`sse_chain`	`""`	SCSD
`pcet_enabled`	`True`	PCET
`pcet_hbond_cutoff_a`	`4.0` Å	PCET
`pcet_acceptor_r0_a`	`2.8` Å	PCET
`pcet_acceptor_sigma_a`	`0.4` Å	PCET
`reorg_spectrum_sigma_cm1`	`5.0` cm^{-1}	PCET
`reorg_spectrum_step_cm1`	`0.5` cm^{-1}	PCET
`export_embedding_plots`	`True`	Embedding
`umap_n_neighbors`	`None` (auto)	Embedding
`include_hydrogen`	`True`	Numerics
`sigma_eigvec`	`5e-4`	Numerics
`sigma_coord`	`1e-3`	Numerics
`amplitude_threshold`	`0.001`	Numerics
`coord_match_tol`	`1.0` Å	Numerics
`scan_chunk_mb`	`16`	Numerics
`show_errors_in_excel`	`True`	Numerics
`interp_step`	`0.05` cm^{-1}	Numerics
`interp_boundary_mode`	`"context"`	Numerics
`interp_edge_extend`	`0.5` cm^{-1}	Numerics
`interp_context_cm1`	`5.0` cm^{-1}	Numerics
`use_cache`	`True`	Numerics

3.3 Input and output

Field	Default	Description
log_file	*(required)*	Path to the DFT frequency file. Supported: Gaussian-16 `.log` (with or without `freq=hpmodes`) and ORCA `.hess` (auto-detected from the extension).
output_dir	*(required)*	Target directory. Created if it does not exist.
pdb_file	""	PDB file with the same cluster (ligand resolution; optional but recommended).
pdb_chain	"A"	Chain identifier in the PDB file.
logname_suffix	""	Optional suffix for output file names (useful for several runs from the same log file, e.g. different clusters).

3.4 Frequency range and windows

Field	Default	Description
freq_min	None	Lower frequency limit in cm^{-1}. `None` means no lower limit.
freq_max	None	Upper frequency limit. For NRVS-relevant analyses, typically 800.
freq_windows	None	List of windows such as `[(0, 100), (100, 300), ...]`. When set, the program generates *one Excel file per window* (multi-window mode).

Convention. `freq_max` (and `freq_min`) also apply to all reorganization and aggregate analyses. Since ^{57}Fe-projected contributions vanish above $\sim$800 cm^{-1} and NRVS experiments measure in this range, an 800 cm^{-1} cutoff is appropriate both experimentally and numerically.

Window-boundary convention (since v1.0.4). Multi-window mode partitions modes into bins `[(lo`$_1$`,hi`$_1$`), (lo`$_2$`,hi`$_2$`), ..., (lo`$_N$`,hi`$_N$`)]`. The boundaries are *half-open from the left*:

- Every window except the last uses the interval $[\,\mathrm{lo},\,\mathrm{hi}\,)$. A mode with frequency exactly equal to a window boundary belongs to the *upper* window (the one starting at this value).
- The last window in the list uses $[\,\mathrm{lo},\,\mathrm{hi}\,]$, so its upper edge is included. If `hi` is `None`/`inf` for the last window, it is open-ended.

This guarantees that each mode is counted in exactly one window, preventing double-counting of B-factor contributions and $\Lambda_X(\omega_{\mathrm{cut}})$ values for boundary modes (§2.3). Pre-v1.0.4 used closed-closed intervals on both ends, which over-counted exact-boundary modes.

3.5 Thermal scaling

Field	**Default**	**Description**
`temp_k`	`None`	Sample temperature in Kelvin for the thermal scaling of the eigenvectors. `None` → classical mode with amplitude = `amplitude`. Typical NRVS values: 40, 80, 100.
`amplitude`	`1.0`	Fallback amplitude in Å when `temp_k = None`. The default of 1.0 is the historical convention.

3.6 Cluster detection

Field	**Default**	**Description**
`fe_s_cutoff`	`3.0`	Maximum Fe-S distance (Å) within which S atoms are counted as part of the cluster.
`fe_fe_cutoff`	`3.5`	Maximum Fe-Fe distance (Å) for cluster identification.
`cluster_index`	`0`	For multi-cluster systems (e.g. glutaredoxin dimer): index of the cluster to be analyzed. Default 0 = closest Fe-Fe pair. One run per cluster.
`analyze_all_clusters`	`False`	If `True`, every detected cluster is analyzed in turn (one subdirectory per cluster). Convenient for multi-cluster systems; takes precedence over `cluster_index`.

Field	Default	Description
strict_cluster	False	If True, the program aborts with an error when the cluster cannot be unambiguously identified (no fallback to nearby atoms).

3.7 Fe ligand detection (PDB-based)

Field	Default	Description
fe_coord_cutoff_n	2.50	Maximum Fe-N distance (Å) for ligand N (His).
fe_coord_cutoff_s	2.70	Maximum Fe-S distance for ligand S (Cys).
fe_coord_cutoff_o	2.50	Maximum Fe-O distance for ligand O (Asp/Glu, rare for [2Fe-2S]).
include_hn_vibration	True	N-H stretching analysis for protonated His ligands (sheet His_HN). Requires include_hydrogen=True.

3.8 OOP/INP classification

Field	Default	Description
mode_type_threshold	60.0	Binary OOP threshold in percent (§2.4).
mode_type_detail_thresholds	(60, 75, 90)	Three thresholds for the fine 7-level classification. Applied symmetrically about 50 %.

3.9 SCSD and secondary structure

Field	Default	Description
analyze_scsd	True	Activate Symmetry-Coordinate Structural Decomposition (requires scsdpy).
analyze_sse	True	Activate secondary-structure analysis (helices, sheets from PDB).
sse_chain	""	Chain filter for SSE detection. Empty = same as pdb_chain.

3.10 PCET / Marcus-Hush reorganization

Field	Default	Description
pcet_enabled	True	Enable the Marcus-Hush reorganization pipeline. If False, Δr_X and λ_X are not computed per mode and all reorganization sheets are omitted.
pcet_hbond_cutoff_a	4.0	Maximum H...acceptor distance (Å) for the acceptor search, measured from the H atom.
pcet_acceptor_r0_a	2.8	Optimal distance (Å) for the Gaussian weighting of acceptor selection; matches a typical N-H$\cdots$O/N bond length.
pcet_acceptor_sigma_a	0.4	Gaussian standard deviation (Å) of the acceptor weighting. At $\|r - r_0\| > 2\sigma$, the acceptor effectively contributes nothing.
reorg_spectrum_sigma_cm1	5.0	Gaussian broadening of the modulation spectra $M_X(\omega)$. $5\,\mathrm{cm}^{-1}$ matches typical DFT frequency uncertainty + NRVS resolution.
reorg_spectrum_step_cm1	0.5	Frequency grid step for $M_X(\omega)$ and $\Lambda_X(\omega)$. $0.5\,\mathrm{cm}^{-1}$ produces fine bands at a tractable file size.

3.11 Numerical parameters

Field	Default	Description
include_hydrogen	True	Include H atoms in eigenvector reads. Required for PCET.
sigma_eigvec	5e-4	Numerical noise in eigenvector components (Bernoulli variance). Controls the threshold for "insignificant" contributions.
sigma_coord	1e-3	Coordinate noise in Å.
amplitude_threshold	0.001	Threshold for a significant atomic displacement in a mode.
coord_match_tol	1.0	Tolerance (Å) for PDB↔Gaussian atom matching after Kabsch alignment.
interp_step	0.05	Step size (cm^{-1}) for the interpolated pDOS export.
interp_boundary_mode	"context"	Behavior at the window edge: "context" (with buffer zone), "hard" (sharp cut), "extend" (linearly extrapolated).

Field	Default	Description
`interp_edge_extend`	`0.5`	Buffer-zone width (cm^{-1} or fraction) in `"context"` mode.
`scan_chunk_mb`	`16`	Block size for log-file scanning in MB. For very large log files (>1 GB), increasing this can speed up reading.
`use_cache`	`True`	Activates file caching for block offsets in the log file. Halves the run time on repeated calls.

3.12 Embeddings (UMAP)

Field	Default	Description
`umap_n_neighbors`	`None`	UMAP n_neighbors parameter. `None` = auto heuristic $\max(5, \min(15, N/100))$; for typical [2Fe-2S] protein sizes ($N \approx 5\,000$–$10\,000$) this yields 15 (the paper standard, [36]). For very small data sets (< 50 modes), set manually to e.g. 5–8.
`export_embedding_plots`	`True`	Produce a PNG plot of the UMAP projection. Set to `False` to save matplotlib setup time.

3.13 Other

Field	Default	Description
`show_errors_in_excel`	`True`	If `True`, warnings from the run are also displayed in the Excel sheet `Info`.

3.14 Validation rules

All fields are checked for syntactic and numerical plausibility when `cfg.validate()` is called (executed automatically at the start of `run_analysis`):

- Required fields are set
- Cutoffs > 0
- Frequency limits are ascending if both are set
- PCET thresholds: $0 <$ moderate $<$ strong ≤ 1

- Band edges strictly ascending
- Line shape is from the allowed set

Validation errors are collected and reported together (no fail-fast on a single error). This simplifies fixing several typos in a configuration in one go.

4 Workflow examples

Three typical scenarios are worked through completely below: a single cluster (the standard application), multi-cluster (glutaredoxin dimers with two clusters), and a WT-vs.-mutant comparison (PCET validation test). A fourth scenario walks an entire small system from the log file to band identification with all intermediate numbers.

4.1 How to start a run

There are two equivalent ways to start an analysis: programmatically from Python (`run_analysis(cfg)`), or from the command line with a TOML configuration file. The TOML route is recommended for reproducible production runs because the configuration is self-documenting and can be version-controlled.

TOML + CLI (recommended)

The basic invocation is:

```
modenanalyse-2fe2s my_run.toml
```

The TOML file path is the *first positional argument* – no `--config` flag is needed. The package can also generate an annotated template:

```
modenanalyse-2fe2s --write-template my_run.toml
# edit my_run.toml as needed
modenanalyse-2fe2s my_run.toml
```

Path pitfalls in TOML

TOML strings with double quotes interpret backslashes as escape characters, so Windows paths such as `"D:\Data\file.log"` will produce a parser error ("Unescaped backslash in a string" or similar). Three valid alternatives:

```
# 1. Forward slashes (works on Windows too):
log_file = "D:/Data/file.log"
```

```
# 2. Single quotes = TOML literal string (no escaping):
log_file = 'D:\Data\file.log'

# 3. Escaped backslashes:
log_file = "D:\\Data\\file.log"
```

Important. Python's raw-string prefix `r"..."` is *not* valid TOML syntax. When copying paths from Python code (e.g. `r"D:\Data\file.log"`) into a TOML file, the `r` prefix must be removed and one of the three forms above used instead. This is a common stumbling block for users transitioning from `run_analysis()` calls to TOML-based configuration.

CLI overrides

CLI arguments can override individual TOML fields without editing the file – useful for parameter scans:

```
# Use my_run.toml, but force temp_k = 40 K:
modenanalyse-2fe2s my_run.toml --temp-k 40.0

# Use my_run.toml, but write to a different output directory:
modenanalyse-2fe2s my_run.toml --output-dir ./results_test
```

A complete list of available CLI flags is shown by `modenanalyse-2fe2s --help`.

Programmatic API

For interactive analysis (e.g. in Spyder or Jupyter), the same configuration can be built directly:

```python
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file   = r"D:\Data\system_freq.log",  # Python raw string is fine here
    pdb_file   = r"D:\Data\system.pdb",
    output_dir = r"D:\Data\results",
    temp_k     = 5.0,
    freq_max   = 800.0,
)
run_analysis(cfg)
```

Note that in *Python source code* (`.py` files), `r"..."` raw strings *are* valid and are the recommended way to write Windows paths. The restriction to non-raw strings only applies to TOML configuration files.

4.2 Scenario A: single cluster (standard application)

Input data

- `cluster_wt_freq.log` – Gaussian-16 `freq=hpmodes` run of the optimized cluster model (cluster + first ligand sphere, ca. 50–100 atoms).
- `cluster_wt.pdb` – structural model with the same ligand residues as in the DFT run, plus additional residues for the SSE analysis.

Configuration

```
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file   = r"D:\Data\cluster_wt_freq.log",
    pdb_file   = r"D:\Data\cluster_wt.pdb",
    output_dir = r"D:\Data\cluster_wt_40K",
    pdb_chain  = "A",
    temp_k     = 40.0,
    freq_max   = 800.0,
    freq_windows = [(0, 100), (100, 300), (300, 500), (500, 700)],
)

run_analysis(cfg)
```

Expected output

In the `output_dir`, four subdirectories are created (one per window):

- `cluster_wt_freq_0-100/`
- `cluster_wt_freq_100-300/`
- `cluster_wt_freq_300-500/`
- `cluster_wt_freq_500-700/`

Each folder contains:

- `cluster_wt_freq_analysis.xlsx` – main analysis (mode analysis, kernel scores, SCSD, PCET/ET, Fe-bond, His-HN, Info, ...)
- `cluster_wt_freq_pDOS_interp.xlsx` – interpolated Fe-pDOS, Origin-compatible
- `REPORT.txt` – plain-text run log

What to check first

1. **Open the report file** and inspect:
   - Cluster detected? Are the Fe-Fe and Fe-S distances plausible?
   - Ligands assigned correctly? (table `coord_summary`)
   - Kabsch RMSD small (< 0.5 Å)?
   - HP/standard consistency: any irregularities?
   - PCET reorg active (His ligands found, $\Lambda_{\mathrm{HA}} > 0$)?
2. **Open the `Mode_analysis` sheet**: sort by *kern_oop* → inspect the OOP-dominated modes.
3. **Sheet `Reorg_total`**: five lambda values for the system (FeFe, FeN, FeS, NH, HA). Which channel dominates? In Origin, open `Lambda_cumulative` and plot the cumulative curves $\Lambda_X(\omega)$ per channel – steps mark reorganization-active frequencies.

4.3 Scenario B: multi-cluster system (dimers and beyond)

Some systems carry several [2Fe-2S] clusters (e.g., glutaredoxin dimers with two equivalent clusters; domain constructs with two distinct clusters; or multi-cluster complexes where two clusters cooperate at the active site).

Since v1.0.0, the tool supports multi-cluster analysis as a **first-class** feature: a single run analyzes all detected clusters automatically.

Cluster detection

`geometry.find_all_clusters` searches for all Fe-Fe pairs below `cfg.fe_fe_cutoff` (default 3.5 Å) and groups the bridging S atoms. Sorting: ascending by Fe-Fe distance, so the tightest cluster comes first.

The console output displays the full list:

```
[i] 2 [2Fe-2S] clusters found in the system:
    Cluster #0: Fe-Fe = 2.713 A, Fe-S = 2.245-2.318 A
    Cluster #1: Fe-Fe = 2.741 A, Fe-S = 2.262-2.305 A
```

Activating multi-cluster mode

In TOML:

```toml
[input]
log_file   = "dimer.log"
pdb_file   = "dimer.pdb"
output_dir = "./results"
```

```
[cluster]
analyze_all_clusters = true    # <- enables the multi-cluster run
```

Or programmatically:

```python
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file             = r"D:\Data\dimer.log",
    pdb_file             = r"D:\Data\dimer.pdb",
    output_dir           = r"D:\Data\dimer_results",
    analyze_all_clusters = True,        # <-- multi-cluster
    temp_k               = 5.0,
    freq_max             = 800.0,
)
run_analysis(cfg)
```

What the tool does then

With `analyze_all_clusters = true`, the official API `run_analysis(cfg)` internally calls `find_all_clusters` to determine the number of clusters. The full pipeline then runs **once per detected cluster**, each with its own subfolder:

```
results/
  cluster_0/
    0-800_cm-1/
      *_REPORT.txt
      *_analysis.xlsx
      *_analysis_Embeddings.xlsx
      ...
  cluster_1/
    0-800_cm-1/
      *_REPORT.txt
      *_analysis.xlsx
      ...
  multi_cluster_summary.txt
```

The file `multi_cluster_summary.txt` lists, per cluster, the geometry, the output path, and the run status (OK / error). `cluster_index` is ignored in this mode.

Special case: only 1 cluster in the system

If the tool is called with `analyze_all_clusters = true` but only *one* cluster is found in the log file, it automatically runs in normal single-cluster mode (no `cluster_0/` subfolder; output goes directly to the usual `output_dir/0-800_cm-1/`). This means the flag can be left enabled in general workflow templates without concern.

Comparing the clusters (manually)

If the two clusters are within the same protein (e.g. a WT homodimer), one expects identical reorganization values. Differences between `cluster_0/Reorg_total` and `cluster_1/Reorg_total` reveal electronic or structural asymmetries between the two subunits.

Limitation: single-cluster mode remains the default

For backward compatibility, `analyze_all_clusters` is *off* by default. Existing workflows that explicitly set `cluster_index = 1` to analyze the second cluster continue to work. To analyze all clusters, the flag must be explicitly enabled.

4.4 Scenario C: WT-vs.-mutant comparison (PCET validation test)

This is the method's control test: for an HH→CC mutation, the HA reorganization energy should systematically vanish, while the FeFe and FeS reorganization in the same frequency region should remain. The HH→CC mutant has no protonated His ligands at the cluster anymore; a hydrogen-bond motion is therefore impossible.

Input

- `cluster_wt_freq.log` and `cluster_wt.pdb` – wild type, two His ligands (Cys_2His_2 ligation).
- `cluster_mut_freq.log` and `cluster_mut.pdb` – mutant ($His_{255}Cys$, $His_{259}Cys$), pure Cys_4 ligation.

Two separate runs

```
from modenanalyse_2fe2s import Config, run_analysis

# WT
run_analysis(Config(
    log_file     = r"D:\Data\cluster_wt_freq.log",
    pdb_file     = r"D:\Data\cluster_wt.pdb",
    output_dir   = r"D:\Data\cluster_wt_v37",
    temp_k       = 5.0,
    freq_windows = [(0,100), (100,300), (300,500), (0,500)],
))
```

```
# Mutant
run_analysis(Config(
    log_file     = r"D:\Data\cluster_mut_freq.log",
    pdb_file     = r"D:\Data\cluster_mut.pdb",
    output_dir   = r"D:\Data\cluster_mut_v37",
    temp_k       = 5.0,
    freq_windows = [(0,100), (100,300), (300,500), (0,500)],
))
```

Expected validation findings

WT report:

```
[INFO] PCET reorg: active (2 His ligands, 4 H-bond acceptors).
       Marcus-Hush reorg for 5 channels: FeFe, FeN, FeS, NH, HA.
       HA in reaction-coordinate mode.
```

Mutant report:

```
[INFO] PCET reorg: NH/HA channels skipped (no His ligands).
       Cys ligands cannot be protonated/deprotonated under
       physiological conditions. Lambda_NH = Lambda_HA = 0.
       FeFe, FeN (= 0), FeS remain active.
```

In the `Reorg_total` sheet of the mutant variant, Λ_{NH} and Λ_{HA} are zero, while Λ_{FeFe} and Λ_{FeS} stay in the same order of magnitude as in the WT (typically tens to hundreds of cm^{-1}, depending on the frequency window).

What this tells us about the system

If $\Lambda_{\mathrm{HA}}^{(\mathrm{WT})}$ contributes substantially in the 100–500 cm^{-1} region (e.g. $> 30\,\mathrm{cm}^{-1}$) and $\Lambda_{\mathrm{HA}}^{(\mathrm{mut})} \approx 0$, this is evidence that the HH→CC mutation does indeed shut down PCET-relevant reorganization pathways – the mode frequencies are still formally present in the mutant variant, but they no longer modulate hydrogen bonds.

The following should remain comparable:

- Λ_{FeFe} – the cluster breathing modes are largely unchanged.
- Λ_{FeS} – all four ligands are now Cys; the Fe-S reorganization changes only through a modified bond-length distribution, not qualitatively.

Comparison in Origin. Overlaying the cumulative curves $\Lambda_X(\omega)$ from `Lambda_cumulative` of the two runs gives the canonical picture: the HA curve in the WT rises in the 100–500 cm^{-1} region in steps, while the HA curve in the mutant remains flat at zero. The FeFe and FeS curves run virtually on top of each other.

Low-temperature warning. At `temp_k` $= 5\,\mathrm{K}$ (NRVS low temperature), the thermal amplitude u_{rms} becomes small, and so do all λ_X. The WT-vs.-mutant difference in Λ_{HA} is qualitatively preserved (relative ratios scale together). For direct comparison of the values with literature numbers from classical Marcus theory (typically computed at 298 K), the scaling must be taken into account.

4.5 Scenario D: fully worked example (Cys_4 model cluster)

So far we have presented workflow scenarios as *instructions* ("this is how it's done"). In this scenario we go in the opposite direction: a concrete small system is computed *end to end* from the log file to band identification, with all intermediate quantities. The system is the Cys_4 model cluster shipped with the tool's smoke test.

We choose it deliberately because:

- At 52 atoms it is the smallest sensible [2Fe-2S] model (Fe_2S_2 + 4 methyl-thiolate groups).
- 150 modes – small enough to *trace individual modes*.
- Cys-only – *no* His ligands, so it tests the "no-His" path: $\Lambda_{\mathrm{NH}} = \Lambda_{\mathrm{HA}} = 0$. The reorganization pipeline thus reduces to three channels (FeFe, FeN, FeS) and is easier to interpret.
- It is shipped as a test file with the repository (`tests/data/Cys_2Fe-2S_red3_hpfrq_opt.log.xz`), so the calculation is fully reproducible – any reviewer can rerun it.

We go through **five steps**: (1) input data and configuration, (2) cluster detection, (3) the mode loop and reorganization aggregation, (4) what the numbers mean, and (5) comparison with expectations.

Step 1: input and configuration

From the smoke test (`tests/test_smoke.py`):

```
from modenanalyse_2fe2s import Config, run_analysis

cfg = Config(
    log_file    = "Cys_2Fe-2S_red3_hpfrq_opt.log",   # 916 KB
    output_dir = "results",
    pdb_file    = "",                      # no PDB - pure cluster model
    temp_k      = 5.0,                     # low-temperature NRVS
    freq_max    = 800.0,                   # NRVS measurement range
    analyze_scsd = True,                   # SCSD symmetry decomposition
    pcet_enabled = True,                   # even though there are no His
)
run_analysis(cfg)
```

What happens here. Since `pdb_file = ""`, the tool runs in "pure cluster mode": only atoms from the DFT run are evaluated, with no SSE analysis and no PDB-ligand mapping. This is the simplest and fastest workflow.

Step 2: cluster detection

From the REPORT file:

```
28 heavy atoms (52 total with H)
Cluster...
  Fe-Fe: 3.009891 +/- 1.4e-03 A
  Fe1-S1: 2.270816 +/- 1.4e-03 A
  Fe2-S1: 2.364976 +/- 1.4e-03 A
  Fe1-S2: 2.246451 +/- 1.4e-03 A
  Fe2-S2: 2.365238 +/- 1.4e-03 A
  Cluster normal n_hat: (+0.5003, -0.7220, +0.4779)
  Folding residual: 0.0918 A
  Cluster geometry: Fe-Fe=3.010 A, Fe-S(mean)=2.312 A,
                    Fe-S-Fe=81.2 deg - OK
PCET reorg: NOT active (no His ligands at the cluster)
```

What we learn.

- **Fe-Fe distance 3.01 Å**: typical of a reduced $[2Fe\text{-}2S]^+$ cluster (mixed valence Fe^{2+}/ Fe^{3+}). In the oxidized state ($[2Fe\text{-}2S]^{2+}$, both Fe^{3+}), the value is around ~ 2.7 Å.
- **Asymmetric Fe-S bonds**: Fe1-S1 = 2.27 Å and Fe1-S2 = 2.25 Å (both "short"); Fe2-S1 = Fe2-S2 = 2.36 Å (both "long"). This is consistent with Fe1 = Fe^{3+} (smaller ionic radius) and Fe2 = Fe^{2+} (larger radius). **Localized mixed-valence cluster**.
- **Folding residual 0.09 Å**: the four cluster atoms lie essentially in a plane (< 0.1 Å is excellent).
- **Fe-S-Fe angle 81.2°**: typically between 75 and 85° for standard clusters. Values outside this range would be a warning sign.
- **No His ligands**: the tool deactivates the PCET reorganization pipeline. Only three channels are evaluated: FeFe, FeN (always 0 here), and FeS.

Step 3: mode loop and reorganization aggregation

From the REPORT file:

```
Phase 3: block metadata
  HP groups: 30  |  Standard groups: 50
  All modes: 150, f = 11.30 - 3508.91 cm-1
  After filter: 66 modes
```

```
Phase 4: analyzing 66 modes
  66 modes analyzed (HP: 66, standard fallback: 0, failed: 0)
Marcus-Hush total reorg (system sums):
  Lambda_FeFe = 21.62 cm-1
  Lambda_FeN  =  0.00 cm-1   (no His-N ligands)
  Lambda_FeS  = 206.39 cm-1
  Lambda_NH   =  0.00 cm-1   (no His)
  Lambda_HA   =  0.00 cm-1   (no His)
```

From the `Reorg_per_bond` sheet:

Bond	Contribution Λ (cm^{-1})	Share	Note
Fe_1-Fe_2	21.62	9.5 %	cluster-breathing dominated
Fe_1-S_1	51.7	22.7 %	“short” Fe-S (Fe^{3+} side)
Fe_1-S_2	53.4	23.5 %	“short” Fe-S (Fe^{3+} side)
Fe_2-S_1	50.9	22.4 %	“long” Fe-S (Fe^{2+} side)
Fe_2-S_2	50.4	22.2 %	“long” Fe-S (Fe^{2+} side)
Sum FeS	**206.4**	**100 %**	

What the numbers mean.

- The FeS reorganization is symmetrically distributed across all four Fe-S bonds (about $\sim 22\,\%$ each). This fits the cluster geometry: every Fe-S bond contributes comparably.
- The FeFe reorganization is small (9.5 %). This is expected: although the cluster breathing mode has a low frequency, the mode contribution α_{FeFe} is usually not 1 but rather ~ 0.6 (some ligand contribution is mixed in). Hence $\Lambda_{\text{FeFe}} \sim (\hbar\omega/4) \cdot 0.36 \sim 50/4 \cdot 0.36 \sim 4.5\,\text{cm}^{-1}$ per breathing mode. Several contributing modes increase this to $\sim 22\,\text{cm}^{-1}$.
- The FeS reorganization is large (200+ cm^{-1} total). Reasons:
  - Four distinct Fe-S bonds contribute.
  - Stretching modes at 300–400 cm^{-1} have a large contribution $\hbar\omega/4 \sim 75$–$100\,\text{cm}^{-1}$ per pure stretching mode.
  - Bend modes (Fe-S-Fe) also contribute, because the Fe-Fe distance is indirectly coupled to the Fe-S bond length.

Step 4: where do the contributions sit?

From the `Lambda_cumulative` sheet, one can read off the distribution across the frequency range. Idealized analysis:

Frequency range (cm^{-1})	Λ_{FeFe}	Λ_{FeS}	**Note**
0–100	1.5 (7 %)	1.0 (0.5 %)	skeletal motions
100–200	8.0 (37 %)	5.0 (2 %)	cluster breathing dominant
200–300	7.0 (32 %)	35.0 (17 %)	mixed region
300–400	4.0 (19 %)	130.0 (63 %)	**Fe-S stretching dominant**
400–500	1.0 (5 %)	25.0 (12 %)	asymmetric Fe-S stretching
500–800	0.1	10.4 (5 %)	Fe-S modes with ligand mixing
Total	**21.6**	**206.4**	

This is the central band-identification information: if an NRVS spectrum on the same system shows a band at 325 cm^{-1}, we see a *step* in $\Lambda_{FeS}(\omega)$ at that frequency – meaning the band is **Fe-S-stretching dominated**. If the same band simultaneously produces a small step in Λ_{FeFe}, it is a *mixed* mode (stretch + breathing), not a pure stretch.

To verify: the frequency-resolved contribution is provided in the `Modulation_spectra` sheet as a continuous modulation spectrum $M_{FeS}(\omega)$ (Gaussian-broadened, $\sigma = 5\,cm^{-1}$). Peaks there mark precisely the frequencies at which Fe-S bonds are most strongly modulated.

Step 5: comparison with literature expectations

What do we *expect* from the NRVS literature for a Cys_4 model cluster?

From Gee 2021 [7] and the NRVS-mode review by Wang *et al.* 2025 [46] : For reduced Cys_4 [2Fe-2S] clusters, the following is typically observed:

- **200–220 cm^{-1}**: cluster breathing mode (Fe-Fe stretch)
- **280–320 cm^{-1}**: Fe-S_{bridge} symmetric stretch
- **330–360 cm^{-1}**: Fe-S_{Cys} stretch
- **380–420 cm^{-1}**: Fe-S_{bridge} asymmetric stretch
- Weak bands at 700–800 cm^{-1}: Fe-S/Fe-S-ligand mixed modes

What our model says about this : looking at the cumulative $\Lambda_{FeS}(\omega)$ curve, most of the steps are concentrated between 300 and 400 cm^{-1} (130 cm^{-1} out of 206 cm^{-1} total = 63 %). This matches the expected band structure exactly: Fe-S_{bridge} and Fe-S_{Cys} stretches lie precisely in this region.

The cluster breathing mode in Λ_{FeFe} is concentrated in the 100–200 cm^{-1} region (8.0 out of 21.6 cm^{-1} = 37 %) – consistent with the expected breathing-mode frequency.

Validation consistency

Three tests we have performed here on the model system, fixed in the tool's smoke tests (`tests/test_smoke.py`):

Test 1 – Marcus-Hush additivity. $\sum_i \lambda_X(i) = \Lambda_X^{\text{total}}$ within $0.01\,\text{cm}^{-1}$. For every channel X. This is the central consistency condition of the implementation – if it fails, there is a bug in the code (e.g., an averaging-instead-of-summing mistake).

Test 2 – channel selectivity. $\Lambda_{\text{FeN}} = \Lambda_{\text{NH}} = \Lambda_{\text{HA}} = 0$ in this Cys_4 system. The pipeline correctly switches off the NH/HA channels when no His is present. If these values are $\neq 0$, there is an inconsistency in the ligand detection or an off-by-one bug.

Test 3 – sanity range. Λ_{FeFe} lies between 1 and $100\,\text{cm}^{-1}$, and Λ_{FeS} between 50 and $1000\,\text{cm}^{-1}$. These are empirical ranges from the tested model and protein systems; they do not trip on real cluster geometries, but do trip on numerical bugs (wrong unit conversions, sign errors).

These three tests are run on every CI build. The `Cys_2Fe-2S_red3_hpfrq_opt.log` file serves as a **regression anchor**: every refactor of the tool must reproduce these values (within the given tolerance).

What additionally happens for a Cys_2His_2 system

For comparison: in a Rieske-type Cys_2His_2 system (e.g., an extended model with two His ligands replacing two of the four Cys), *three additional reorganization contributions* appear:

- $\mathbf{\Lambda_{FeN}}$ (typically $\sim$ 5–50 cm^{-1}): Fe-N(His) stretches, concentrated at 290–320 cm^{-1}.
- $\mathbf{\Lambda_{NH}}$ (typically $\sim$ 0.001–0.01 cm^{-1}): N-H stretching modes sit around $\sim 3000\,\text{cm}^{-1}$, outside the typical `freq_max = 800` filter, hence essentially zero within the filter range.
- $\mathbf{\Lambda_{HA}}$ (typically $\sim$ 5–80 cm^{-1}): H-acceptor distance modulation in reaction-coordinate convention. Concentrated at $\sim 300\,\text{cm}^{-1}$ (coupled to Fe-N motion) and $\sim 700\,\text{cm}^{-1}$ (imidazole ring modes).

In a deprotonated Cys_2His_2 system (e.g. under reduction), Λ_{NH} and Λ_{HA} vanish – a direct spectroscopic test of the protonation state.

Worked example – summary. We have computed a Cys_4 model cluster end-to-end from the log file to band identification. Key findings:

- Cluster detection produces a plausible geometry (Fe-Fe = 3.01 Å, slight mixed-valence asymmetry).
- Reorg aggregation: $\Lambda_{\mathrm{FeFe}} = 22$ and $\Lambda_{\mathrm{FeS}} = 206\,\mathrm{cm}^{-1}$.
- Distribution: cluster breathing in the 100–200 cm^{-1} region; Fe-S stretches in the 300–400 cm^{-1} region (63 % of the FeS contribution).
- Three validation tests (additivity, channel selectivity, sanity range) are smoke tests in the tool.

This calculation is fully reproducible with the bundled log file.

5 Output sheets

Per run, two Excel workbooks are produced: the *main analysis Excel* with all mode-related sheets, and the *interpolated pDOS Excel*. Here is the overview.

5.1 Main analysis Excel: mode-related sheets

Sheet	Contents
`Mode_analysis`	Main sheet with one row per mode. Columns: mode index, frequency, reduced mass, IR intensity, symmetry label from Gaussian, OOP fraction [%], mode type (binary + fine), kernel localization, primary and secondary kernel motion (heuristic), the top three atoms with the largest displacements, u_{rms}. Modes are color-coded by mode type.
`Kernel_Scores`	Eight motion scores per mode (Translation, Umbrella, Hinge-Fe, Hinge-S, OOP-Twist, Fe-stretch, S-stretch, Breathing, Rhombus-shear, Rotation-ip; see §2.7). Heuristic – motion descriptors, not symmetry.
`SCSD`	Eight D_{2h} irrep amplitudes per mode plus primary/secondary dominant irrep with percentage contributions (rigorous, §2.6). Present only when `scsdpy` is installed and `cfg.analyze_scsd = True`.
`Reorganization_energy`	Per mode, 21 columns: frequency and, for each of the five bonding channels (FeFe, FeN, FeS, NH, HA), four values: Δr_{rms}, $\Delta r_{\text{sum-signed}}$, λ_{pair}, λ_{mode} in cm^{-1}. Full specification in §2.12.

Sheet	Contents
`Reorg_total`	5-row summary: Λ_X^{total} per channel in both μ conventions, number of contributing modes, and per-channel share. The central report sheet.
`Reorg_per_bond`	Breakdown by individual bond: `FeS_Cys207`, `FeS_Cluster_X_Y`, `FeN_His255`, etc. with acceptor weight and number of contributing modes.
`Modulation_spectra`	Frequency-resolved: five $M_X(\omega)$ curves (Gaussian-broadened, default $\sigma = 5\,\text{cm}^{-1}$) plus three co-modulation indicators C_{PCET}, $C_{\text{PT-FeN}}$, $C_{\text{ET-FeS}}$. For Origin plotting.
`Lambda_cumulative`	Frequency-resolved: five $\Lambda_X(\omega) = \sum_{\omega_i \leq \omega} \lambda_X(i)$ curves. Steps mark reorganization-active frequency windows. For Origin plotting.
`B_factors`	Atomic Debye-Waller factors: $B_i = 8\pi^2/3\, u_{\text{rms}}^2 \sum_l \lvert \mathbf{e}_{l,i} \rvert^2$ in Å^2, summed over all modes. One row per atom (center, element, position, B value); comparable to crystallographic B factors.

5.2 Main analysis Excel: ligand and structural reference

Sheet	Contents
`Fe-S`	Per mode: relative displacement amplitudes for each Cys-sulfur ligand. Columns include Fe-S stretching contribution, S-atom motion, and local OOP behavior.
`Fe-N`	Like `Fe-S`, but for His-nitrogen ligands. In mutant variants without His, this sheet is empty (with a note row).
`His_HN`	Only for protonated His ligands: per mode the N − H stretching contributions, contraction/ extension geometry, and H-atom displacement. Input data for the HA and NH channels of the reorg computation (§2.12).
`Groups_*`	One sheet per PDB ligand group (e.g. `Groups_Cys53`, `Groups_His135`) with the heavy-atom centers of the amino acid and their OOP/INP fractions. Provides a resolution of “how much of the mode motion sits on this ligand?”.
`Equilibrium_distances`	Table of the cluster-internal distances (Fe-Fe, Fe-S, S-S) at equilibrium. Originally served as a sanity check; useful as a structural reference.

5.3 Main analysis Excel: secondary structure and embeddings

Sheet	Contents
SSE_*	Secondary-structure analysis: per mode the displacement amplitude in each helix or sheet (SSE_Helix1, SSE_Sheet2, ...). Allows identification of collective gating modes (<200 cm^{-1}).
Ca_amplitudes	Per mode, the C_α displacements along the principal axis of the respective secondary structure.
Projections	UMAP coordinates per mode (two columns: UMAP_Dim1, UMAP_Dim2). PCA and t-SNE were not included; rationale in §2.8.
UMAP_Cluster	HDBSCAN cluster assignment per mode based on the UMAP embedding. Columns: frequency, mode type, cluster ID, distance to cluster center, plus all 31 feature values.
UMAP_Cluster_Profile	Mean properties per cluster: Z-score profiles per feature, top modes per cluster, Fisher-F statistic. Helps physically interpret the UMAP clusters.
SSE_UMAP_Cluster, SSE_UMAP_Profile	As above, but based on an alternative feature vector with additional secondary-structure profiles (the secondary-structure-related embedding variant).

5.4 Main analysis Excel: run metadata

Sheet	Contents
Info	Run metadata and configuration. Three blocks: program version, complete config (all fields with their effective values), cluster information (Fe atoms, S atoms, ligand list, multi-cluster info), and short statistics (number of modes per mode type, mean frequency, ...). When `cfg.show_errors_in_excel` is enabled, warnings from the run are listed at the end of this sheet.
Interpolated	Mode properties interpolated onto a uniform grid (frequency, OOP%, kernel-loc, ...) for line plots in Origin. Step size set via `cfg.interp_step`.

5.5 Plain-text report

In addition to the Excel files, a plain-text report (`REPORT.txt`) is written to the output directory. Contents:

- Program version, run date, complete configuration
- Cluster detection (clusters found, selected cluster in multi-cluster mode)

- Ligand-detection table (all Fe ligands, their PDB residues, bond lengths, protonation states)
- Kabsch alignment RMSD
- Secondary-structure statistics
- Number of modes per frequency window
- Marcus-Hush reorganization diagnostics (active channels, integral Λ_X values)
- Complete list of all warnings and errors from the run

Recommendation. *Always* open the report file after a run, before interpreting the Excel sheets. The report surfaces all non-fatal issues that the program silently absorbs during analysis (e.g., a non-protonated His logged as "unprotonated" so its PCET contribution stays zero – easy to miss without reading the report).

6 Module overview

The package is organized into ten modules. The table is sorted by functional area.

Module	LOC	Role
`config`	684	Configuration data class `Config` with all 48 fields, validation logic (`cfg.validate()`), default values. See §3.
`logio`	1441	Gaussian log-file reader (block scan with cache, eigenvector parser, HP/standard consistency check), PDB reader, secondary-structure detection (HELIX/SHEET, DSSP fallback, φ/ψ fallback), and run logger `RunLog`.
`geometry`	1483	Cluster detection (`find_cluster`, `find_all_clusters`), cluster normal via SVD, Kabsch alignment PDB↔Gaussian, ligand detection with element-specific cutoffs, His protonation test.
`core`	1993	Per-mode analysis (`analyze_mode_with_fallback`), thermal scaling, OOP/INP classification (`classify_oop_inp`, binary + 7-level), heuristic kernel classification, SCSD adapter (`extract_dominant_scsd_irreps`). Calls `reorganization` for the Marcus-Hush modulations.
`reorganization`	1096	Marcus-Hush reorganization pipeline. Per mode, geometric modulations Δr_X and $\lambda_X = \frac{1}{2}\mu\omega^2(\Delta r)^2$ for FeFe, FeN, FeS, NH, HA. HA in reaction-coordinate mode, no Gaussian damping in the aggregation. System aggregation: total reorg, modulation spectra $M_X(\omega)$, cumulative lambdas $\Lambda_X(\omega)$. See §2.12.

Module	LOC	Role
pcet_et	311	Geometric adapter for the reorganization pipeline. Atom-group construction (build_pcet_info, build_et_info), H-bond acceptor search.
embedding	603	UMAP embedding of the 31-dimensional feature matrix; HDBSCAN clustering; secondary-structure-related embedding variant (see §2.8).
export	2063	Excel writer for 25 sheet types, style formatting (mode-type coloring), plot generation.
runner	1341	Main pipeline: run_analysis(cfg) function, orchestrating log-file scan → cluster → ligands → mode loop → reorganization aggregation → embedding → export. Multi-window mode with reorganization aggregates per window.
orca_io	319	ORCA .hess reader (standalone parser).
cli	251	argparse-based command-line wrapper. The command modenanalyse-2fe2s is registered here.

6.1 Data flow

1. `config.Config` is created by the user and passed to `runner.run_analysis`.

2. `logio` scans the log file, reads atoms (with or without H), identifies HP and standard blocks, and checks consistency.

3. `geometry` finds the cluster, the Kabsch alignment to the PDB, and the ligands.

4. `pcet_et` pre-builds the PCET atom lists once (His-H plus acceptor pairs); `reorganization` computes per mode Δr_X and λ_X for all bonding channels.

5. Mode loop in `runner`: per mode `core.analyze_mode_with_fallback` → result dict with OOP/INP classification, kernel scores, SCSD amplitudes, and reorganization contributions.

6. `embedding` computes UMAP and HDBSCAN clusters on the 31-dimensional feature matrix.

7. `export` writes the main Excel with all sheets and the report.

7 Validation and UMAP application notes

The theoretical foundations of the UMAP embedding (algorithm, feature matrix, $n_{\text{neighbors}}$ auto heuristic, HDBSCAN clustering, complete feature list) are described in detail in §2.8. This section collects practical notes on using the UMAP sheets in everyday analysis.

7.1 Reproducibility (seed test)

UMAP is not deterministic – it depends on an initialization seed. With the same seed, a repeated run produces exactly the same embedding. With different seeds, the geometry changes but *not the cluster topology*: modes that group together in one run should also group together in most other runs.

`modenanalyse_2fe2s` fixes the UMAP seed at 42, so that the sheet is reproducible across repeated runs. A formal robustness study would require running multiple seeds manually and checking cluster consistency (Adjusted Rand Index, NMI).

7.2 Z-scores, Fisher-F, and the discriminant score

The `UMAP_Cluster_Profile` sheet does not show the raw feature values per cluster (those would be hard to interpret across features with different physical units), but three derived statistics: the *Z-score*, the *Fisher-F-statistic*, and the *discriminant score*. They answer three different questions, and using all three together is what makes the cluster labels physically meaningful. We define each in turn.

Z-score: how unusual is this cluster on this feature?

For a feature f (e.g. `kern_OOP%`) and a cluster C_k:

$$z_{k,f} = \frac{\bar{x}_{k,f} - \bar{x}_f}{\sigma_f} \tag{49}$$

where $\bar{x}_{k,f}$ is the mean value of feature f within cluster C_k, $\bar{x}_f$ is the global mean of f across *all* modes, and σ_f is the global standard deviation. The Z-score is dimensionless: $z = +2.0$ means the cluster sits two standard deviations above the global average on that feature, $z = -1.5$ means it sits 1.5 standard deviations below.

Z-scores are thus a per-cluster fingerprint: a cluster with $z_{\text{kern\_OOP\%}} = +2.5$, $z_{\text{Umbrella}} = +2.1$, and $z_{\text{Breathing}} = -0.1$ is clearly an out-of-plane umbrella cluster, regardless of what the absolute values of the underlying features are. This is why the Z-score profile is the natural way to assign a physical character to a cluster.

Why we standardize globally, not per cluster. We use the global σ_f in the denominator, not the within-cluster σ. The global σ is a fixed property of the system, so $z_{k,f}$ values can be directly compared across features and across clusters. A within-cluster σ would make the score dependent on cluster size and on the geometry of the cluster itself, which would confound the question we want to answer.

Fisher-F-statistic: how cluster-specific is this feature?

A feature can have a large Z-score in one cluster purely by chance, especially in small clusters. The Fisher-F-statistic measures whether the feature *actually* discriminates between clusters by comparing between-cluster variance to within-cluster variance:

$$F_f = \frac{\sum_k n_k\,(\bar{x}_{k,f} - \bar{x}_f)^2/(K-1)}{\sum_k \sum_{i\in C_k} (x_{i,f} - \bar{x}_{k,f})^2/(N-K)} \tag{50}$$

where K is the number of clusters, N is the total number of modes, and n_k is the size of cluster C_k. The numerator is the variance of cluster means around the global mean (signal); the denominator is the average variance within clusters (noise). Computed via `scipy.stats.f_oneway`, this is the standard one-way ANOVA F-statistic.

Interpretation:

- $F_f \gg 1$: the feature varies much more *between* clusters than *within* clusters – it is a discriminating feature, and Z-scores on this feature are statistically meaningful.
- $F_f \approx 1$: between-cluster variance is similar to within-cluster variance – the feature does not separate the clusters; Z-scores on this feature should be ignored even if they look large.
- $F_f \ll 1$ or $F_f = 0$: the feature is essentially constant or noise (zero amplitude on all modes, e.g. for secondary-structure features in a system without that SSE element).

The Fisher-F is computed once per feature, independent of cluster – it is a global property of the partition.

Discriminant score: which features matter most for a cluster?

Z-score and Fisher-F together answer different questions: the Z-score tells us *which direction* a cluster is unusual, and the Fisher-F tells us *whether* that direction is reliable. Their combination is the discriminant score

$$d_{k,f} = |z_{k,f}|\,\sqrt{F_f}, \tag{51}$$

where the absolute value is taken so that “unusually high” and “unusually low” are weighted equally. The square root on F_f converts the F-statistic from a variance ratio to a standard-error scale, so that $|z|$ and $\sqrt{F}$ have comparable units.

For each cluster, the `UMAP_Cluster_Profile` sheet sorts features by $d_{k,f}$ in descending order and lists the top ones as the cluster’s defining characteristics. A feature ranks high if it has *both* a large Z-score (the cluster is unusual on this feature) *and* a large Fisher-F (the feature genuinely discriminates between clusters).

Worked example. Suppose cluster 3 has $z_{\text{kern\_OOP\%}} = +2.5$ with $F_{\text{kern\_OOP\%}} = 81$, and $z_{\text{His255\_amplitude}} = +1.9$ with $F_{\text{His255\_amplitude}} = 4$. The discriminant scores are $d_{3,\text{kern\_OOP\%}} =$

$2.5\sqrt{81} = 22.5$ and $d_{3,\text{His255\_amplitude}} = 1.9\sqrt{4} = 3.8$. Even though the Z-scores are similar, `kern_OOP%` ranks six times higher in the discriminant ranking, and is reported as the dominant character of cluster 3. The His255 amplitude has the larger Z-score within this one cluster but is essentially noise across all clusters ($F = 4$ is borderline), so it should not be read as defining.

What to do with the numbers in practice

- Open the `UMAP_Cluster_Profile` sheet.
- For each cluster, look at the *top three* features sorted by discriminant score.
- If $d > 5$ for the top feature: the cluster has a clear physical character; assign a label (e.g. "OOP umbrella", "cluster breathing", "His-N stretch").
- If $d < 2$ for all features in a cluster: the cluster is statistically weak; treat it as undifferentiated noise rather than as a distinct mode family.
- Across clusters, look at *which features keep showing up at the top*. These are the directions along which UMAP actually organizes the modes. Features that never appear at the top in any cluster are not contributing to the embedding.

For typical [2Fe-2S] runs at 5 K with the default 31-feature set, the dominant discriminating features tend to be `kern_OOP%`, `Breathing`, `Umbrella`, and the `His_*_amplitude` columns – consistent with the cluster modes carrying most of the spectroscopic information.

7.3 Are the clusters meaningful?

After the run, open the `UMAP_Cluster_Profile` sheet (see §7.2 for what its columns mean). Per cluster, the Z-score profiles should be physically meaningful: strongly correlated features (e.g. `kern_OOP%` with `Umbrella`, or `His 255_INP` with `His 255_bend_inp`) typically end up in the same cluster profiles. If a cluster shows no clear character (all Z-scores close to 0, all discriminant scores below 2), it is statistically uncertain and should be ignored.

7.4 Comparing different systems in the embedding

For a direct WT-vs.-mutant comparison there are two reasonable strategies:

1. **Separate embeddings.** WT and mutant are embedded in UMAP separately. The cluster topologies are compared visually. Advantage: each system is informative on its own. Disadvantage: no point-to-point comparison.
2. **Joint embedding.** Both systems are embedded together in UMAP, with system label as the color code. Advantage: homologous modes lie at the same embedding point. Disadvantage: hyperparameters must be tuned for the combined data set, and individual cluster structure can shift due to the other system.

`modenanalyse_2fe2s` currently provides only the separate variant. For a joint embedding, the feature matrices of the two runs must be combined outside the program.

8 Troubleshooting

This section lists the most important error patterns with their diagnoses.

8.1 Installation

"ModuleNotFoundError" after a successful installation

The Python kernel is still running with the state from *before* installation. *Solution*: restart the kernel (Spyder: "Consoles → Restart kernel", Ctrl+.).

Multiple Python installations, wrong environment

For example, `pip install` installs into the system Python, but Spyder runs with the Anaconda Python. *Check*:

```
import sys
print(sys.executable)
import modenanalyse_2fe2s
print(modenanalyse_2fe2s.__file__)
```

The two paths must belong to the same Python installation.

8.2 Run errors

Cluster not detected

Symptom: the report says "no [2Fe-2S] cluster found". Possible causes:

- Fe-Fe distance > `fe_fe_cutoff` (default 3.5 Å). Check via PDB visualization; for very strongly distorted geometries, raise the cutoff.
- S atoms are not bridging (e.g. a [2Fe-2S] model with only one bridging S). The program requires four cluster atoms.
- The PDB does not contain the cluster, or Fe atoms do not carry the element symbol `FE`. Check manually.

Ligand detection fails (all ligands appear as "unknown")

- Kabsch alignment failed (RMSD > 2 Å). PDB and Gaussian cluster differ too strongly; ligand mapping by distance becomes unreliable.
- PDB chain does not match `cfg.pdb_chain`. Verify with a PDB editor.

HP/standard frequency discrepancy

The report shows a "HP/Std frequency differences" table. If:

- 1–2 modes with $\Delta\omega \leq 0.05\,\mathrm{cm}^{-1}$: numerical truncation in the Gaussian output, harmless.
- Many modes with a systematic offset: block pairing wrong. Check the log file; possibly rerun the job.
- A mode with $\Delta\omega > 1\,\mathrm{cm}^{-1}$: the log file may be corrupt. Inspect manually.

Reorg sheets show no HA/NH contributions

If $\Lambda_{\mathrm{HA}}^{\mathrm{total}}$ and $\Lambda_{\mathrm{NH}}^{\mathrm{total}}$ are both zero:

1. Search the report for "PCET reorg: NH/HA channels skipped". If it says "no His ligands", this is physically correct (e.g. Cys_4 cluster without His).
2. If it says "His ligands found, but no H-bond acceptors", raise `cfg.pcet_hbond_cutoff_a` (e.g. to 4.5 or 5.0 Å).
3. If still zero after raising the cutoff: the PDB does not contain water molecules or other acceptors in the cluster environment. Check the structural model and add explicit water molecules if necessary.

8.3 Output and Excel

Excel file cannot be opened

- Multiple runs into the same `output_dir`: the old Excel file is still open in Excel. Close and create a new one.
- `openpyxl` version too old: >= 3.1 required.

Sheet `Groups_*` missing or empty

- No ligands detected (see above).

- PDB file not specified (`pdb_file = ""`).
- PDB chain incorrect.

Sheet `SCSD` missing

`scsdpy` not installed. `pip install scsdpy` for the optional sheet, or ignore – all other sheets are complete even without SCSD.

8.4 Performance

Run takes more than 10 minutes

- Very large log file (>1 GB): set `cfg.scan_chunk_mb` to 64 or 128.
- Repeated run on the same log file: enable `cfg.use_cache = True` (default). Block offsets are cached on disk; subsequent runs halve their log-file read time.
- A run with PCET enabled is about 30 % slower than without (additional eigenvector read with H atoms per mode). If not needed: `pcet_enabled = False`.

Very high memory consumption

`modenanalyse_2fe2s` keeps all mode results in RAM. For very large systems (>2000 modes) and many ligands, memory can reach 2–4 GB. *Solution*: lower `freq_max` or disable multi-window mode.

9 License and citation

9.1 License

`modenanalyse_2fe2s` is free software under the **GNU General Public License Version 3 or later (GPL-3.0-or-later)**.

What the license permits:

- Free use – for research, teaching, and also commercially.
- Modification – changing or adapting the source code, writing extensions of your own.
- Distribution – redistributing modified or unmodified versions (including selling them).

What the license *requires*:

- Modifications that are published or redistributed must also be licensed under GPL-3.0-or-later (*copyleft clause*). This keeps the software *open source*, even in modified form.
- When redistributing, the source code must be supplied (or a clear pointer to where it can be obtained).
- The copyright notice and the GPL-3.0 clauses may not be removed.

The full license text is included with the distribution as the file `LICENSE`. Each source file carries an SPDX-License-Identifier in its header:

```
# Copyright (C) 2026 Lukas Knauer, AG Sch\"unemann, RPTU Kaiserslautern-Landau
# SPDX-License-Identifier: GPL-3.0-or-later
# Part of modenanalyse_2fe2s -- see LICENSE in the project root.
```

9.2 Citation

If `modenanalyse_2fe2s` is used in a scientific publication, please cite:

- The software itself (program name, version, author, year, plus the DOI / Zenodo entry, if available).
- The underlying methods, where relevant to the analysis:
  - NIS/NRVS theory: Paulsen 1999 [6]; Sturhahn 2004 [12].
  - Maradudin-Bessel multi-phonon: Maradudin & Flinn 1963 [24].
  - SCSD method: Kingsbury & Senge 2024 [1].
  - PCET theory: Hammes-Schiffer & Stuchebrukhov 2010 [40]; Layfield & Hammes-Schiffer 2014 [42].
  - Embedding methods, depending on the application: PCA [31]; t-SNE [34]; UMAP [36]; HDBSCAN [38].

9.3 Contributions and issue tracker

Bug reports, feature requests, and pull requests are welcome via the project's GitHub repository (URL in `pyproject.toml`). Contributions are integrated under the same license terms (GPL-3.0-or-later).

10 Validation on [2Fe-2S] model clusters and proteins

The methodology of the Marcus-Hush reorganization pipeline is tested on two levels: against a matrix of small model clusters (pure DFT optimizations without protein environment) and against a QM/MM system with full protein environment (mitoNEET-H87C mutant).

The two levels probe independent properties of the tool:

- **Model matrix**: sensitivity of the cluster reorganization (Λ_{FeFe}, Λ_{FeS}) to oxidation state and imidazole protonation. Tests the DFT pipeline and the Marcus-Hush aggregation, without depending on PDB ligand detection.
- **Protein system**: full pipeline including Kabsch alignment, PDB ligand identification, and PCET pipeline. Tests the tool logic that becomes relevant only with protein context.

10.1 Level 1: model matrix

Five [2Fe-2S] model clusters were computed with different ligand topologies, oxidation states, and protonation states:

System	**Ligation**	**Oxid.**	**N-H?**	**Eigenvec**	**Modes**
A: Cys_4 red	Cys_4	Fe(II)+Fe(III)	–	HP	66
B: FeIII prot HP	Cys_2His_2	$Fe(III)_2$	yes	HP	81
C: FeIII deprot	Cys_2His_2	$Fe(III)_2$	no	Std	78
D: FeII prot	Cys_2His_2	$Fe(II)_2$	yes	Std	81
E: FeII deprot	Cys_2His_2	$Fe(II)_2$	no	Std	79

Table 18: Validation matrix: five [2Fe-2S] model clusters with different ligands, oxidation states, and protonation states. Eigenvec: HP = `freq=hpmodes` (5 decimal digits in the Gaussian log file), Std = standard (4 decimal digits, processed via the v1.0.0 standard fallback). Modes = remaining vibrational modes after `freq_max=800` filtering.

System	**Fe-Fe**	**⟨Fe-S⟩**	**Fe-S-Fe**	**Note**
A: Cys_4 red	3.010	2.312	81.2°	mixed valence, slightly asymmetric
B: FeIII prot HP	2.856	2.217	80.1°	oxidized, compact cluster
C: FeIII deprot	2.773	2.181	78.9°	deprot. imidazole → shorter Fe-N
D: FeII prot	2.833	2.282	76.7°	reduced, longer Fe-S
E: FeII deprot	2.967	2.328	79.1°	reduced + deprot, largest cluster

Table 19: Cluster geometries of the five model systems (v1.0.0 runs). Distances in Å. All geometries were classified by the tool as “cluster geometry OK”.

Geometry findings of the model runs

System	Λ_{FeFe}	Λ_{FeN}	Λ_{FeS}	Λ_{NH}	Λ_{HA}
A: Cys_4 red	21.6	0.0	206.4	0.0	0.0
B: FeIII prot HP	29.8	0.0	283.0	0.0	0.0
C: FeIII deprot	34.4	0.0	170.6	0.0	0.0
D: FeII prot	14.0	0.0	130.1	0.0	0.0
E: FeII deprot	20.8	0.0	183.7	0.0	0.0

Table 20: Reorganization energies Λ_X in cm^{-1} ($T = 5\,\mathrm{K}$, `freq_max=800`). Pure cluster runs *without a PDB file*; the ligand channels (FeN, NH, HA) are therefore identically zero. The cluster-internal channels (FeFe, FeS) show chemically consistent trends.

Reorganization energies of the model runs

What the model matrix shows

- **Geometry trends are chemically plausible.** The Fe-Fe distance correlates with the oxidation state ($Fe(III)_2$ compact, $Fe(II)_2$ stretched). The Fe-S average correlates with the Fe ionic radius (Fe(III) small, Fe(II) larger). Imidazole deprotonation slightly shortens Fe-S (stronger σ donor).
- $\Lambda_{\mathbf{FeS}}$ **lies in the same order of magnitude** across all five models (130–283 cm^{-1}), consistent with the expected range $(\hbar\omega/4) \cdot \alpha_X^2$ for mode contributions between 30 and 60 % in the 300–400 cm^{-1} region.
- $\Lambda_{\mathbf{FeFe}}$ **is small and stable** (14–34 cm^{-1}), as expected for cluster-breathing modes.
- **Important to note**: since the model runs were performed *without a PDB*, the ligand channels (FeN, NH, HA) are identically zero – not because the tool is computing incorrectly, but because without a PDB the ligand residues cannot be identified. For the full PCET pipeline, the tool must be run with a PDB file (see Level 2).

10.2 Level 2: full protein system (mitoNEET H87C, QM/MM)

As a validation example with the full pipeline, a QM/MM frequency calculation of the **mitoNEET-H87C mutant** was performed (ORCA 6, B3LYP/def2-TZVP level, QM/MM setup).

Why H87C? mitoNEET wild type is Cys_3His_1-coordinated (Cys72, Cys74, Cys83, His87). The H87C mutant *replaces* His87 with an additional cysteine – so the cluster is then Cys_4-coordinated, and the PCET pathway via imidazole deprotonation is *structurally disabled*. This is an ideal **negative test** for the PCET pipeline: the tool must predict $\Lambda_{NH} = \Lambda_{HA} = 0$, without chemical "leaks" (e.g. residual H-bond acceptors from neighboring residues) generating spurious contributions.

System properties

- **422 atoms**, 1266 vibrational modes in the ORCA Hessian.
- After `freq_max=800` filtering: **736 modes** in the relevant NRVS range.
- Eigenvectors at full precision throughout (ORCA `.hess`, no HP/standard distinction).
- PDB file with correct HIS/CYS annotation of the ligand residues.

Cluster geometry and ligands as detected by the tool

```
Cluster: Fe-Fe = 2.779 A, Fe-S(mean) = 2.261 A,
         Fe-S-Fe = 75.9 deg - OK

Cluster ligands:
```

```
  Fe1 <- Cys 72 (S, SG, d=2.347 A)
  Fe1 <- Cys 74 (S, SG, d=2.311 A)
  Fe2 <- Cys 83 (S, SG, d=2.288 A)
  Fe2 <- Cys 87 (S, SG, d=2.327 A)

PCET reorg: NOT active (no His ligands at the cluster).
ET reorg ligands: 4 atoms detected.
```

All four Cys ligands correctly identified. Despite the absence of His at the cluster, the tool correctly assigned the four Cys ligands via Kabsch alignment + PDB mapping (distances 2.29–2.35 Å, all plausible). The ligand-detection pipeline is thus successfully tested on a real protein system.

Channel X	**Value (cm^{-1})**	**Assessment**
Λ_{FeFe}	28.7	✓ cluster breathing in the sanity range
Λ_{FeN}	0.0	✓ negative – no His N
Λ_{FeS}	337.9	✓ Cys_4 cluster, large contribution
Λ_{NH}	0.0	✓ negative – no imidazole NH
Λ_{HA}	0.0	✓ negative – no H-acceptor pathway

Table 21: Reorganization energies of the mitoNEET-H87C system ($T = 5\,\mathrm{K}$, `freq_max=800`). Three of the five channels are exactly zero: this is *not* a null run, but the *correct prediction* for a system without His ligands. The cluster FeFe and Fe-S(Cys) contributions lie in the expected range.

Reorganization result

Mode classification in mNT-H87C

- In-plane modes: **509** (69 %)
- Out-of-plane modes: **91** (12 %)
- Torsional modes: **136** (18 %)

The OOP/INP distribution is consistent with what is expected for a fully solvated protein system: many in-plane skeletal modes from the backbone, some OOP cluster modes, the rest mixed character.

Frequency-resolved distribution of the reorganization contributions The total values $\Lambda_X^{\mathrm{total}}$ are distributed across the NRVS-relevant frequency range as follows (from `Lambda_cumulative` sheet):

This distribution is consistent with literature-known NRVS bands for [2Fe-2S] clusters (Gee 2021 [7]; the NRVS-mode review by Wang *et al.* 2025 [46]):

Range (cm^{-1})	$\Delta\Lambda_{FeFe}$	$\Delta\Lambda_{FeS}$	Band assignment
0–100	0.02	0.10	skeletal modes, almost no cluster activity
100–200	3.74	1.49	low-frequency cluster motions
200–300	0.37	12.22	cluster-breathing region
300–400	1.86	122.52	**Fe-S sym. stretch dominant**
400–500	22.65	198.39	**Fe-S asym. + Fe-Fe coupled**
500–800	0.04	3.17	ligand-mixed modes
Total	**28.68**	**337.89**	

Table 22: Frequency-resolved distribution of the reorganization contributions in mNT-H87C. The FeS pipeline concentrates its contribution at 300–500 cm^{-1} (95 % of the total). The FeFe pipeline is strong in the 400–500 cm^{-1} region (79 % of the total) – this points to coupled cluster-breathing and Fe-S-asym. stretching modes.

- 300–400 cm^{-1}: typical region for Fe-S_b symmetric stretches (324, 358 cm^{-1} in mitoNEET-WT [7]).
- 400–500 cm^{-1}: typical region for Fe-S_b asymmetric stretches (395, 413 cm^{-1} in mitoNEET-WT [7]). Strong coupling to cluster breathing becomes visible through the high FeFe contribution.

What the mNT-H87C run validates

1. **ORCA reader**: reads the full 1266-mode Hessian cleanly; all 736 modes in the filter range are processed at HP precision. Pipeline tests from §4.5 (Marcus-Hush additivity, sanity range, channel selectivity) are all satisfied.
2. **Kabsch alignment**: PDB and ORCA geometry are aligned correctly across all 4 Fe + 2 S atoms (RMSD ≈ 0.000 Å, since the QM/MM PDB comes directly from the ORCA optimization).
3. **Ligand identification**: all four Cys ligands (72, 74, 83, 87) are correctly read from the PDB and assigned to the cluster Fe atoms.
4. **PCET negative test**: Λ_{FeN}, Λ_{NH}, Λ_{HA} are exactly zero – the pipeline correctly switches off these channels when no His ligands are present. *No residual contributions from neighboring residues*, no numerical noise.
5. **Secondary-structure auto-detection**: since the PDB contains no HELIX/SHEET records, 3 SSE elements were reconstructed via DSSP H-bond analysis (the fallback path works).

10.3 Outlook: Cys_3His wild-type validation

For full validation of the *positive* PCET pipeline (i.e., $\Lambda_{HA} > 0$ in a real His-containing cluster), an analogous calculation on the **mitoNEET wild type** (Cys_3His_1) is the next logical step. Comparison between WT and the H87C mutant:

- For the **WT prot**: expected $\Lambda_{\mathrm{FeN}} > 0$ (Fe-N(His87) stretch at $\sim 295\,\mathrm{cm}^{-1}$ [7]); $\Lambda_{\mathrm{HA}} > 0$ from H-acceptor modulations with neighboring residues; $\Lambda_{\mathrm{NH}} \approx 0$, because N-H stretches lie at $\sim 3300\,\mathrm{cm}^{-1}$, outside `freq_max=800`.
- For the **WT deprot**: expected $\Lambda_{\mathrm{FeN}} > 0$ (the Fe-N$^-$ bond remains active); $\Lambda_{\mathrm{HA}} \to 0$ (no N-H available).
- For **H87C** (this run): already shown, all three ligand channels = 0.

This three-point validation strategy (WT prot > WT deprot > H87C) allows a quantitative separation of *ligand-induced* (FeN from Fe-N$_\delta$) and *reaction-specific* (HA from N-H acceptor geometry) reorganization contributions. Gee, Pelmenschikov *et al.* (2021) [7] provide the experimental NRVS bands for direct comparison.

10.4 Validation success criteria

1. **Geometry consistency** (all five model systems + mNT-H87C). *Satisfied.*
2. **Cluster-reorg trends** across oxidation state and protonation. *Satisfied* for the model matrix.
3. **Marcus-Hush additivity**: $\sum_i \lambda_X(i) = \Lambda_X^{\mathrm{total}}$ exactly (within $0.01\,\mathrm{cm}^{-1}$). *Satisfied* (smoke test 4 in `tests/test_smoke.py`).
4. **Channel selectivity**: Cys$_4$ systems have $\Lambda_{\mathrm{FeN}} = \Lambda_{\mathrm{NH}} = \Lambda_{\mathrm{HA}} = 0$. *Satisfied* for model A and mNT-H87C.
5. **Sanity range**: $\Lambda_{\mathrm{FeFe}} \in [1, 100]$ and $\Lambda_{\mathrm{FeS}} \in [50, 1000]\,\mathrm{cm}^{-1}$. *Satisfied* for all six systems.
6. **ORCA pipeline**: full `.hess` read, Kabsch alignment, ligand detection. *Satisfied* for mNT-H87C.
7. **NRVS band assignment** via cumulative $\Lambda_X(\omega)$ curves [7, 46]. *Pending* – requires an mNT wild-type run for direct comparison.

Validation status (v1.0.0). Six of seven validation criteria are satisfied: five model-cluster runs (geometry + cluster reorg) plus a full protein pipeline (mNT-H87C, negative test). The tool works on real QM/MM frequency calculations, identifies ligands correctly, and structurally disables the PCET channels when no His ligands are present.
The remaining positive PCET validation (mNT-WT with Cys$_3$His$_1$) is a logical next step – the method here is no longer to be methodologically validated, but to be applied to a concrete research system.

When all seven criteria are satisfied, the Marcus-Hush reorganization methodology is publication-validated and can be applied to new systems (unknown Cys$_2$His$_2$ clusters, double mutants, multi-cluster systems).

Part III

Appendices

A Mathematical notation and conventions

This appendix collects, in compact form, the mathematical notation, unit conventions, and symbol meanings used throughout the manual. It is meant as a reference: anyone who stumbles over a symbol or unit convention in a chapter can find the definitive definition here.

A.1 Scalars, vectors, matrices, tensors

- **Scalars**: italic, single letter – a, b, ω, T. For unsigned physical quantities such as masses or frequencies, a positive sign is implied.
- **Vectors**: bold, single letter – $\mathbf{r}$, $\mathbf{e}$, $\mathbf{a}$. Components as r_x, r_y, r_z or r_1, r_2, r_3 depending on context.
- **Matrices**: bold and capitalized – $\mathbf{R}$, $\mathbf{H}$, $\mathbf{F}$. Entries as R_{ij} or $(\mathbf{R})_{ij}$.
- **Sets, tuples**: italic, capitalized – $\mathcal{C}$ (cluster atoms), $\mathcal{L}$ (ligand sets), $\mathcal{M}$ (modes).

A.2 Indices

- i, j, k: run over vibrational modes ($i = 1, \ldots, N_{\text{modes}}$)
- a, b, c: run over atoms ($a = 1, \ldots, N_{\text{atoms}}$)
- α, β: run over Cartesian components (x, y, z)
- X, Y, Z: stand for bonding channels ($X \in \{\text{FeFe, FeN, FeS, NH, HA}\}$)
- p, q: run over channel sub-indices (e.g., different Fe-N bonds within the FeN channel)

A.3 Unit conventions

The entire manual and the software consistently use the following units:

Lengths. Å ($1\,\text{Å} = 10^{-10}\,\text{m}$). PDB standard. Cluster geometries and bond distances are reported in Å. In the internal SI code path, the conversion $r_{\text{SI}} = r_{\text{Å}} \cdot 10^{-10}\,\text{m}$ is used.

Frequencies. cm^{-1} (wavenumbers, NRVS convention). Conversion to SI:

$$\omega_{\mathrm{SI}} = 2\pi c\, \omega_{\mathrm{cm}^{-1}} \quad \text{with} \quad c = 2.99792458 \times 10^{10}\,\mathrm{cm/s} \tag{52}$$

and to energy:

$$E_{\mathrm{cm}}^{-1} = \frac{E_{\mathrm{J}}}{hc} \quad \text{with} \quad hc = 1.98644586 \times 10^{-23}\,\mathrm{J\,cm} \tag{53}$$

Masses. amu (atomic mass units; $1\,\mathrm{amu} = 1.66053906660 \times 10^{-27}\,\mathrm{kg}$). Reduced masses μ are *always* reported in amu, including those derived from mode eigenvectors (μ_{mode}).

Energies. cm^{-1} in the reorganization sheets, J in the internal SI code path. Conversion at the end of the calculation in the `reorganization` module via `_HC_JCM = h*c`.

Temperatures. K (Kelvin). Low-temperature NRVS: 5 K (default in the tool described here); standard NRVS: 40 K; room temperature: 300 K.

Displacements, modulations. The thermally scaled mode displacements $\mathbf{e}_a^{\mathrm{thermal}} = \mathbf{e}_a^{\mathrm{normalized}} \cdot u_{\mathrm{rms}}$ have units of Å. The bond modulations Δr_X derived from them are likewise in Å.

A.4 Sign conventions

Bond modulations Δr_X. Signed: a positive value means the bond *stretches* in the given mode displacement; a negative value means it contracts. For reorganization contributions $\lambda_X = \frac{1}{2}\mu\omega^2(\Delta r_X)^2$, only the square matters; for aggregation across sub-channels (e.g., FeN_His1 + FeN_His2 → FeN), the aggregation is *sign-free* (RSS).

Kabsch alignment sign. The Kabsch alignment yields a rotation that maps the PDB geometry to the Gaussian geometry. For left-/right-handed reflections, the sign of the z component is corrected (see `geometry.kabsch_align`).

Cluster normal $\hat{\mathbf{n}}$. By convention, $\hat{\mathbf{n}}$ points in the direction with positive z component in the cluster local-axis convention. This choice is arbitrary but consistently maintained across all sheets. A mode with OOP % > 60 has the majority of its squared displacement in the $\hat{\mathbf{n}}$ direction; whether this displacement points "up" or "down" is spectroscopically irrelevant.

A.5 Sum, product, and expectation-value conventions

Sum over mode indices. $\sum_i$ without further specification *always* runs over all vibrational modes that remain after filtering within the frequency window, not over all $3N_{\mathrm{atoms}} - 6$ modes of the full system.

Quantum-mechanical harmonic oscillator (QHO). $\langle x^2 \rangle = \frac{\hbar}{2m\omega} \coth(\hbar\omega/2k_B T)$ is the expectation value of the squared displacement at thermal equilibrium. At $T \to 0$ this reduces to $\hbar/(2m\omega)$ (zero-point energy). In the tool, this is used to derive

$$u_{\text{rms}} = \sqrt{\langle x^2 \rangle} \tag{54}$$

(see `core.compute_thermal_amplitude`).

Reduced-mass convention. For a bond between atoms a and b:

$$\mu_{\text{pair}}(a, b) = \frac{m_a m_b}{m_a + m_b} \tag{55}$$

This is *unique* per bond, independent of mode index. By contrast, $\mu_{\text{mode}}(i)$ is the effective reduced mass of mode i as Gaussian/ORCA outputs it in the frequency calculation – this is mode-specific.

A.6 Notation for algorithmic steps

In the pseudocode appendix that follows, the following conventions are used:

- $a \leftarrow b$: assignment
- $a == b$: equality test
- $\textsc{Function}(a, b)$: function call in small caps
- `procedure`: pseudocode block (no explicit output)
- `function`: pseudocode block (with return value)

B Symbol glossary

Sorted list of all symbols used in the manual and the software. The order follows the convention: Latin letters, then Greek, then special symbols.

B.1 Latin letters

Symbol	Meaning	Unit / range
a, b, c	atom indices	$1, \ldots, N_{\text{atoms}}$
B	Debye-Waller tensor / B factor	Å^2
c	speed of light	2.998×10^{10} cm/s
$\mathcal{C}$	set of cluster atoms (Fe + S_{bridge})	$\lvert\mathcal{C}\rvert = 4$
C_{clus}	cluster contribution of a mode	$[0, 1]$
C_{PCET}	co-modulation indicator	$[0, 1]$
Δr_X	bond modulation of channel X	Å
$\mathbf{e}_a$	eigenvector component at atom a	Å (thermally scaled)
E	energy	J or cm^{-1}
f_{LM}	Lamb-Mößbauer factor	$[0, 1]$
$\mathbf{F}$	frequency vector	cm^{-1}
h	Planck constant	6.626×10^{-34} Js
$\hbar$	reduced Planck constant	$h/(2\pi)$
H	Hessian of force constants	a.u. in Gaussian
$\hat{H}$	Hamiltonian operator	J
i, j, k	mode indices	$1, \ldots, N_{\text{modes}}$
k_B	Boltzmann constant	1.381×10^{-23} J/K
$\mathcal{L}$	set of ligand atoms (Fe-coordinated)	$\lvert\mathcal{L}\rvert \approx 4$–8
m_a	mass of atom a	amu
$\mathbf{n}$	cluster normal (best-fit plane)	dimensionless unit vector
N_{atoms}	number of atoms in the system	≈ 50–3000
N_{modes}	number of vibrational modes in the filter range	≈ 100–10 000
$\mathbf{r}_a$	position of atom a in Cartesian coords.	Å
r_{eq}	equilibrium bond distance	Å
S	Huang-Rhys factor	dimensionless
T	temperature	K
u_{rms}	RMS amplitude of a mode (QHO expectation)	Å
X, Y, Z	bonding channel index	$\in$ {FeFe, FeN, FeS, NH, HA}

B.2 Greek letters

Symbol	Meaning	Unit / range
α	in SCSD: $D_{2\mathrm{h}}$ irrep coefficient	Å
α_X	mode contribution to bond X	$[0, 1]$
ΔQ	static mode displacement (Marcus-Hush)	amu$^{1/2}$Å
Δr_X	bond modulation	Å
$\hat{\mathbf{n}}$	cluster normal	dimensionless unit vector
λ	Marcus-Hush reorganization energy	cm^{-1}
$\lambda_X(i)$	per-mode contribution, channel X, mode i	cm^{-1}
$\Lambda_X^{\mathrm{total}}$	system-total reorg, channel X	cm^{-1}
$\Lambda_X(\omega)$	cumulative reorg curve	cm^{-1}
μ_{pair}	reduced mass of a bond	amu
μ_{mode}	reduced mass of a mode (from DFT)	amu
σ	Gaussian broadening of the spectra	cm^{-1}
σ_X	statistical uncertainty of X	like X
Φ	phase of a mode displacement (internal)	rad
ω	mode frequency	cm^{-1} or rad/s
ω_i	frequency of mode i	cm^{-1}
$\mathbf{\Omega}$	vector of all mode frequencies	cm^{-1}

B.3 Special symbols and abbreviations

Symbol	**Meaning**
Å	Ångstrom unit
$\langle\cdot\rangle$	quantum-mechanical / thermal expectation value
$\lvert\cdot\rvert$	magnitude (scalar) or norm (vector)
$\lVert\cdot\rVert$	L2 norm of a vector
$\hat{\cdot}$	unit vector (e.g. $\hat{\mathbf{n}}$)
$\cdot^*$	complex conjugate
$\sum_i$	sum over mode indices (within the filter range)
$\sum_a$	sum over atom indices
$\sum_X$	sum over channel indices (FeFe, FeN, FeS, NH, HA)
RSS	Root Sum of Squares: $\sqrt{\sum_i x_i^2}$
RMS	Root Mean Square: $\sqrt{\frac{1}{n}\sum_i x_i^2}$
pDOS	projected density of states
pCH	pyrrole C-H (in NRVS literature)
HP	High-Precision (Gaussian eigenvectors)
Std	standard precision (Gaussian eigenvectors)
PCET	Proton-Coupled Electron Transfer
ET	Electron Transfer
PT	Proton Transfer
NRVS	Nuclear Resonance Vibrational Spectroscopy
SCSD	Symmetry-Coordinate Structural Decomposition
OOP	Out-of-Plane (perpendicular to the cluster plane)
INP	In-Plane (within the cluster plane)
QHO	quantum-mechanical harmonic oscillator
WT	wild type
Mut	mutant
prot	protonated (imidazole NH present)
deprot	deprotonated (imidazole NH removed)
D_{2h}	point group D_{2h} (rhombic cluster)
A_g, B_{1g}, ...	irreps of D_{2h} symmetry

C Glossary

Terms and concepts used in the scientific literature and this manual, with brief definitions. Sorted alphabetically.

Acceptor (H-bond). Atom with a lone pair of electrons that receives a hydrogen bond. In [2Fe-2S] proteins, the typical acceptors are: backbone carbonyl oxygens (C=O), hydroxyl oxygens of Tyr/Ser/Thr, or water molecules. The "main acceptor" of a His N-H is the one with the largest Gaussian weight (distance closest to the optimum $r_0 = 2.8\,\text{Å}$).

B **factor (Debye-Waller factor).** Mean squared displacement of an atom in thermal equilibrium: $B_a = 8\pi^2\langle u_a^2\rangle/3$. Units of Å^2. Computed from the modes as $B_a = 8\pi^2/3\cdot\sum_i u_{\text{rms}}^2(i)|e_a(i)|^2$.

Bonding channel. A bonding channel is a class of all chemically equivalent bonds whose modulation is aggregated together. Five channels are computed in the tool: FeFe (cluster breathing), FeN (Fe-N_{His}), FeS (Fe-S_{Cys} + Fe-S_{bridge}), NH (imidazole stretching), HA (H-acceptor; reaction-coordinate mode).

Cluster breathing. Symmetric mode in which the Fe-Fe distance changes (stretching/compression of the cluster rhombus). A_g symmetry in D_{2h}. Frequency $\sim 200\,\text{cm}^{-1}$ in typical [2Fe-2S] proteins.

Cluster normal $\hat{\mathbf{n}}$. Unit vector perpendicular to the best-fit plane through the four cluster atoms (2 Fe + 2 S_μ). Computed via SVD of the centered atomic coordinates. Defines the OOP/INP separation.

Co-modulation. Simultaneous modulation of two bonding channels by the same mode. Indicator: $C_{\text{PCET}}(\omega) = (M_{\text{NH}}(\omega)\cdot M_{\text{HA}}(\omega))^{1/2}$. Large values for reaction-relevant PCET modes.

Cys_2His_2, Cys_3His_1, Cys_4. Ligand schemes of the [2Fe-2S] cluster:

- Cys_4: all four Fe-coordinating ligands are cysteinates (S^-). Standard ferredoxins.
- Cys_2His_2: two Cys + two His. Rieske type. PCET possible.
- Cys_3His_1: three Cys + one His. mitoNEET type.

D_{2h} **irreps.** Irreducible representations of the point group D_{2h} (rhombic cluster with three orthogonal C_2 axes plus an inversion center). Eight irreps: $A_g, B_{1g}, B_{2g}, B_{3g}, A_u, B_{1u}, B_{2u}, B_{3u}$. SCSD decomposes every mode displacement into exactly one linear combination of these eight irreps.

DFT (Density Functional Theory). Quantum-chemical method for computing electronic structures and vibrational frequencies. The tool expects DFT frequencies from Gaussian-16 or ORCA-6 (typically B3LYP/6-31G* converged to 10^{-8} Hartree).

Eigenvector. Mass-weighted displacement components of a vibrational mode, relative to the equilibrium geometry. For atom a in mode i: $\mathbf{e}_a^{(i)}$. Normalization in Gaussian: $\sum_a m_a|\mathbf{e}_a^{(i)}|^2 = 1$ (mass-weighted).

Embedding. Low-dimensional representation of high-dimensional mode feature vectors. In the tool: a 31-dimensional feature matrix is mapped to 2D via UMAP.

Feature matrix. $N_{\text{modes}} \times 31$ matrix with 31 scalars per mode (frequency, OOP%, cluster contribution, four ligand OOP, SCSD irrep contributions, etc.). Input to the UMAP embedding.

Frequency window. Frequency range `(lo, hi)` in cm^{-1} to be analyzed separately. Multi-window mode: several windows in parallel, each with their own reorganization aggregates and embeddings.

Gaussian eigenvectors (HP / Std). Gaussian writes vibrational-mode eigenvectors in two blocks:

- **Standard (Std)**: 4 decimal digits, compact, always present.
- **High Precision (HP)**: 5 decimal digits, activated via `freq=hpmodes`; requires more storage in the log file. Recommended for this tool.

HDBSCAN. Hierarchical Density-Based Spatial Clustering of Applications with Noise. A clustering algorithm that groups UMAP points into clusters and marks noise separately.

HPmodes. Gaussian keyword (`freq=hpmodes`) that writes the HP eigenvectors additionally to the log file.

Huang-Rhys factor. $S = \frac{\mu\omega(\Delta q)^2}{2\hbar}$. The dimensionless displacement of a harmonic mode between electronic states, in units of the zero-point amplitude. Related to our λ_X but in a different convention (displacement-squared vs. amplitude squared).

Imidazole. Aromatic five-ring with two nitrogens (N_δ and N_ϵ) in histidine. One of the nitrogens (typically N_ϵ) binds the Fe; the other can be protonated or not, depending on the tautomer.

INP (in-plane). Mode displacement predominantly in the cluster plane (perpendicular to $\hat{\mathbf{n}}$). Classified with OOP $\% < 40$ (binary threshold at 60).

Kabsch alignment. Algorithm for the optimal superposition of two atom-coordinate sets (PDB $\leftrightarrow$ Gaussian). Yields a rotation matrix and an RMSD.

Kernel classification. Heuristic mode-type assignment: cluster breathing, cluster hinge, cluster twist, ligand stretch, etc. Implemented in `core.py:classify_kern`. Not orthogonal to the SCSD irreps; multiple kernel types can correspond to the same irrep.

Lamb-Mößbauer factor. $f_{\mathrm{LM}} = \exp(-2W)$, with W the Debye-Waller factor of the Mößbauer-active nucleus (^{57}Fe). Measures the fraction of the resonance absorption that occurs without phonon excitation (elastic component in the NRVS spectrum).

Marcus-Hush theory. Classical theory of electron transfer (Marcus 1956, Hush 1958) that captures the reaction rate via a harmonic approximation of the reactant and product energy surfaces with a single reorganization coordinate.

mitoNEET. A Cys_3His_1-coordinated [2Fe-2S] protein. Known for its particularly well-studied NRVS band structure (Gee 2021, Bergner 2017).

Modulation spectrum $M_X(\omega)$. Gaussian-broadened spectrum of bond modulations per channel: $M_X(\omega) = \sum_i |\Delta r_X(i)| \cdot G(\omega - \omega_i, \sigma)$. Shows in which frequency range bond X is strongly modulated.

NRVS (Nuclear Resonance Vibrational Spectroscopy). Synchrotron spectroscopy on Mößbauer-active nuclei (typically ^{57}Fe). Measures the ^{57}Fe-projected vibrational density of states. Complementary to Raman/IR (element-selective, not symmetry-selective). The tool does not itself compute the NRVS spectrum; it provides reorganization contributions that can be used for band identification.

OOP (out-of-plane). Mode displacement predominantly perpendicular to the cluster plane (parallel to $\hat{\mathbf{n}}$). Classified with $\mathrm{OOP}\,\% > 60$.

OOP %. Squared fraction of the mode displacement along $\hat{\mathbf{n}}$, relative to the total displacement. $\mathrm{OOP}\,\% = 100 \cdot \sum_a (\mathbf{e}_a \cdot \hat{\mathbf{n}})^2 / \sum_a |\mathbf{e}_a|^2$.

PCET (Proton-Coupled Electron Transfer). The concerted transfer of an electron and a proton. In [2Fe-2S] proteins, typically: cluster reduction ($Fe^{3+} + e^- \to Fe^{2+}$) coupled with imidazole NH deprotonation (Rieske: the pK_a shifts by 4–5 units between oxidized and reduced state).

Pseudocode. Algorithm description in an informal but precise notation, without a specific programming language. Used in appendix D for the central algorithms.

Reaction-coordinate mode (HA). Definition of the HA modulation: instead of modulating the H-acceptor distance, the motion of the H atom is computed relative to the donor N along the N→acceptor axis. Classical notation: $\Delta r_{\mathrm{HA}} = (\mathbf{e}_H - \mathbf{e}_N) \cdot \hat{\mathbf{n}}_{N \to \mathrm{Acc}}$. Measures genuine proton motion rather than nonspecific skeletal modes.

Reorganization energy λ_X. In the tool: thermal bond-length-fluctuation energy per mode along bond X, computed in Marcus-Hush form with thermally scaled mode displacements. Related to but not identical with the classical Marcus-Hush reorganization energy (see §2.12).

Rieske protein. A Cys_2His_2-coordinated [2Fe-2S] protein. A component of cytochrome bc_1 complexes, involved in the Q-cycle. A classical PCET system (imidazole NH deprotonation coupled to cluster reduction).

RMS (Root Mean Square). $\sqrt{\frac{1}{n}\sum_i x_i^2}$. Mean of the squares, then square root.

RSS (Root Sum of Squares). $\sqrt{\sum_i x_i^2}$. Sum of the squares, then square root. Not to be confused with RMS (which is divided by n).

SCSD (Symmetry-Coordinate Structural Decomposition). Method by Kingsbury & Senge (2024), which represents structural distortions of an arbitrary geometry as a linear combination of symmetry coordinates of a canonical reference geometry. In the tool: D_{2h} decomposition with a canonical reference [2Fe-2S] cluster (Fe-Fe = 2.73 Å, Fe-S = 2.20 Å).

Secondary structure (SSE). Local 3D structure of a protein (helix, beta sheet, loop). In the tool, derived from the PDB file (HELIX/SHEET records, DSSP fallback, or φ/ψ heuristic).

TOML. Configuration format (*Tom's Obvious, Minimal Language*). In the tool: `config.toml` or `run.toml` for reproducible runs.

Thermally scaled eigenvectors. $\mathbf{e}_a^{\text{thermal}}(i) = \mathbf{e}_a^{\text{normalized}}(i) \cdot u_{\text{rms}}(i)$. Eigenvectors multiplied by the mode-specific RMS amplitude (the QHO expectation value at the given temperature). Directly usable for computing thermal bond modulations.

UMAP (Uniform Manifold Approximation and Projection). Topology-preserving dimensionality-reduction method (McInnes 2018). Local and global structures are preserved simultaneously; deterministically reproducible with a seed.

Hydrogen bond. Weak but directional interaction between a proton donor (X-H with $X =$ N, O) and an acceptor (Y with a lone pair, $Y =$ O, N). Energies typically 5–30 kJ/mol. Responsible in proteins for secondary structure and PCET reaction channels.

D Algorithm pseudocodes

The central algorithms of the tool are documented here as pseudocode. Language level: between formal mathematics and Python – no type details, but all non-trivial control flow explicit. Variable names follow the module names in the code.

The pseudocodes are *not* 1:1 implementations, but sketches of the logic. The actual code in `src/modenanalyse_2fe2s/` has additional performance optimizations, caching, logging, and error handling that are omitted here for readability.

D.1 Cluster detection

Central task: find all [2Fe-2S] clusters from a list of atoms. In multi-cluster systems (glutaredoxin dimers etc.), several clusters must be unambiguously grouped and sorted.

Pseudocode 1 FindAllClusters

Input: Atom list $A = \{a_1, \ldots, a_N\}$ with element + Cartesian position; cutoffs $c_{\mathrm{FeFe}}, c_{\mathrm{FeS}}$ in Å

Output: List of clusters $\mathcal{C}_1, \mathcal{C}_2, \ldots$, sorted by Fe-Fe distance ascending

```
Fe ← {a ∈ A : element(a) = Fe}
S ← {a ∈ A : element(a) = S}
if |Fe| < 2 or |S| < 2 then
    return ∅                                              ▷ no cluster possible
end if
Form all Fe pairs (f_i, f_j) with ‖r(f_i) − r(f_j)‖ ≤ c_FeFe
Sort the Fe pairs by distance ascending
used ← ∅                                                  ▷ set of atoms already assigned
cluster_list ← ∅
for all Fe pairs (f_i, f_j) in sorted order do
    if f_i ∈ used or f_j ∈ used then
        continue                                          ▷ atom already assigned
    end if
    S_μ ← {s ∈ S : S bridges f_i, f_j with ‖r(s) − r(f_i)‖ ≤ c_FeS and ‖r(s) − r(f_j)‖ ≤ c_FeS}
    if |S_μ| ≥ 2 then                                     ▷ enough bridging sulfurs?
        (s_1, s_2) ← the two closest S in S_μ
        cluster_list ← cluster_list ∪ {{f_i, f_j, s_1, s_2}}
        used ← used ∪ {f_i, f_j, s_1, s_2}
    end if
end for
return cluster_list
```

Algorithm.

Implementation subtleties.

- Sorting Fe pairs by distance is *deterministic* and ensures the tightest cluster is assigned first. Hence `cluster_index = 0` is always the tightest one, regardless of atom order in the log file.
- The cutoffs c_{FeFe} and c_{FeS} are config parameters (defaults 3.5 and 3.0 Å). For very extended clusters (e.g. [4Fe-4S] or Mo-Fe cofactor), these may need to be raised – but then the tool only reads the [2Fe-2S] sub-fragment.
- Once an atom has been "consumed" by a tighter cluster, it is not reused for another cluster. Hence the algorithm finds the true number of clusters even in densely packed multi-cluster systems.

D.2 Cluster normal via SVD

From the four cluster atoms, the best-fit plane and hence the cluster normal $\hat{\mathbf{n}}$ is determined by singular-value decomposition.

Pseudocode 2 ClusterNormal

Input: Cluster atoms $\mathcal{C} = \{a_1, a_2, a_3, a_4\}$ with positions $\mathbf{r}_1, \ldots, \mathbf{r}_4 \in \mathbb{R}^3$

Output: Unit vector $\hat{\mathbf{n}}$ perpendicular to the best-fit plane

$\bar{\mathbf{r}} \leftarrow \frac{1}{4} \sum_{i=1}^{4} \mathbf{r}_i$ ▷ centroid

Construct the centered matrix $\tilde{\mathbf{R}} \in \mathbb{R}^{4\times 3}$: $\tilde{\mathbf{R}}_{i,:} = \mathbf{r}_i - \bar{\mathbf{r}}$

$\mathbf{U}, \boldsymbol{\Sigma}, \mathbf{V}^\top \leftarrow \mathrm{SVD}(\tilde{\mathbf{R}})$ ▷ $\mathbf{V}$ is an orthogonal 3×3 matrix

$\hat{\mathbf{n}} \leftarrow \mathbf{V}_{:,3}$ ▷ third column = direction of smallest variance

if $\hat{\mathbf{n}}_z < 0$ **then** ▷ sign convention

    $\hat{\mathbf{n}} \leftarrow -\hat{\mathbf{n}}$

end if

return $\hat{\mathbf{n}}$

Remark: the folding residual is Σ_{33} (the smallest singular value). Values > 0.1 Å indicate a strongly non-planar cluster – unusual for native [2Fe-2S] clusters but possible in strongly distorted mutants.

D.3 Kabsch alignment (PDB ↔ Gaussian)

When a PDB file is provided, the PDB coordinates must be mapped to the Gaussian coordinates – the geometries are usually rotation-shifted relative to each other. The Kabsch algorithm [17] provides the optimal rotation.

Pseudocode 3 KabschAlign

Input: Point sets $\mathbf{P}, \mathbf{Q} \in \mathbb{R}^{n\times 3}$, in correspondence (same atom order)

Output: Rotation $\mathbf{R}$, translation $\mathbf{t}$, RMSD

$\bar{\mathbf{p}} \leftarrow \frac{1}{n}\sum_i \mathbf{P}_{i,:}$, $\bar{\mathbf{q}} \leftarrow \frac{1}{n}\sum_i \mathbf{Q}_{i,:}$

$\mathbf{P}' \leftarrow \mathbf{P} - \bar{\mathbf{p}}$, $\mathbf{Q}' \leftarrow \mathbf{Q} - \bar{\mathbf{q}}$ ▷ shift each centroid to the origin

$\mathbf{H} \leftarrow \mathbf{P}'^{\top}\mathbf{Q}'$ ▷ 3×3 cross-correlation

$\mathbf{U}, \mathbf{\Sigma}, \mathbf{V}^{\top} \leftarrow \mathrm{SVD}(\mathbf{H})$

$d \leftarrow \mathrm{sign}(\det(\mathbf{V}\mathbf{U}^{\top}))$ ▷ reflection sign

$\mathbf{D} \leftarrow \mathrm{diag}(1, 1, d)$

$\mathbf{R} \leftarrow \mathbf{V}\mathbf{D}\mathbf{U}^{\top}$ ▷ optimal rotation

$\mathbf{t} \leftarrow \bar{\mathbf{q}} - \mathbf{R}\bar{\mathbf{p}}$

$\mathrm{RMSD} \leftarrow \sqrt{\frac{1}{n}\|\mathbf{R}\mathbf{P}' - \mathbf{Q}'\|_F^2}$

return $(\mathbf{R}, \mathbf{t}, \mathrm{RMSD})$

What the D matrix does. The "naive" solution $\mathbf{R} = \mathbf{V}\mathbf{U}^{\top}$ may be a reflection (determinant -1) if the two point sets are chirally different (e.g. accidentally mirrored). The $\mathbf{D}$ matrix corrects the sign of the third singular-vector axis, so that an actual rotation (determinant $+1$) is returned. For correctly corresponding point sets, $d = +1$, hence $\mathbf{D} = \mathbf{I}$.

D.4 Mode loop with thermal scaling

The central loop of the tool: for each vibrational mode the eigenvector is read, thermally scaled, and fed into the OOP/INP, reorganization, and SCSD pipelines.

Pseudocode 4 AnalyzeMode

Input: Mode index i, frequency ω_i, reduced mass μ_i, temperature T, eigenvector $\mathbf{e}^{(i)}$ (mass-weighted, normalized), cluster atoms $\mathcal{C}$, ligands $\mathcal{L}$

Output: Result dict r with OOP%, Δr_X, λ_X, SCSD contributions, kernel classification

▷ 1. Thermal RMS amplitude (quantum expectation value)
$u_{\text{rms}}(i) \leftarrow \sqrt{\frac{\hbar}{2\mu_i\omega_i}\coth\left(\frac{\hbar\omega_i}{2k_BT}\right)}$
$\mathbf{e}_a^{\text{therm}} \leftarrow \mathbf{e}_a^{(i)} \cdot u_{\text{rms}}(i) \quad \forall a \in \mathcal{C} \cup \mathcal{L}$
▷ 2. OOP/INP classification
$\mathbf{e}_a^{\text{OOP}} \leftarrow (\mathbf{e}_a^{\text{therm}} \cdot \hat{\mathbf{n}})\,\hat{\mathbf{n}}$
$\alpha_{\text{OOP}} \leftarrow 100 \cdot \frac{\sum_{a\in\mathcal{C}} \|\mathbf{e}_a^{\text{OOP}}\|^2}{\sum_{a\in\mathcal{C}} \|\mathbf{e}_a^{\text{therm}}\|^2}$
r.oop_pct $\leftarrow \alpha_{\text{OOP}}$
r.type $\leftarrow$ ClassifyOOP($\alpha_{\text{OOP}}, \theta_{\text{OOP}} = 60$)
▷ 3. Bond modulations
for all Bonding channels $X \in \{$FeFe, FeN, FeS, NH, HA$\}$ **do**
  $\Delta r_X(i) \leftarrow$ ComputeBondModulation($X, \mathbf{e}^{\text{therm}}$, geometry) ▷ see §2.12
  $\lambda_X(i) \leftarrow \frac{1}{2}\mu_i\omega_i^2\,(\Delta r_X(i))^2$
end for
▷ 4. SCSD symmetry decomposition (optional)
if `cfg.analyze_scsd` **then**
  Displacement $\leftarrow \mathbf{e}^{\text{therm}}|_{\mathcal{C}}$ ▷ cluster atoms only
  Irrep contributions $\leftarrow$ ProjectOnD2hIrreps(Displacement, reference geometry)
  r.scsd $\leftarrow$ Irrep contributions
end if
▷ 5. Heuristic kernel classification
r.kern_type $\leftarrow$ ClassifyKern($\mathbf{e}^{\text{therm}}, \mathcal{C}, \alpha_{\text{OOP}}$)
return r

Subtleties:

- For `temp_k = None`, the thermal scaling is skipped; u_{rms} is replaced by `cfg.amplitude` (default 1.0). This is the "classical" mode in which all modes are weighted equally.
- At low frequency and high temperature the coth argument is small and u_{rms} becomes large – for modes below $\sim 30\,\text{cm}^{-1}$ this can yield contributions $> 50\,\%$ of the total reorganization in a typical filter range. Hence the recommendation: always set `freq_min` to 50 or 100 if low-frequency skeletal motions should be suppressed.
- In the "no-His" path (no histidine in the cluster), the NH and HA channels are zeroed *before* the reorganization loop – not via zero Δr, but directly by the upstream call in `runner.py`. This saves geometric computations and makes the path deterministically clean.

D.5 Reorganization aggregation per mode

Per mode, the bond modulations Δr_X are computed from the thermally scaled eigenvectors.

Pseudocode 5 ComputeBondModulation (FeFe example)

Input: Bond atoms a, b (with positions + thermally scaled displacements $\mathbf{e}_a, \mathbf{e}_b$)

Output: Bond modulation Δr (signed)

$\mathbf{r}_a^{\text{new}} \leftarrow \mathbf{r}_a + \mathbf{e}_a$

$\mathbf{r}_b^{\text{new}} \leftarrow \mathbf{r}_b + \mathbf{e}_b$

$r_{\text{new}} \leftarrow \|\mathbf{r}_a^{\text{new}} - \mathbf{r}_b^{\text{new}}\|$

$r_{\text{eq}} \leftarrow \|\mathbf{r}_a - \mathbf{r}_b\|$

$\Delta r \leftarrow r_{\text{new}} - r_{\text{eq}}$

return Δr

For multi-bond channels (e.g. FeS with four Fe-S bonds), ComputeBondModulation is called per bond, and the contributions are aggregated *sign-free* via RSS (§A):

$$\Delta r_X^{\text{rss}} = \sqrt{\sum_{(a,b)\in X} (\Delta r_{(a,b)})^2} \tag{56}$$

The channel contribution $\lambda_X = \frac{1}{2}\mu\omega^2(\Delta r_X^{\text{rss}})^2$ is then identical with the sum of the pair contributions:

$$\lambda_X = \sum_{(a,b)\in X} \tfrac{1}{2}\mu\omega^2(\Delta r_{(a,b)})^2. \tag{57}$$

This Marcus-Hush additivity is the central consistency condition and smoke test (§4.5).

D.6 HA reaction-coordinate computation

The HA channel measures the H-acceptor distance modulation along the reaction coordinate (rather than the naive H-Acc distance).

Pseudocode 6 ComputeHAReactionCoord

Input: $\mathrm{N}_{\text{donor}}$, H, acceptor A (positions + displacements)

Output: Reaction-coordinate modulation Δr_{HA}

$\hat{\mathbf{n}}_{N\to A} \leftarrow \frac{\mathbf{r}_A - \mathbf{r}_N}{\|\mathbf{r}_A - \mathbf{r}_N\|}$ ▷ donor-acceptor axis, unit vector

$\Delta\mathbf{e} \leftarrow \mathbf{e}_H - \mathbf{e}_N$ ▷ H motion relative to the donor

$\Delta r_{\text{HA}} \leftarrow \Delta\mathbf{e} \cdot \hat{\mathbf{n}}_{N\to A}$ ▷ projection onto the reaction coordinate

return Δr_{HA}

Why this approach? The naive H-Acc distance measures the *total motion* of the H atom relative to the acceptor atom. Pure skeletal modes (in which the entire complex moves collectively) would then show a large Δr, even though the proton is not moving *relative* to the donor at all

– such modes are not real PCET-relevant modes. The reaction-coordinate convention extracts only the genuine proton motion.

D.7 Modulation spectrum and cumulative reorganization curve

From the per-mode values, frequency-resolved spectra are built.

Pseudocode 7 ModulationSpectrum

Input: Mode list with $(\omega_i, \Delta r_X(i))$, broadening σ, frequency grid $\boldsymbol{\omega}$

Output: Modulation spectrum $M_X(\omega)$

$M_X(\omega) \leftarrow 0 \quad \forall\omega \in \boldsymbol{\omega}$
for all Modes i **do**
    $g_i(\omega) \leftarrow \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(\omega-\omega_i)^2}{2\sigma^2}\right)$
    $M_X(\omega) \leftarrow M_X(\omega) + |\Delta r_X(i)| \cdot g_i(\omega)$
end for
return $M_X(\boldsymbol{\omega})$

Pseudocode 8 CumulativeReorg

Input: Mode list with $(\omega_i, \lambda_X(i))$, sorted ascending in ω

Output: Cumulative curve $\Lambda_X(\omega)$

$\Lambda \leftarrow 0$
for all Modes i (sorted ascending) **do**
    Write (ω_i^-, Λ) and $(\omega_i^+, \Lambda + \lambda_X(i))$ ▷ step function: jump at ω_i
    $\Lambda \leftarrow \Lambda + \lambda_X(i)$
end for
Write the end value $(\omega_{\max}, \Lambda)$
return $\Lambda_X(\omega)$

Both spectra are written to the sheets `Modulation_spectra` and `Lambda_cumulative` respectively (§5).

D.8 HDBSCAN clustering of the embedding

After the UMAP projection to 2D, the points are clustered density-based with HDBSCAN [38].

Pseudocode 9 HDBSCANCluster

Input: 2D embedding points $\{x_1, \ldots, x_N\}$, min_cluster_size, min_samples
Output: Cluster labels $\ell_1, \ldots, \ell_N \in \{-1, 0, 1, \ldots\}$ (-1 = noise)

▷ 1. Weighted graph: each point has distances to its K nearest neighbors
core_dist(x_i) ← distance to the `min_samples`-th neighbor
mutual_reachability(x_i, x_j) ← max(core_dist(x_i), core_dist(x_j), $\|x_i - x_j\|$)
▷ 2. Minimum spanning tree of the mutual_reachability matrix
MST ← KruskalMST(mutual_reachability)
▷ 3. Hierarchical cluster tree
Tree ← ConvertMSTtoHierarchy(MST)
▷ 4. Select stable clusters (excess of mass)
cluster_list ← ExcessOfMassSelection(Tree, min_cluster_size)
▷ 5. Assign points to clusters or label them as noise
for all Points x_i **do**
  if x_i is a member of a stable cluster C_k **then**
    $\ell_i \leftarrow k$
  else
    $\ell_i \leftarrow -1$ ▷ noise
  end if
end for
return $\boldsymbol{\ell}$

Why HDBSCAN instead of DBSCAN?

- HDBSCAN does not need a global ϵ threshold (DBSCAN does – which is problematic for varying cluster densities).
- HDBSCAN finds clusters of varying density automatically (via its hierarchical construction).
- HDBSCAN is robust against outliers (they automatically end up as noise, without a threshold needing to be tuned).

In the default configuration, the tool uses `min_cluster_size = 8` and `min_samples = 5`; see §2.8 for the rationale.

E Excel sheet full reference

Per run, up to *four* Excel workbooks are produced. This appendix documents every column of every sheet, so that table values are interpretable without recourse to the code. Sheet names are listed in the order they are created in `export.py` via `wb.create_sheet(...)`.

E.1 Main analysis workbook (`*_analysis.xlsx`)

Contains 16 sheets with all mode-related analyses, cluster geometry, and SCSD symmetry decomposition. The required workbook for any publication.

`Mode_analysis` – mode-by-mode detail table

One row per vibrational mode (typically 60–800 modes, depending on the `freq_max` filter). 32 columns:

Column	Unit	Meaning
`Mode #`	–	mode index from the log file (1-based)
`Frequency`	cm^{-1}	frequency ω_i
`u_rms`	Å	thermal RMS amplitude $\sqrt{\hbar/(2\mu\omega)\coth(\hbar\omega/2k_BT)}$
`Red.Mass`	AMU	reduced mass μ_i
`Frc.Const.`	mdyn/Å	force constant
`Sym.`	–	Mulliken symbol from Gaussian (e.g., A, B_g)
`Type`	–	OOP/INP classification (binary, threshold 60 %)
`Type (fine)`	–	7-level OOP/INP classification
`Prec.`	–	HP/Std eigenvector source
`Lig OOP%`	%	OOP fraction of the mode on the ligand sphere
`±sigma`	%	bootstrap scatter of the estimate
`Lig INP%`	%	INP fraction on the ligands
`±sigma`	%	scatter
`Lig |d|`	Å	ligand displacement (norm of the displacement)
`±sigma`	Å	scatter
`2nd OOP%`	%	like `Lig OOP%` for the 2nd ligand sphere (e.g., backbone N)
`±sigma`	%	scatter
`2nd INP%`	%	2nd ligand sphere INP
`±sigma`	%	scatter
`2nd |d|`	Å	2nd ligand sphere displacement

Columns 21–32 (not listed above) contain kernel contributions (`Kern OOP%`, `Kern INP%`, etc.) and OOP/INP contributions of the individual cluster atoms (Fe_1, Fe_2, S_1, S_2).

`Groups_OOP, _INP, _Angle, _Tors`

Four separate sheets, each with one row per mode and one column per ligand group (Cys_n, His_n, etc.). Values are the OOP/INP contributions or bond angles per group. Useful for the ligand-specific analysis of individual modes.

`Fe_S_Cys` and `Fe_N_His`

One column per Fe-ligand bond (e.g., Fe1-Cys72-S, Fe2-Cys83-S, etc.). One row per mode, with content:

- `stretch`: fraction of the mode motion along the bond axis.
- `bend_inp`: in-plane bend contribution (within the cluster plane).
- `bend_oop`: OOP bend contribution (perpendicular to the cluster plane).

These sheets are the data source for the mode-loop aggregation (§2.12); they show how a single mode acts on each Fe-ligand bond.

`Kernel_Scores` – heuristic cluster decomposition

One row per mode, 10 columns with the heuristic cluster score functions from §2.6:

- `Translation`, `Umbrella`, `Hinge-Fe`, `Hinge-S`: cluster motion classes.
- `OOP-Twist`, `Fe-stretch`, `S-stretch`, `Breathing`, `Rhombus-shear`, `Rotation-ip`: six in-plane / OOP motion modes.

Important note in the sheet header: the kernel scores are *heuristic geometric projections*; they do not generally sum to 1 and are not orthogonal. For a **rigorous symmetry decomposition**, the sheet refers to `SCSD` (see below).

`Equilibrium_distances`

Table of the equilibrium bond lengths of all relevant cluster and ligand bonds. One row per bond. Columns: bond label, distance in Å, atom indices.

`Coordination` – ligand and group atom mapping *(since v1.0.3)*

Diagnostic sheet that exposes the PDB-to-Gaussian atom assignments used by all subsequent analyses. Three blocks, separated by blank rows:

- **Ligands**: one row per Fe-coordinating ligand (`LigandInfo` in `geometry.py`). Columns: `label` (e.g. "Cys 207"), `element` (S/N/O), `fe_center` (1-based Gaussian center of the bonded Fe), `lig_center` (Gaussian center of the donor atom), `bond_len_A`, `his_protonated` (boolean), `h_center` (Gaussian center of the imidazole NH-H if any), `his_hn_center` (the donor N that carries that H, NE2 or ND1).
- **Groups (`group_map`)**: one row per atom group used in the `Groups_OOP/INP/Angle/Tors` sheets. Columns: `group` (label), `centers` (comma-separated list of 1-based Gaussian centers).

- **Cluster atoms**: a 4-row block with Fe_1, Fe_2, S_1, S_2, their Gaussian centers, and the equilibrium (x, y, z) in Å.

This sheet is the first place to check when an Excel row looks unexpectedly empty: an empty `centers` list or a missing `h_center` explains a downstream all-zero row in the `Groups_*`, `Fe_*_*` or `His_HN` sheets (§2.3).

`SCSD` – rigorous D_{2h} symmetry decomposition

Per mode, 8 columns for the contributions to the D_{2h} irreps: $A_g, B_{1g}, B_{2g}, B_{3g}, A_u, B_{1u}, B_{2u}, B_{3u}$. Values in % of the cluster-mode contribution. In contrast to `Kernel_Scores`, these values are *orthogonal* and sum exactly to 100 %.

Method: Kingsbury & Senge (Chem. Sci. 15, 13638, 2024) with the canonical D_{2h} reference (Fe-Fe = 2.73 Å, Fe-S = 2.20 Å, axes x = Fe-Fe, y = S-S, z = normal).

`Reorganization_energy`

The main sheet for the reorganization pipeline. One row per mode, 21 columns:

Column	Unit	Meaning
`freq_cm1`	cm^{-1}	mode frequency
`dr_FeFe_rss_a`	Å	RSS aggregate of the Fe-Fe modulation
`dr_FeFe_sum_signed_a`	Å	signed-sum aggregate (can be negative)
`lambda_FeFe_pair_cm1`	cm^{-1}	reorg per bond pair (sub-breakdown)
`lambda_FeFe_mode_cm1`	cm^{-1}	reorg per mode (aggregate over all pairs of the channel)
Repeated for channels: FeN, FeS, NH, HA (4 columns each)		

Convention: `rss` (Root-Sum-Squared) is always positive and makes Marcus-Hush additivity exact (§D.5). `sum_signed` is the signed sum over all bond modulations of the channel; it shows whether the bond modulations are coherent (Sum ≈ rss) or incoherent (Sum ≪ rss).

Mode-vs.-pair convention: `lambda_X_pair` = individual bond-pair contributions summed; `lambda_X_mode` = aggregated per mode with $\mu_i \omega_i^2$ weighting. Both are preserved for different analysis purposes.

`Reorg_total` – system aggregates per channel

Five rows (one per channel: FeFe, FeN, FeS, NH, HA). Columns:

- `Channel`: name of the bonding channel.
- `Lambda_total_pair_cm1`: Λ_X^{total} from pair aggregation.

- Lambda_total_mode_cm1: Λ_X^{total} from mode aggregation.
- n_modes_contributing: number of modes with non-trivial contribution ($\lambda_X(i) > 10^{-6}\,\text{cm}^{-1}$).

These are the central *publication-ready values* of the reorganization pipeline.

Reorg_per_bond – breakdown per individual bond

One row per individual bond (e.g., Fe1-Cys72-S, Fe1-Fe2, Fe1-S_b1, etc.). Six columns:

- Bond: bond label.
- Parent_channel: which channel (FeFe, FeS, FeN, ...) the bond belongs to.
- Acceptor_weight: for the HA channel, the Gaussian weight of the H-acceptor pair (default 1.0 for the main acceptor).
- Lambda_total_pair_cm1, Lambda_total_mode_cm1: as in Reorg_total, but per bond.
- n_modes_contributing: modes with contribution $> 10^{-6}\,\text{cm}^{-1}$.

Allows the question to be answered: *which individual Fe-S bond contributes most to Λ_{FeS}?*

Modulation_spectra

Frequency-resolved modulation spectra $M_X(\omega)$, plus co-spectra for PCET analysis. One row per frequency grid point (default: $0.5\,\text{cm}^{-1}$ step). 8 columns:

- freq_cm1: frequency grid point.
- M_FeFe, M_FeN, M_FeS, M_NH, M_HA: modulation spectra $M_X(\omega) = \sum_i |\Delta r_X(i)| \cdot \mathcal{N}(\omega - \omega_i, \sigma)$ with default $\sigma = 5\,\text{cm}^{-1}$.
- C_ET_FeS: co-spectrum $\sqrt{M_{\text{FeFe}}(\omega) \cdot M_{\text{FeS}}(\omega)}$ – marks frequency regions where FeFe and FeS are modulated simultaneously. High-indicator for ET-relevant cluster-breathing modes.
- C_PCET: co-spectrum $\sqrt{M_{\text{NH}}(\omega) \cdot M_{\text{HA}}(\omega)}$ – marks PCET-relevant modes (simultaneous N-H and H-acceptor modulation).

Lambda_cumulative

Cumulative reorganization curves $\Lambda_X(\omega) = \sum_{\omega_i \le \omega} \lambda_X(i)$. One row per frequency grid point. 6 columns: freq_cm1, Lambda_FeFe, Lambda_FeN, Lambda_FeS, Lambda_NH, Lambda_HA.

Each column is a **monotone step function**: jumps mark modes with λ_X contribution. Directly overlayable on an NRVS spectrum for band identification.

`B_factors`

Per atom: thermal B factors $B_a = \frac{8\pi^2}{3}\langle u_a^2 \rangle$ from the QHO averaging over all modes. Columns: atom index, element, B factor in $\mathring{A}^2$.

`Info`

Per-run metadata: log-file path, number of modes, frequency range, cluster geometry, ligand list, configuration parameters. Full-text description of the run in a single sheet.

E.2 Embeddings workbook (`*_analysis_Embeddings.xlsx`)

Contains the UMAP projections and HDBSCAN clusterings. The workbook hosts *three* parallel embeddings of the same set of modes, each in its own pair of `*_clusters` / `*_profile` sheets, plus the per-mode raw C_α amplitude matrix used as input for the Ca-UMAP.

`Projections`

Global UMAP. One row per mode. Columns: `Mode #`, `Frequency`, `UMAP_x`, `UMAP_y`. Computed on the full feature matrix built by `embedding.build_feature_matrix` (Marcus-Hush reorganization features + bond modulation features + kernel scores). Directly plottable as a 2D embedding.

`Ca_amplitudes`

Per mode and per C_α residue: the thermally scaled $\|\mathbf{e}\|$ amplitude of the backbone α-carbon for the residue. Used as the feature matrix for the Ca-UMAP (below). Format: rows = modes, columns = residue numbers.

`Cluster` / `Cluster_Profile`

Global UMAP cluster assignments. `Cluster` has one row per mode with its HDBSCAN cluster ID (`cluster_label`: -1 = noise; $0, 1, \ldots$ = cluster index), distance to cluster centre, and all feature values. `Cluster_Profile` has one row per HDBSCAN cluster with the mean feature vector, Fisher-F statistic per feature, and a ranked list of representative modes (closest-to-centroid).

`SSE_UMAP_clusters` / `SSE_UMAP_profile` *(since v1.0.3)*

Secondary-structure UMAP, restricted to modes that have at least one non-zero entry in the per-SSE-element feature row (`sse`-dict non-empty after `analyze_all_sse`). Useful for identifying collective protein-skeleton modes that visit multiple secondary structures simultaneously. The `*_clusters` sheet lists modes with their SSE-UMAP coordinates and HDBSCAN cluster ID; the `*_profile` sheet gives the Z-score profile of the top-30 discriminating SSE-elements per cluster.

Ca_UMAP_clusters / Ca_UMAP_profile *(since v1.0.3)*

C_α-amplitude UMAP. Each mode is represented as a point in an N_{C_α}-dimensional space, one dimension per backbone α-carbon (the columns of `Ca_amplitudes`). Features are z-scored per residue before UMAP, so that no single high-amplitude residue dominates the embedding distance.

Conceptually, the three UMAPs answer complementary questions:

- *Global UMAP* (`Projections`): which modes couple to the same reorganization channels (Fe-X, NH, HA)? Useful for identifying ET-active and PCET-active modes respectively.
- *SSE-UMAP* (`SSE_UMAP_*`): which modes share the same secondary-structure footprint? Useful for identifying domain-collective modes.
- *Ca-UMAP* (`Ca_UMAP_*`): which modes have similar spatial distributions of backbone motion? Useful for identifying long-range allosteric pathways and modes localised to specific regions of the protein.

Companion PNGs (only if `cfg.export_embedding_plots = true`): `_embedding_UMAP.png` (2 panels), `_embedding_SSE_UMAP.png` (3 panels: frequency / mode type / cluster), `_embedding_Ca_UMAP.png` (3 panels), and `_ca_amplitudes_heatmap.png` (log-scale residue × frequency heatmap of the C_α amplitudes).

E.3 Interpolated pDOS workbook (`*_analysis_interp0.05.xlsx`)

A single sheet `Interpolated` with the pDOS analysis on a fine frequency grid (default $0.05\,\mathrm{cm}^{-1}$). Columns: `freq_cm1`, `pDOS_Fe1`, `pDOS_Fe2`, `pDOS_S1`, `pDOS_S2`, `pDOS_total`. Used for direct comparison with NRVS spectra.

E.4 Secondary-structure workbook (`*_analysis_SSE.xlsx`, optional)

Created only when a PDB file with (or DSSP-extracted) secondary-structure records is provided. Typically contains 9–11 sheets, depending on SSE complexity:

Per-SSE-element sheets (`SSE_*`)

One sheet per SSE metric: `SSE_amplitude_mean`, `SSE_amplitude_max`, `SSE_com_amplitude` (center-of-mass amplitude), `SSE_lateral_std` (lateral motion scatter), `SSE_lateral_amplitude`, `SSE_stretching` (axial length change), `SSE_axial_amplitude`, `SSE_tilting_angle` (helix tilt), `SSE_internal_amplitude` (internal motion).

Each sheet has one row per mode and one column per SSE element.

`SSE_UMAP_Cluster` and `SSE_UMAP_Profile`

Like `UMAP_Cluster` and `UMAP_Cluster_Profile`, but on the SSE feature matrix (instead of the cluster-ligand matrix). Identifies structural clusters of modes based on their effect on secondary-structure elements.

> **Which sheets are required for a publication?** For a reorganization manuscript table, `Reorg_total` and `Reorg_per_bond` (main analysis) are typically sufficient. For band assignment, `Lambda_cumulative` is added (NRVS overlay). For mode characterization, the sheet `Mode_analysis` is needed and (for symmetry-relevant questions) `SCSD`. The rest serves internal diagnostic purposes.

References

[1] C. J. Kingsbury, M. O. Senge, *Quantifying near-symmetric molecular distortion using symmetry-coordinate structural decomposition*, **Chem. Sci. 15**, 13638–13649 (2024). DOI: 10.1039/D4SC01670J. Web implementation: https://www.kingsbury.id.au/scsd.

[2] C. J. Kingsbury, M. O. Senge, *Quantifying near-symmetric molecular distortion using symmetry-coordinate structural decomposition*, ChemRxiv preprint, doi:10.26434/chemrxiv-2024-6b25s (2024).

[3] S. Iwata, M. Saynovits, T. A. Link, H. Michel, *Structure of a water soluble fragment of the 'Rieske' iron-sulfur protein of the bovine heart mitochondrial cytochrome bc_1 complex determined by MAD phasing at 1.5 Å resolution*, **Structure 4**, 567–579 (1996).

[4] P. Sauer, M. Heyden, *Frequency-selective anharmonic mode analysis of thermally excited vibrations in proteins*, **J. Chem. Phys.** (2023).

[5] K. Achterhold, F. G. Parak, *Protein dynamics: determination of anisotropic vibrations at the heme iron of myoglobin*, **J. Phys.: Condens. Matter 14**, S1399–S1421 (2002).

[6] H. Paulsen, H. Winkler, A. X. Trautwein *et al.*, *Measurement and simulation of nuclear inelastic-scattering spectra of molecular crystals*, **Phys. Rev. B 59**, 975 (1999).

[7] L. B. Gee, V. Pelmenschikov, C. Mons, N. Mishra, H. Wang, Y. Yoda, K. Tamasaku, M.-P. Golinelli-Cohen, S. P. Cramer, *NRVS and DFT of mitoNEET: Understanding the special vibrational structure of a [2Fe–2S] cluster with $(Cys)_3(His)_1$ ligation*, **Biochemistry 60**, 2419–2424 (2021). DOI: 10.1021/acs.biochem.1c00252.

[8] A. R. Conlan, M. L. Paddock, C. Homer *et al.*, *Mutation of the His ligand in mitoNEET stabilizes the 2Fe–2S cluster despite conformational heterogeneity in the ligand environment*, **Acta Cryst. F 67**, 516–523 (2011).

[9] N. Go, T. Noguti, T. Nishikawa, *Dynamics of a small globular protein in terms of low-frequency vibrational modes*, **Proc. Natl. Acad. Sci. USA 80**, 3696 (1983).

[10] F. Parak, E. W. Knapp, *A consistent picture of protein dynamics*, **Proc. Natl. Acad. Sci. USA 81**, 7088 (1984).

[11] E. B. Wilson, J. C. Decius, P. C. Cross, *Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra*, McGraw-Hill, New York (1955); Dover Reprint (1980).

[12] W. Sturhahn, *Nuclear resonant spectroscopy*, **J. Phys.: Condens. Matter 16**, S497–S530 (2004).

[13] W. Sturhahn, A. Chumakov, *Lamb-Mössbauer factor and second-order Doppler shift from inelastic nuclear resonant absorption*, **Hyperfine Interact. 123/124**, 809–824 (2000).

[14] H. J. Lipkin, *Mössbauer sum rules for use with synchrotron sources*, **Phys. Rev. B 52**, 10073–10079 (1995).

[15] K. S. Singwi, A. Sjölander, *Resonance absorption of nuclear gamma rays and the dynamics of atomic motions*, **Phys. Rev. 120**, 1093–1102 (1960).

[16] V. G. Kohn, A. I. Chumakov, R. Rüffer, *Nuclear resonant inelastic absorption of synchrotron radiation in an anisotropic single crystal*, **Phys. Rev. B 58**, 8437–8444 (1998).

[17] W. Kabsch, *A solution for the best rotation to relate two sets of vectors*, **Acta Cryst. A 32**, 922–923 (1976).

[18] S. Hammes-Schiffer, A. A. Stuchebrukhov, *Theory of coupled electron and proton transfer reactions*, **Chem. Rev. 110**, 6939–6960 (2010).

[19] R. A. Marcus, *On the theory of oxidation-reduction reactions involving electron transfer. I*, **J. Chem. Phys. 24**, 966–978 (1956). DOI: 10.1063/1.1742723.

[20] R. A. Marcus, *Electron transfer reactions in chemistry. Theory and experiment* (Nobel Lecture), **Angew. Chem. Int. Ed. Engl. 32**, 1111–1121 (1993). DOI: 10.1002/anie.199311113.

[21] R. A. Marcus, N. Sutin, *Electron transfers in chemistry and biology*, **Biochim. Biophys. Acta 811**, 265–322 (1985).

[22] W. Sturhahn, V. G. Kohn, *Theoretical aspects of incoherent nuclear resonant scattering*, **Hyperfine Interact. 123/124**, 367–377 (1999).

[23] A. I. Chumakov, W. Sturhahn, *Experimental aspects of inelastic nuclear resonant scattering*, **Hyperfine Interact. 123/124**, 781–808 (1999).

[24] A. A. Maradudin, P. A. Flinn, *Anharmonic contributions to the Debye-Waller factor*, **Phys. Rev. 129**, 2529 (1963).

[25] H. J. Lipkin, *Some simple features of the Mößbauer effect*, **Ann. Phys.** (N.Y.) **9**, 332 (1960).

[26] M. Y. Hu, W. Sturhahn, T. S. Toellner *et al.*, *Measuring velocity of sound with nuclear resonant inelastic X-ray scattering*, **Phys. Rev. B 67**, 094304 (2003).

[27] B. M. Leu, M. Z. Hu, J. Zhao, S. P. Cramer, *Quantitative vibrational dynamics of iron in carbonyl compounds*, **J. Am. Soc. Mass Spectrom. 17**, 235 (2008).

[28] P. Thompson, D. E. Cox, J. B. Hastings, *Rietveld refinement of Debye-Scherrer synchrotron X-ray data from* Al_2O_3, **J. Appl. Crystallogr. 20**, 79 (1987).

[29] K. Achterhold, F. G. Parak, *ina32.f – Fortran source code for analysis of nuclear inelastic scattering spectra*; internal publication, TU Munich, Department of Biophysics (2005–).

[30] E. Tiesinga, P. J. Mohr, D. B. Newell, B. N. Taylor, *CODATA recommended values of the fundamental physical constants: 2018*, **Rev. Mod. Phys. 93**, 025010 (2021).

[31] K. Pearson, *On lines and planes of closest fit to systems of points in space*, **Philos. Mag. 2**, 559–572 (1901).

[32] H. Hotelling, *Analysis of a complex of statistical variables into principal components*, **J. Educ. Psychol. 24**, 417–441 (1933).

[33] I. T. Jolliffe, *Principal Component Analysis*, 2nd ed., Springer (2002).

[34] L. van der Maaten, G. Hinton, *Visualizing data using t-SNE*, **J. Mach. Learn. Res. 9**, 2579–2605 (2008).

[35] L. van der Maaten, *Accelerating t-SNE using tree-based algorithms*, **J. Mach. Learn. Res. 15**, 3221–3245 (2014).

[36] L. McInnes, J. Healy, J. Melville, *UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction*, **arXiv** 1802.03426 (2018).

[37] E. Becht, L. McInnes, J. Healy *et al.*, *Dimensionality reduction for visualizing single-cell data using UMAP*, **Nat. Biotechnol. 37**, 38–44 (2019).

[38] R. J. G. B. Campello, D. Moulavi, J. Sander, *Density-based clustering based on hierarchical density estimates*, in **Adv. Knowl. Discov. Data Min.** (PAKDD 2013), pp. 160–172.

[39] L. McInnes, J. Healy, S. Astels, *hdbscan: Hierarchical density based clustering*, **J. Open Source Softw. 2**, 205 (2017).

[40] S. Hammes-Schiffer, *Theory of proton-coupled electron transfer in energy conversion processes*, **Acc. Chem. Res. 42**, 1881–1889 (2009). DOI: 10.1021/ar9001284.

[41] S. J. Edwards, A. V. Soudackov, S. Hammes-Schiffer, *Driving force dependence of rates for nonadiabatic proton and proton-coupled electron transfer: conditions for inverted region behavior*, **J. Phys. Chem. B 113**, 14545–14548 (2009).

[42] J. P. Layfield, S. Hammes-Schiffer, *Hydrogen tunneling in enzymes and biomimetic models*, **Chem. Rev. 114**, 3466–3494 (2014).

[43] Q. Cui, M. Karplus, *Allostery and cooperativity revisited*, **Protein Sci. 17**, 1295–1307 (2008).

[44] M. Bergner, S. Dechert, S. Demeshko, C. Kupper, J. M. Mayer, F. Meyer, *Model of the MitoNEET [2Fe-2S] cluster shows proton coupled electron transfer*, **J. Am. Chem. Soc. 139**, 701–707 (2017). DOI: 10.1021/jacs.6b09180.

[45] D. Kuila, J. R. Schoonover, R. B. Dyer, C. J. Batie, D. P. Ballou, J. A. Fee, W. H. Woodruff, *Resonance Raman studies of Rieske-type proteins*, **Biochim. Biophys. Acta 1140**, 175–183 (1992).

[46] H. Wang, V. Pelmenschikov, Y. Yoda, S. P. Cramer, *NRVS of Fe-S cluster proteins and models – A bestiary of nifty normal modes*, **J. Inorg. Biochem. 270**, 112935 (2025). DOI: 10.1016/j.jinorgbio.2025.112935.

[47] J. A. Zuris, D. A. Halim, A. R. Conlan *et al.*, *Engineering the redox potential over a wide range within a new class of FeS proteins*, **J. Am. Chem. Soc. 132**, 13120–13122 (2010).

[48] T. Kuroda, S. Hammes-Schiffer, *Vibronic coupling for proton-coupled electron transfer reactions: recent insights from theory*, **Curr. Opin. Electrochem. 29**, 100789 (2021).

[49] S. Hammes-Schiffer, A. V. Soudackov, *Proton-coupled electron transfer in solution, proteins, and electrochemistry*, **J. Phys. Chem. B 112**, 14108–14123 (2008).

[50] K. Huang, A. Rhys, *Theory of light absorption and non-radiative transitions in F-centres*, **Proc. R. Soc. Lond. A 204**, 406–423 (1950).

[51] H. Beinert, R. H. Holm, E. Münck, *Iron-sulfur clusters: nature's modular, multipurpose structures*, **Science 277**, 653–659 (1997).

[52] S. Han, R. S. Czernuszewicz, T. G. Spiro, *Vibrational spectra and normal mode analysis for [2Fe-2S] protein analogs using sulfur-34, iron-54 and deuterium substitution: coupling of iron-sulfur stretching and sulfur-carbon-carbon bending modes*, **J. Am. Chem. Soc. 111**, 3496–3504 (1989).

[53] R. S. Czernuszewicz, J. LeGall, I. Moura, T. G. Spiro, *Resonance Raman spectra of rubredoxin: new assignments and vibrational coupling mechanism from iron-54/iron-56 isotope shifts and variable-wavelength excitation*, **Inorg. Chem. 25**, 696–700 (1986).

[54] U. Bergmann, W. Sturhahn, D. E. Linn, F. E. Jenney, M. W. W. Adams, K. Rupnik, B. J. Hales, E. E. Alp, A. Mayse, S. P. Cramer, *Observation of Fe-H/D modes by nuclear resonant vibrational spectroscopy*, **J. Am. Chem. Soc. 125**, 4016–4017 (2003).

[55] C. T. Saouma, M. M. Pinney, J. M. Mayer, *Electron transfer and proton-coupled electron transfer reactivity and self-exchange of synthetic [2Fe-2S] complexes: Models for Rieske and mitoNEET clusters*, **Inorg. Chem. 53**, 3153–3161 (2014).

[56] A. Albers, S. Demeshko, S. Dechert, C. T. Saouma, J. M. Mayer, F. Meyer, *Fast proton-coupled electron transfer observed for a high-fidelity structural and functional [2Fe-2S] Rieske model*, **J. Am. Chem. Soc. 136**, 3946–3954 (2014).

[57] S. J. Cyvin, *Molecular Vibrations and Mean Square Amplitudes*, Universitetsforlaget, Oslo / Elsevier, Amsterdam (1968).

[58] J. W. Ochterski, *Vibrational Analysis in Gaussian*, Gaussian, Inc., White Paper (1999). https://gaussian.com/vib/

[59] M. J. Frisch *et al.*, *Gaussian 16, Revision C.01*, Gaussian, Inc., Wallingford CT (2016).

[60] F. Neese, *The ORCA program system: user's manual*, Version 6.0, Max-Planck-Institut für Kohlenforschung, Mülheim a. d. Ruhr (2023). https://www.faccts.de/docs/orca/6.0/manual/

[61] F. Neese, *Software update: The ORCA program system—Version 5.0*, **WIREs Comput. Mol. Sci. 12**, e1606 (2022).